Zoo507 Mid Term Merged Updated PPTs (1 to 60)

Effort By Taaha Khan

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Principles of animal Ecology

Lecture No. 1

Module 1
Introduction

Principles of Animal Ecology

Agenda

1. Introduction.
2. What is meant by Ecology
3. Importance of Ecology
4. Scope of Ecology
5. Its relevance with man kind

Ecology and Environment: Introduction

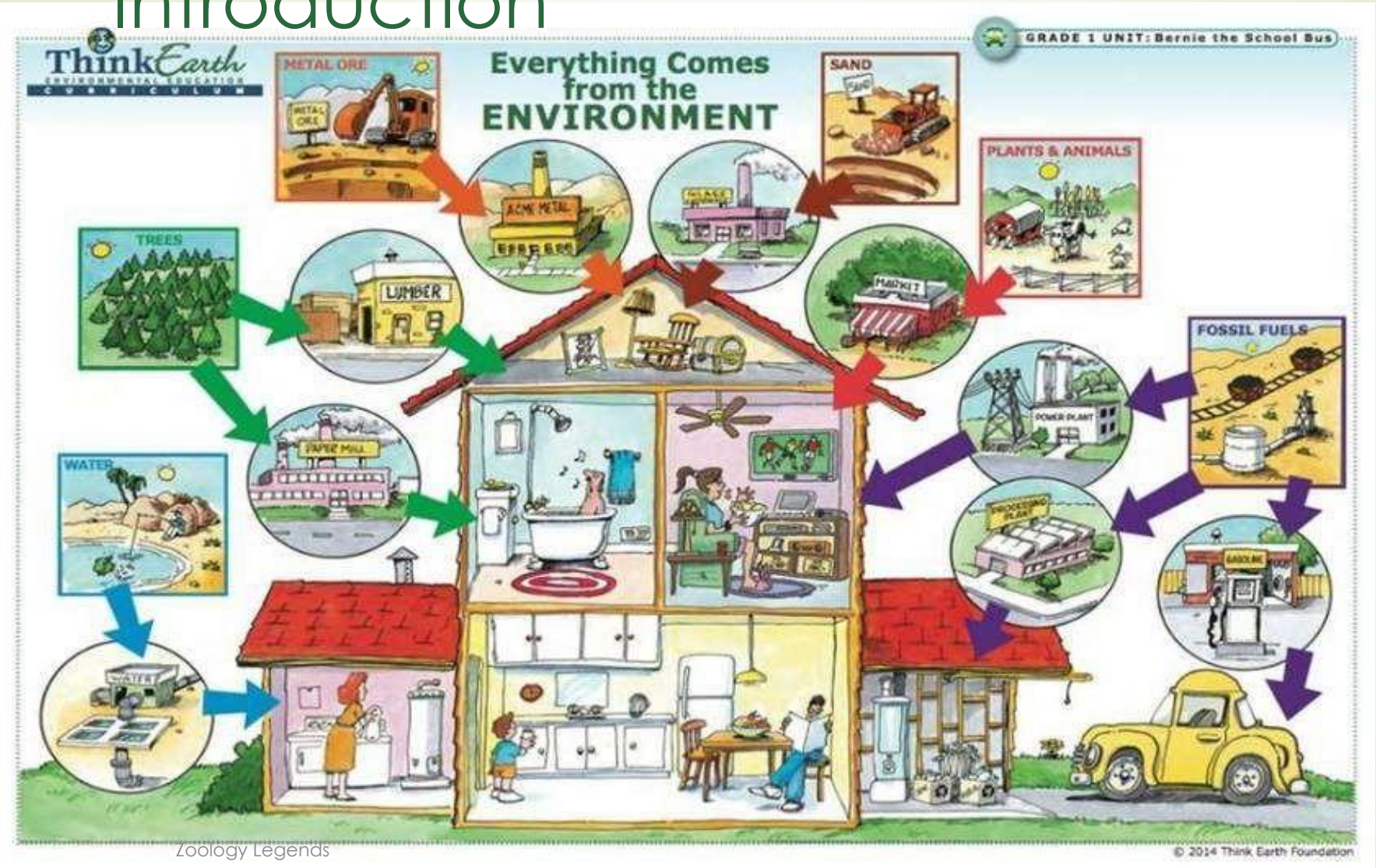
Topic 2

Ecology and Environment: Introduction

1. **Ecology** (from Greek word: οἶκος, "house", or "environment“ and logos means "study of")

2. **Ecology** is the scientific study of the interactions between organisms and their environment

Ecology and Environment: Introduction



Ecology and Environment: Introduction

a) **Autecology**-individual's life history, behavior, adaptation to environment

b) **Synecology**-group of organisms associated together as a unit

END

Ecology and Environment: Introduction

- **Topic 3**

Ecology and Environment: Importance

3. Why it is important to study ecology? Or Importance of Ecology

i. To understand the distribution and abundance of living things in the physical environment.

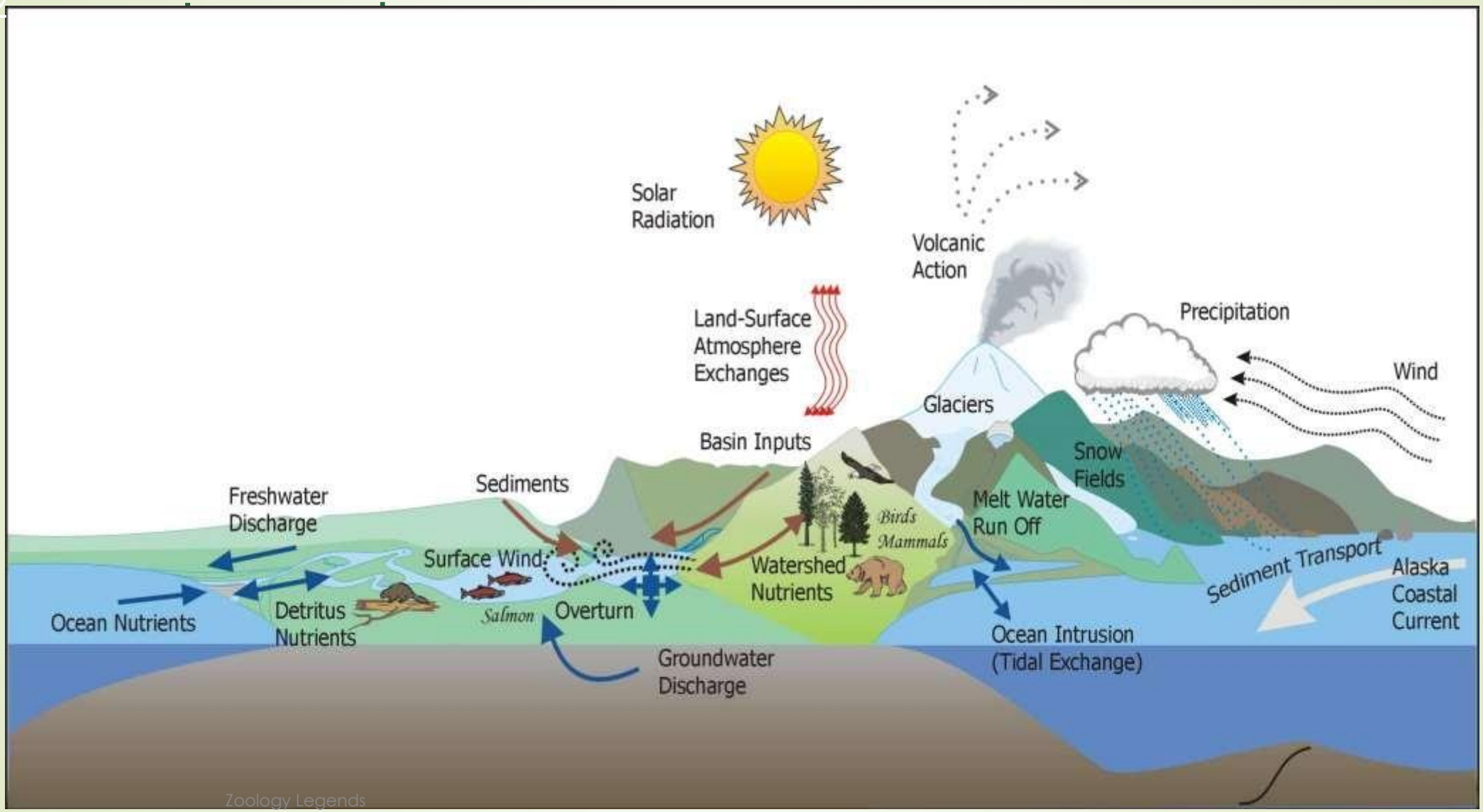
Ecology and Environment: Importance

3. Why it is important to study ecology? Or Importance of Ecology

ii. It includes the integration of scientific disciplines inside and outside of biology

iii. Such as biochemistry, physiology, evolution, biodiversity, molecular biology, geology, and climatology.

Ecology and Environment:



Ecology and Environment: Branches

- **Topic 4**

Ecology and Environment: Branches

Branches of Ecology

i. Applied ecology, (includes agroecology and conservation biology)
ii. Biogeochemistry
iii. Biogeography
iv. Conservation ecology
v. Ecological succession
vi. Evolutionary ecology

Ecology and Environment: Branches

Branches of Ecology

vii. Functional ecology,
viii. Global ecology
ix. Marine ecology, and fresh water ecology
x. Microbial ecology
xi. Paleoecology
xii. Restoration ecology
xiii. Soil ecology
xiv. Urban ecology,

Ecology and Environment: Relevance

Topic 5

Ecology and Environment: Relevance

Relevance to man kind

The more you know about the past, the better you prepared for the future

1. Start of civilization-Pure Environment- dependency on nature

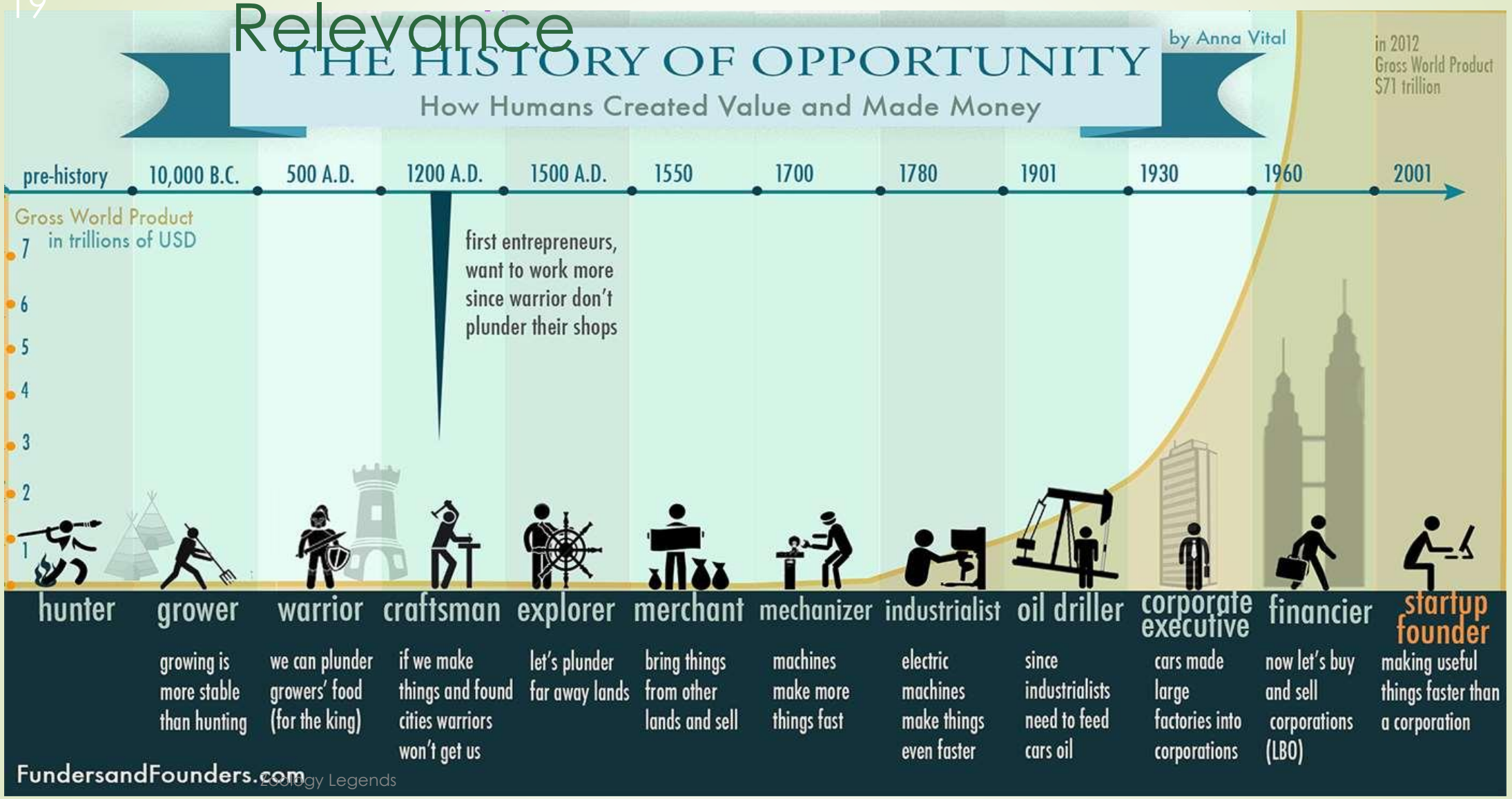
Ecology and Environment: Relevance

Relevance to man kind

The more you know about the past, the better you prepared for the future

2. Use of fire and other tools to modify surroundings
3. Advances in technology-less natural items
4. Result-Pollution

Ecology and Environment: Relevance



Ecology and Environment: History

Topic 6

Ecology and Environment: History

History of Ecology

1. Aristotle and ancient Greece philosophers clearly point out ecological topics

2. German Biologist-Ernst Haeckel -1869-**“Ecology” first used**

Ecology and Environment: History

History of Ecology

3. Early 1700s-Leeuwenhoek- pioneer in study of **food chain**

4. Botanist's- Richard Bradley- **Biological Productivity**

Ecology and Environment: History

History of Ecology

5. Frederick Clements, Victor shelford, Raymond L, Edward A- establish **“field of Ecology”**

6. **1968-1970-** picture of Earth- Pollution

7. 1970s- “**Decade of Environment**”

Ecology and Environment: History

History of Ecology

8. 22nd April 1970- First " **Earth Day**"

9. **1980s-1990s-** Environmental issues with political backward

END

Ecology and Environment: Scope

Topic 7

Ecology and Environment: Scope

2. Scope of ecology

i) all organisms living on Earth

ii) all physical and chemical surroundings.

iii) for this reason,

iv) the field is usually divided into different levels of study including: organismal ecology, population ecology, community ecology and ecosystem ecology

END

Level of organization

Topic 8

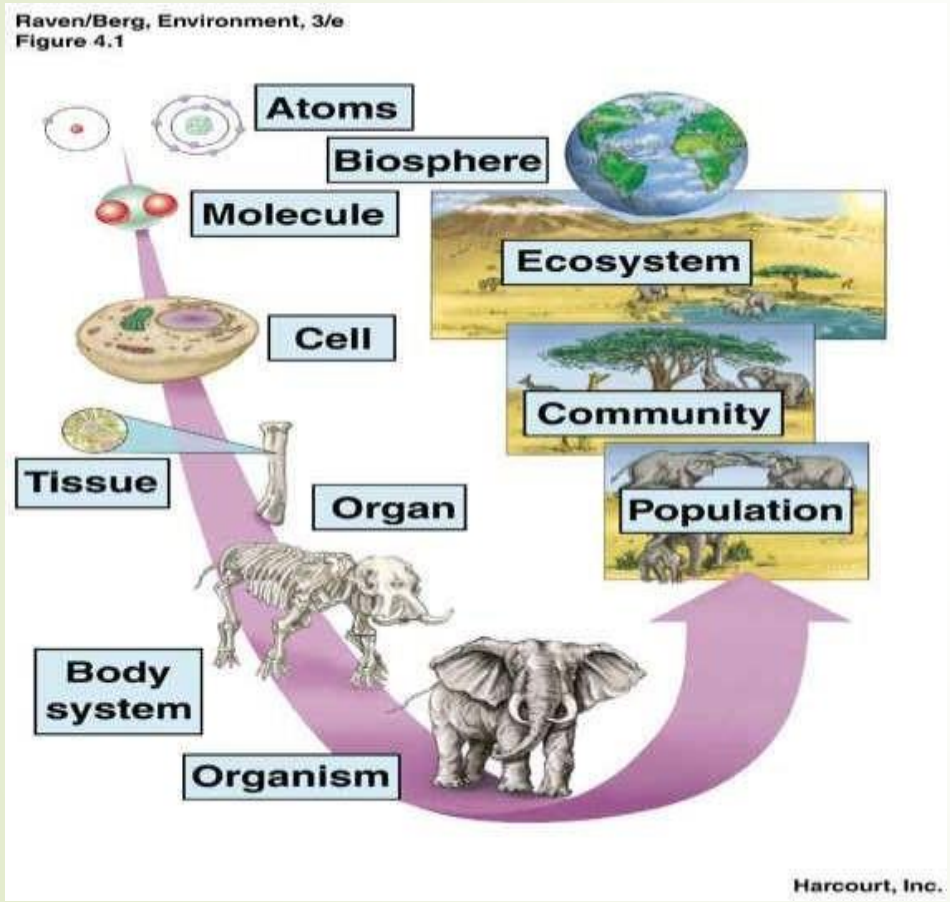
Level of organization

Four levels of biological organization—

a) **organismal**, organisms interactions with their environments

b)Population, how populations changes over time

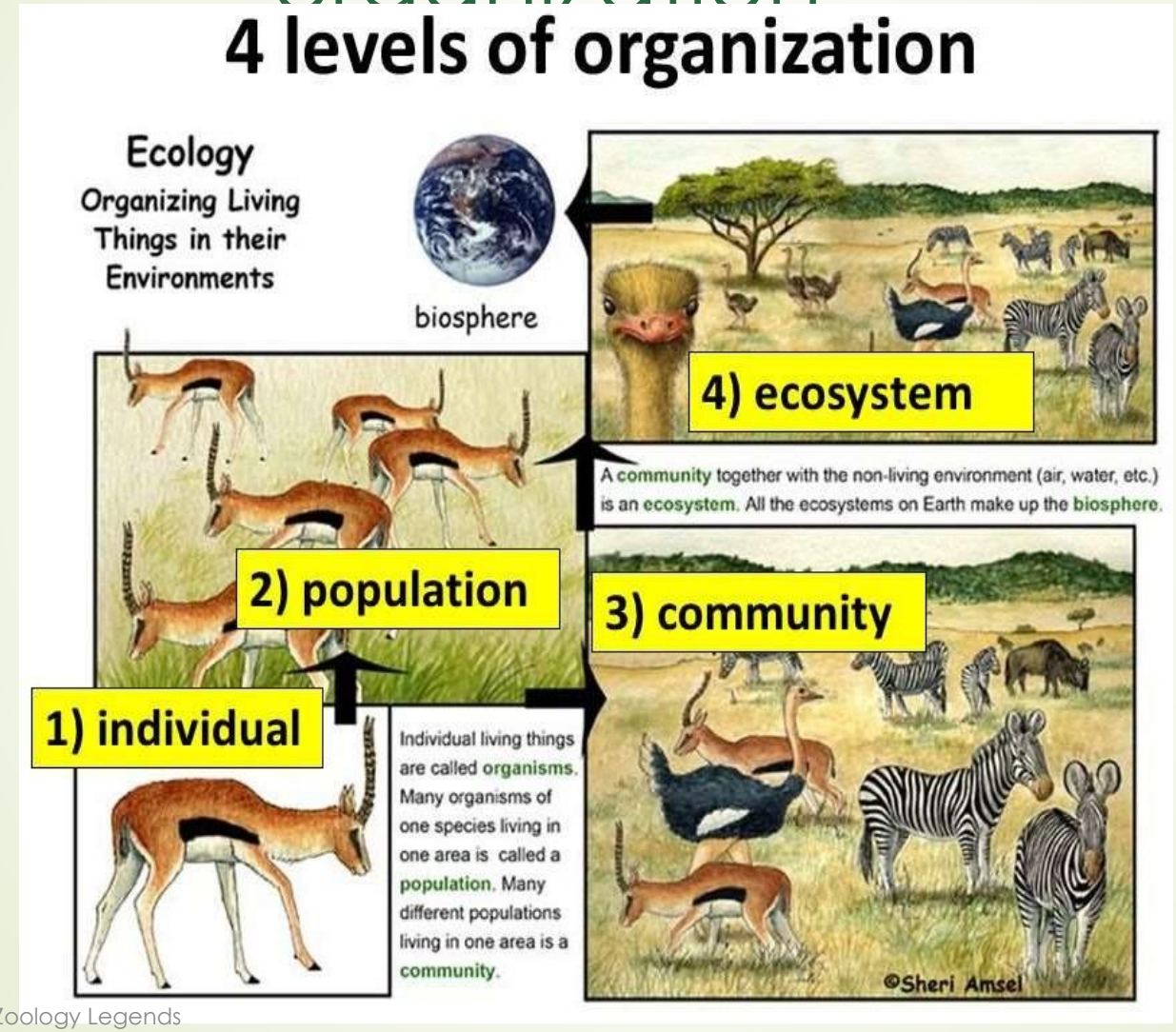
Level of organization



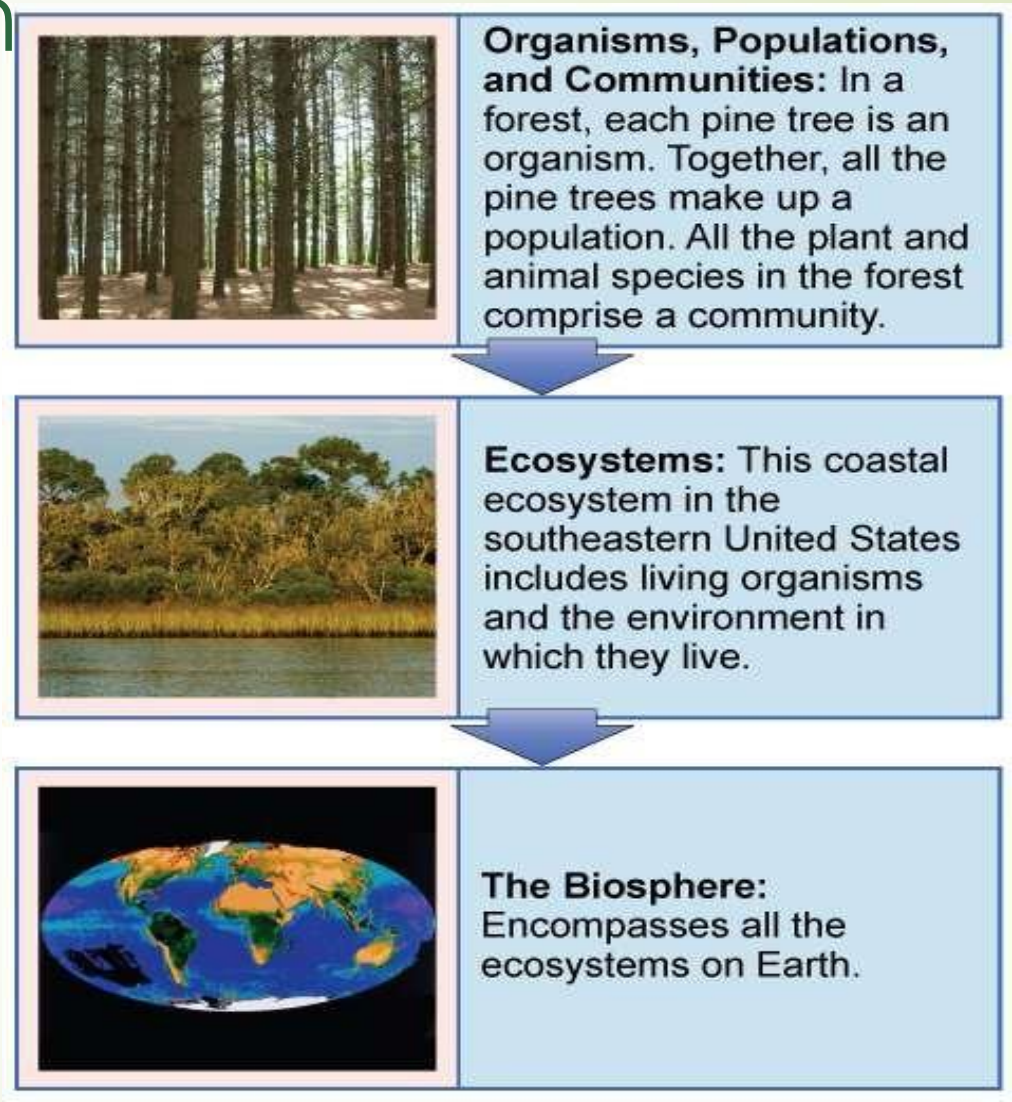
c)Community, different populations interactions in a community

d)Ecosystem, examine the living species (the biotic components) of the ecosystem and the nonliving portions (abiotic components), such as air, water, and soil, of the environment.

Level of organization



Level of organization



END

Level of organization

Topic 9

Level of organization: Homeostasis

Definition:

"tendency of system, especially the physiological system of higher animals, to maintain internal stability owing to the coordinated response of its parts to any situation or stimulus that would tend to disturb its normal condition or function"

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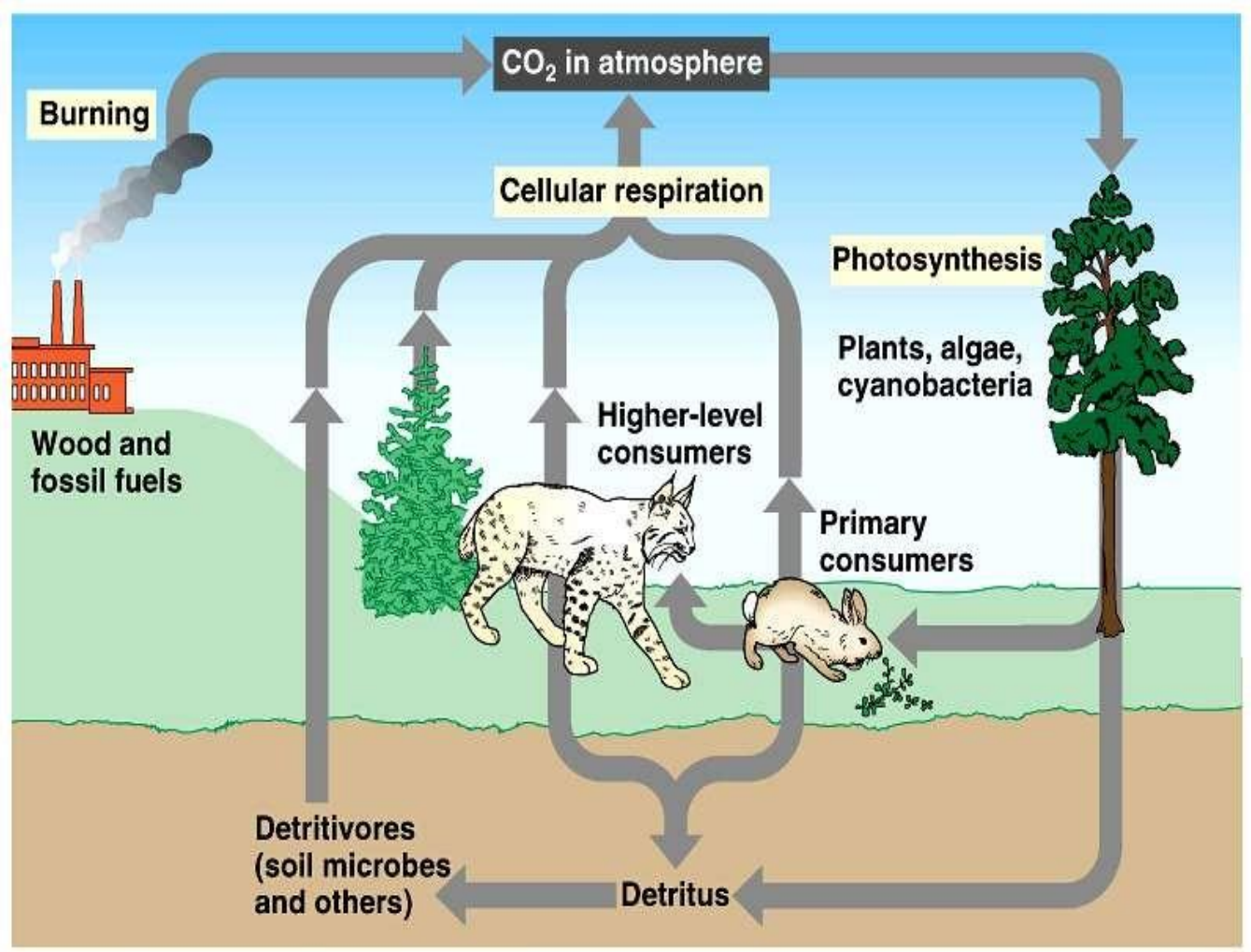
Level of organization

Topic 10

Level of organization: Characteristics of Ecosystem

Ecosystem- living (biotic) and non-living (abiotic) environments
-inseparable
-interrelated
-interact with each other
e.g. small as puddle, Large as ocean or forest or desert

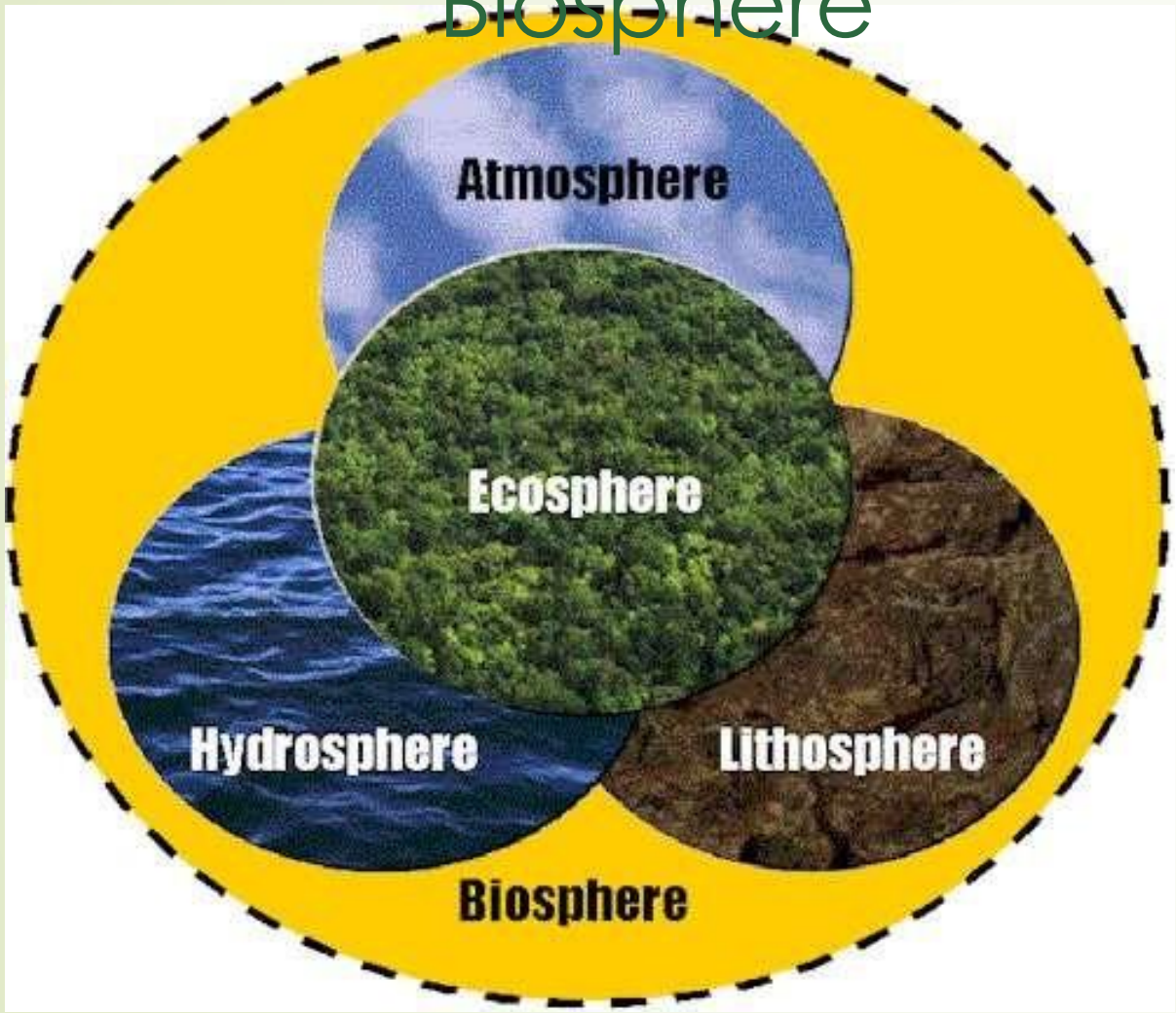
Level of organization: Characteristics of Ecosystem



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Ecosystem- living (biotic) and non-living (abiotic) environments
-inseparable
-interrelated
-interact with each other
-flow of energy (inputs) and cycling of materials (outputs) between living and non-living components
-functional unit/system of Ecology
-flow of energy follow-nevery used again
thermodynamics laws
-materials can be reused

Level of organization: Biosphere



Ecosphere- largest and nearly self sufficient biological system Including all living organisms of earth interacting with the physical environment
(Lithosphere-the Soil
Hydrosphere-water
Atmosphere-air
whereas
Biosphere- life zone of the earth

Ecosphere = Biosphere + Lithosphere + Hydrosphere + Atmosphere)

Level of organization

Topic 11

Level of organization: Types of Ecosystem

Types of Ecosystem

Natural

i) Terrestrial

e.g. Grassland (Savana), Forest, Desert ecosystems

ii) Aquatic

a.Lentic (Stagnant water) like lake, ponds etc.

b.Lotic (Flowing water) like river, ocean, sea, etc.

Level of organization

Types of Ecosystem

Natural

Artificial

A crop land, garden, aquarium, park, kitchen garden.

END

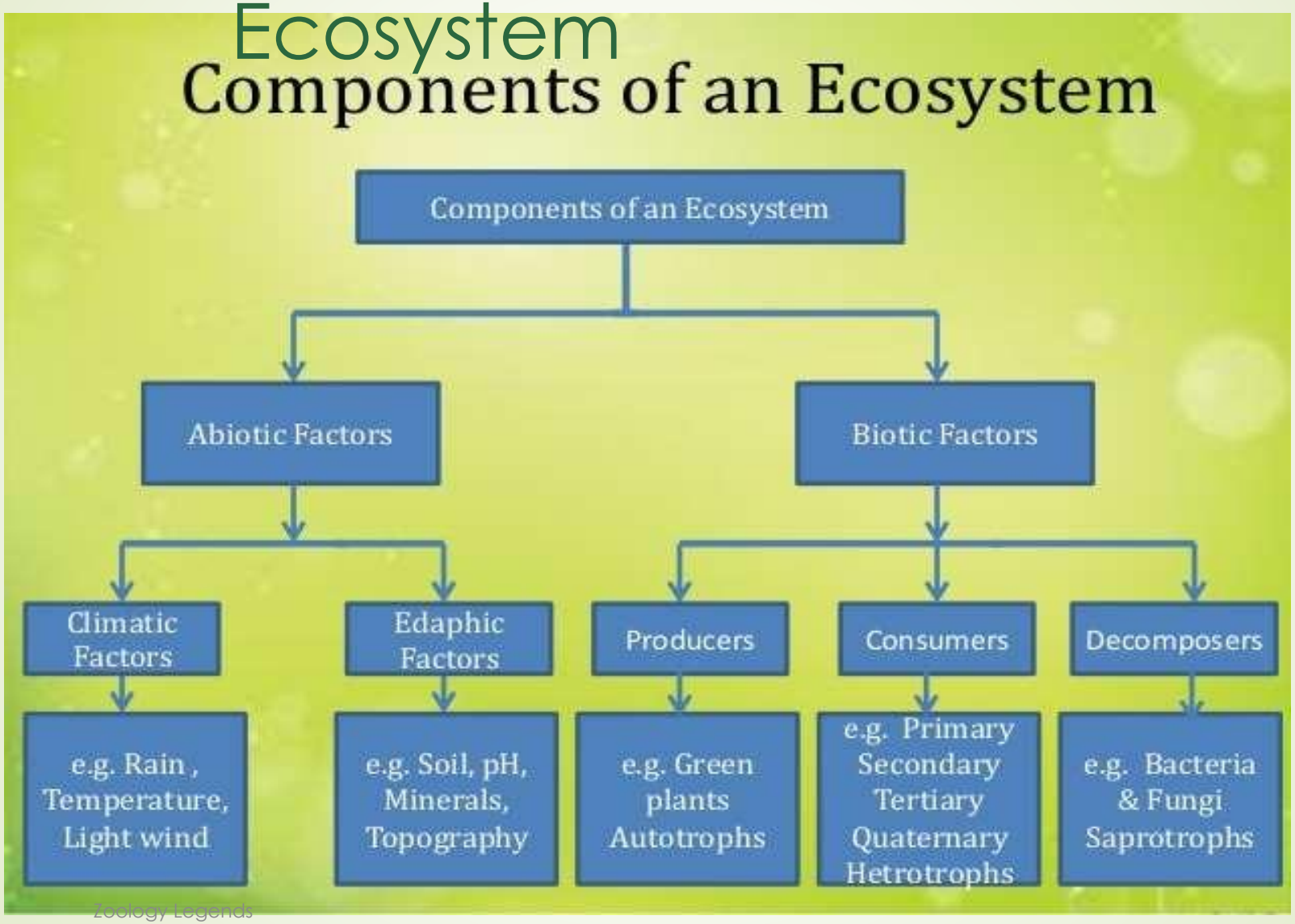
Components of Ecosystem

Topic 12

Principles of Anima Ecology

Components of Ecosystem

Components of Ecosystem



Components of Ecosystem- Abiotic

Tolerance range-tolerance level for temperature range
e.g. a human being would die if he stood out in minus 50◦C for any length of time
Optimum range- certain range within tolerance range
e.g. human body works at 37◦C
Limiting factor-abiotic factor out of animal tolerance range

Components of Ecosystem- Abiotic

- Abiotic Components
- Non-living components of Ecosystem-
  - -can not be generated by living organisms by themselves
    - three basic categories:
      - Climatic (humidity, sunlight, Air & temperature)
      - Edaphic (Soil and geography of land)
      - Social (use of land and water)

Components of Ecosystem- Abiotic

Abiotic Components- act as

-Resources

- Soil
- Water
- Atmosphere/Air

-Regulatory abiotic factors

- Light
- Temperature
- Pressure
- Humidity etc

END

Components of Ecosystem-Abiotic

Topic 13

Components of Ecosystem- Abiotic

Light

i) Sun light-ultimate source

ii) Photosynthesis-glucose-light energy converts into chemical energy

iii) Light intensity and duration depends on degree of latitude

Components of Ecosystem-Abiotic

Topic 14

Components of Ecosystem- Abiotic

Light

iii) Light intensity and duration depends on degree of latitude

iv) Equator-direct sunlight- warm

v) Tropics-less light than equator- diverse life forms

vi) Temperate cold zones

Components of Ecosystem- Abiotic

Components of Ecosystem-Abiotic

Topic 15

Components of Ecosystem- Abiotic

Components of Ecosystem- Abiotic

Atmosphere/Air

i. **Mixture of gases**
ii. **Layers of atmosphere**
iii. **Air pressure**
iv. **Role of atmosphere(air) for life on earth**

END

Components of Ecosystem-Abiotic

Topic 16

Components of Ecosystem- Abiotic

Temperature

i)Temperature is the function of light

ii)temp-affect body's metabolic rate

a) **Torpor**
b) **Hibernation**
c) **Winter sleep**
d) **Aestivation**

END

Components of Ecosystem-Abiotic

Module 17

Components of Ecosystem- Abiotic

Soil

i)Basic medium for land based ecosystems

ii) Natural reservoir of inorganic mineral elements

iii) Contain humus- material formed by decomposition of dead plants and animals

iv) More humus-diverse life forms

Components of Ecosystem- Abiotic

Soil Types

a)Sandy soil-do not hold- desert

b) Silt Soil

c) Clay soil- hold water – grasslands and forests

d) Loamy soil-fertile-agricultural land

Components of Ecosystem-Abiotic

Module 18

Components of Ecosystem- Abiotic

Water

1. Life originator
   Source
2. Ice, glaciers, rainfall
3. Living bodies are almost made of water
4. **70%** of earth

Components of Ecosystem- Abiotic

5. Helps in the process of photosynthesis and other cycles in nature
6. Water bodies-heat sink- regulate temperature
7. Water need varies among different animals

Components of Ecosystem- Abiotic

Water

To categories ecosystems

1. Aquatic Ecosystems

a) Ocean ecosystem

b) Freshwater ecosystem

2. Terrestrial Ecosystem

a) Tropical rain Forests

b) Desert Ecosystem

Components of Ecosystem- Abiotic

Water as limiting factor

1. In terrestrial environment
2. In water environments in which amount is subjected to great fluctuation (temporary ponds)
3. High salinity raises water loss from organisms by osmosis

END

Components of Ecosystem-Abiotic

Module 19

Components of Ecosystem- Abiotic

Humidity and Temperature

1. Invisible vapour (humidity)
2. Absolute humidity
3. Ounce/cubic yard
4. Relative humidity
5. Temperature governs humidity
6. Effect on intensity of solar rays
7. Effect on transpiration vs evaporation
8. Source of soil moisture

Components of Ecosystem-Abiotic

Module 20

Components of Ecosystem- Abiotic

Flood and Fire

Flood

"Excess of water flows Seasonal"

a) Caused sever damaged to soil
b) Removal of fertile top soil layer
c) Increase moisture in soil
d) Unfit for crop cultivation

Components of Ecosystem- Abiotic

Flood and Fire

Fire

a) Itself neither abiotic nor biotic

b) Caused by burning of biotic and abiotic chemicals

c) product abiotic compounds, including charcoal that may continue as part of carbon cycling, and carbon dioxide that enters the atmosphere and is used by plants.

Components of Ecosystem-Biotic

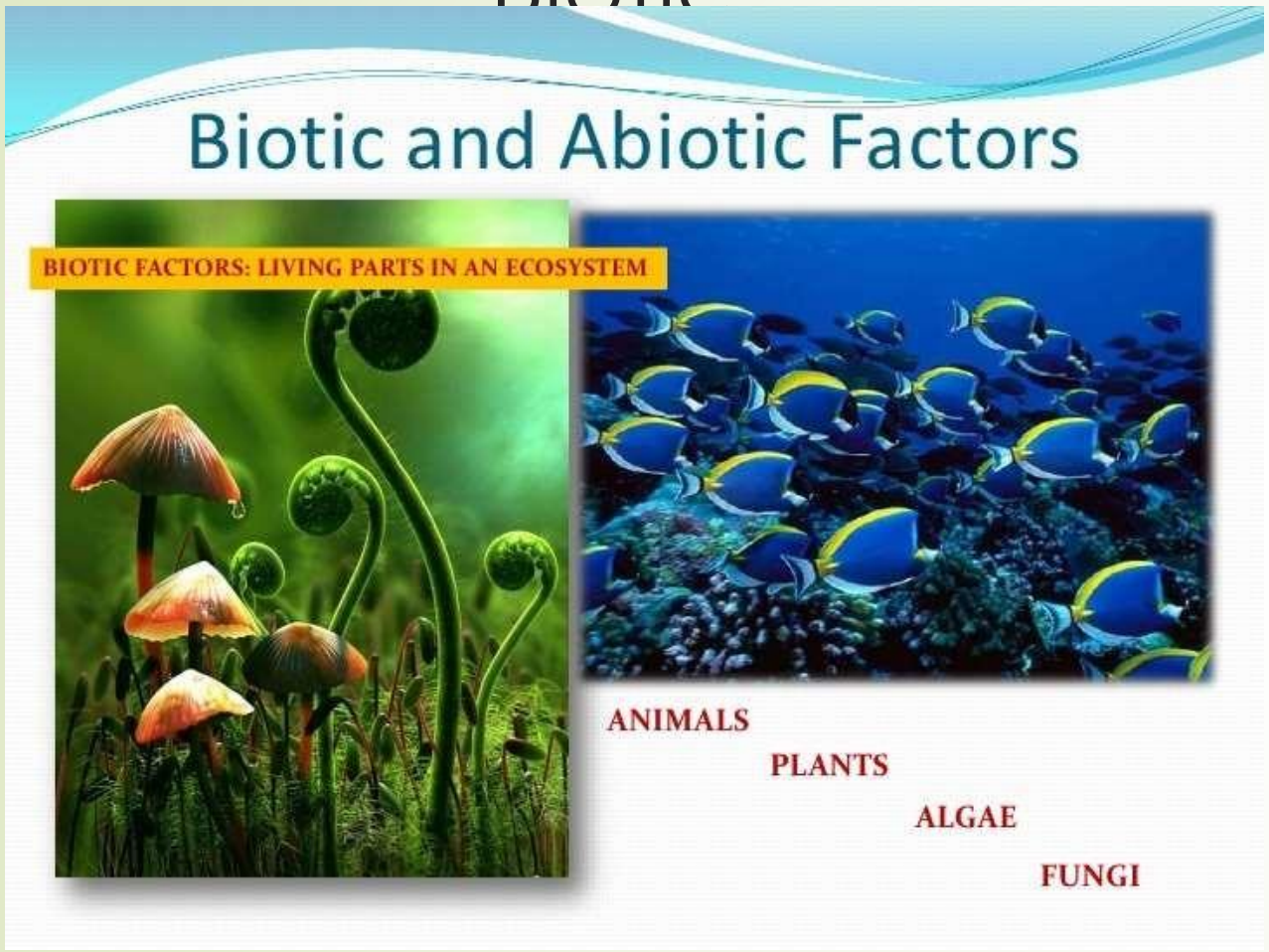
Module 21

Components of Ecosystem- Biotic

Biotic or Living components

All living organisms found in the environment including plants, animals and microorganisms

Components of Ecosystem-Biotic



Biotic or Living components-

Categories of biotic component

1. Producers or Autotrophs-Plants
2. Consumers or Heterotrophs-Animals
3. Decomposers-Microorganisms

Components of Ecosystem-Biotic

Module 22

Components of Ecosystem- Biotic

Producers or Autotrophs-

Green plants (Land environment)

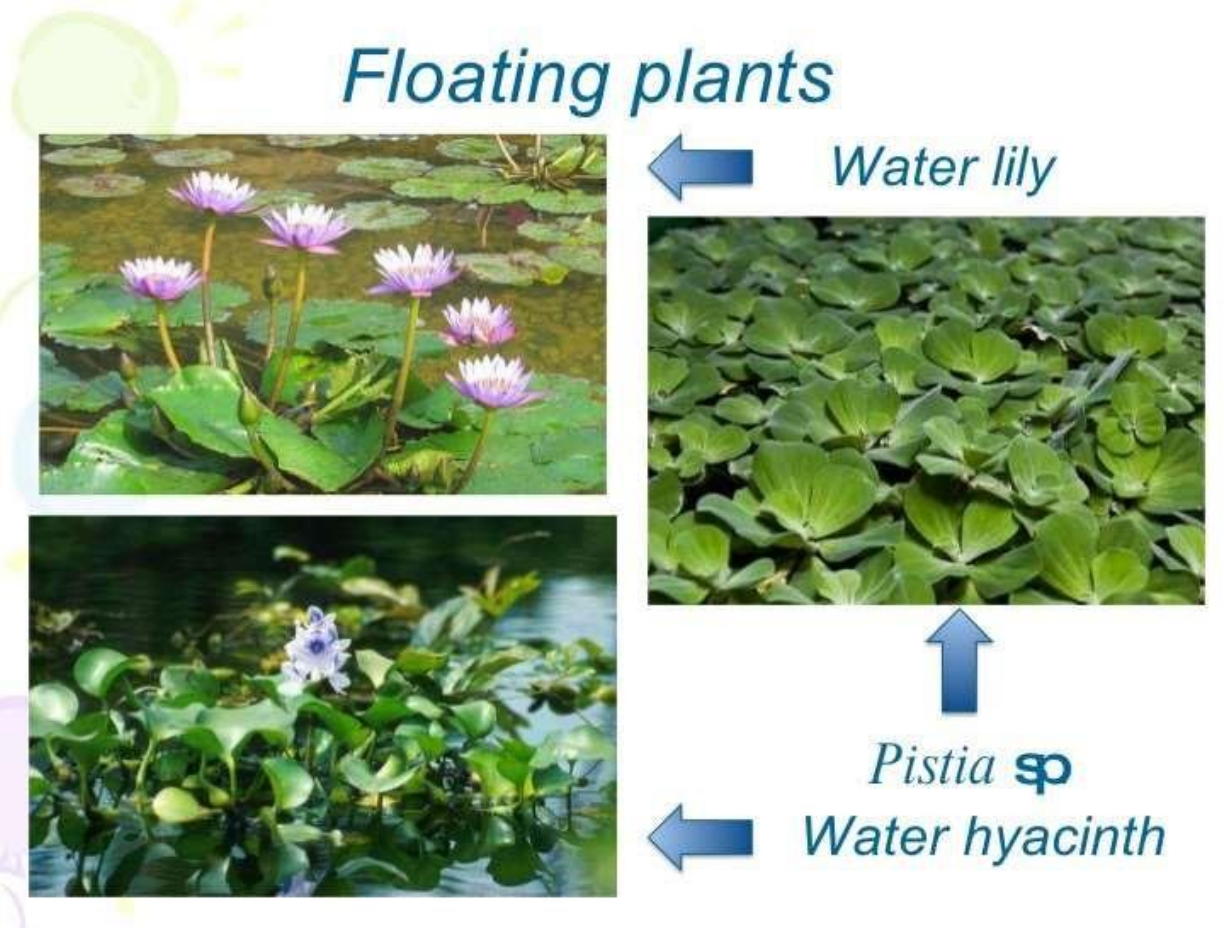
Components of Ecosystem- Biotic



Producers or Autotrophs- Green plants (Land environment)

1. They synthesize food for all the organisms of ecosystem
2. Convert sunlight or solar energy to chemical energy
3. converts inorganic substrate into organic food by the process of photosynthesis

Components of Ecosystem- Biotic



Producers or Autotrophs-

(Aquatic environment)

Two Types

Floating plants : called **phytoplankton** ("phyto"- plants, "plankton" - floating.) for example, *Spirogyra*, *Ulothrix*, diatoms and *Volvox*.

Components of Ecosystem- Biotic



Producers or Autotrophs-

(Aquatic environment)

Two Types

(b) **Rooted plants :** These plants occur in concentric layers from periphery to the deeper zones. examples of rooted plants are *Typha bulrushes*, *Sagittaria*, *Hydrilla*, *Rupia*, *Chara.*

Components of Ecosystem- Biotic

4. 1st trophic level
5. rate at which the radiation energy is stored by photosynthesis in green plant is called gross primary productivity (GPP).

END

Components of Ecosystem-Biotic

Module 23

Components of Ecosystem- Biotic

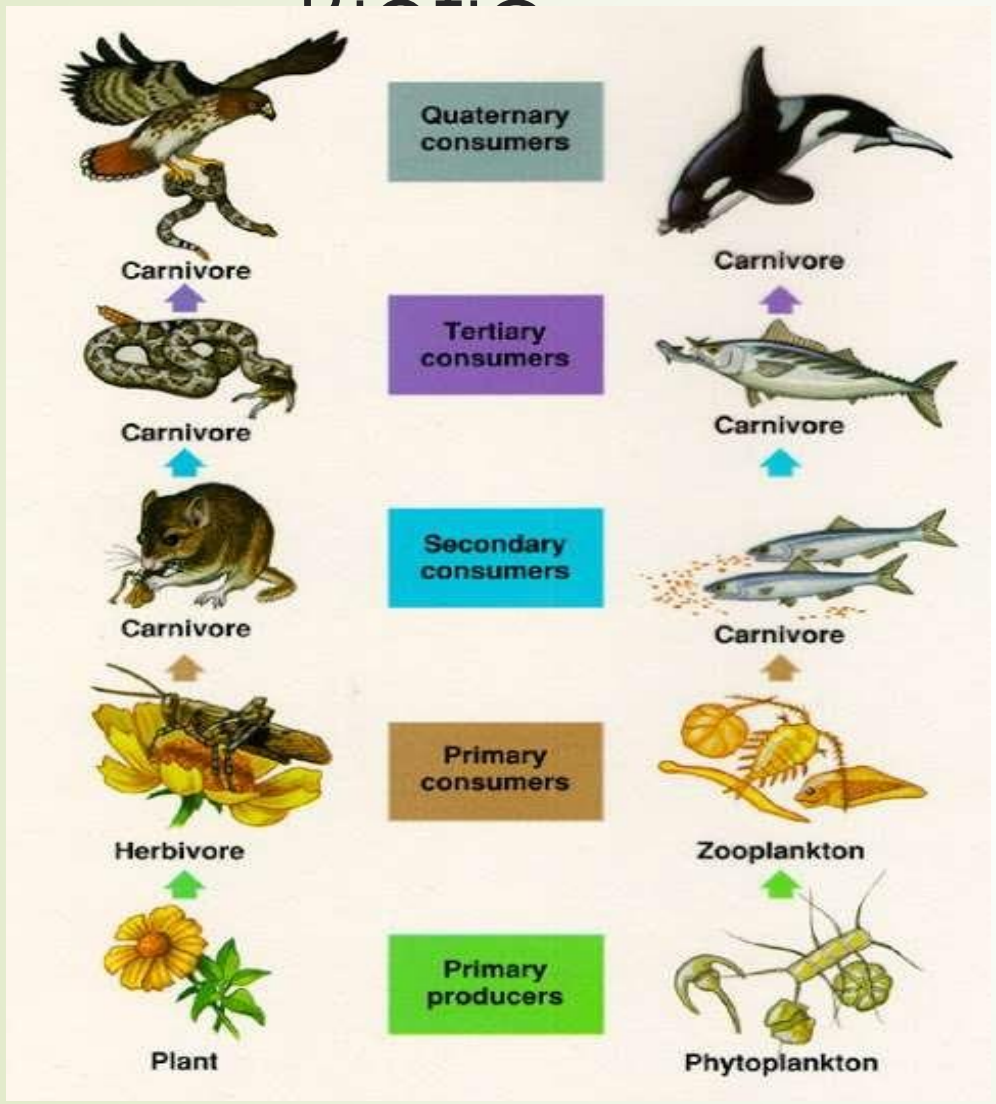
Consumers:

They are heterotrophs which obtain energy from producers directly or indirectly.

They can be further divided as

i. **Primary consumers**
ii. **Secondary consumers**
iii. **Tertiary consumers**

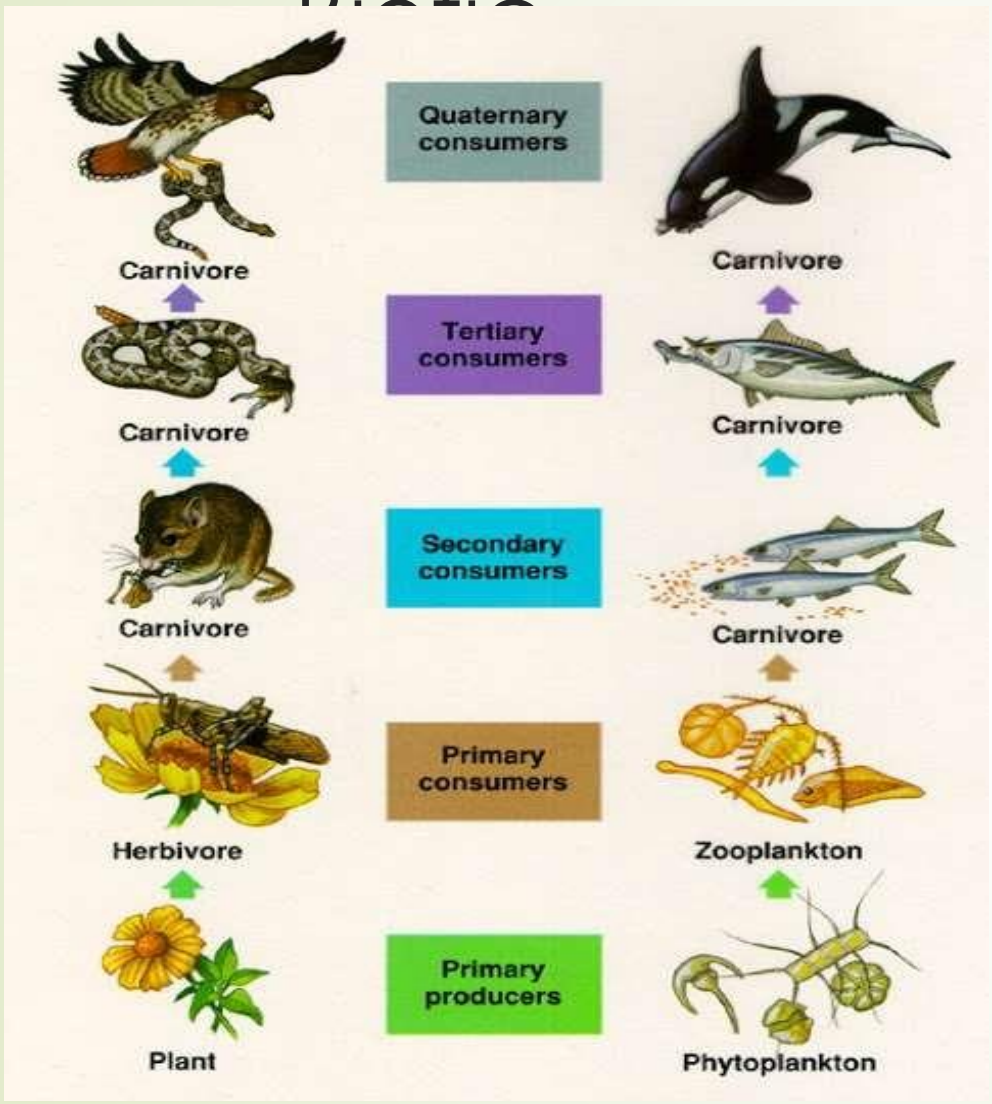
Components of Ecosystem- Biotic



Zoology Legends

- Consumers:
  - i. Primary consumers- (herbivores) in land ecosystem- animals which feeds directly on plants.
  - At 1st level consumers, called as primary consumers,

Components of Ecosystem-Biotic



Consumers:

i. **Secondary consumers**

animals that feeds on other animals.

They are omnivores and carnivores animals.

e.g: Bear, wolf, jackel, and snake etc.

Components of Ecosystem- Biotic

iii. Tertiary consumers:

These animals get their food from all consumers. They are top carnivores.

e.g: lion, tiger, and eagle etc.

Components of Ecosystem- Biotic

Special feeding groups (Consumers)

Scavengers : These are the animals that feed on the dead plants and animals.

e.g. termites and beetles feed on the decaying wood

Marine invertebrates, Vultures and hyena are other examples of scavengers

Components of Ecosystem- Biotic

Special feeding groups (Consumers)

Omnivores : Omnivores consume both plants and animals as source of their food

e.g. human beings.

the red fox feeds on berries, small rodents as well as on dead animals.

Thus it is a herbivore, carnivore and also a scavenger.

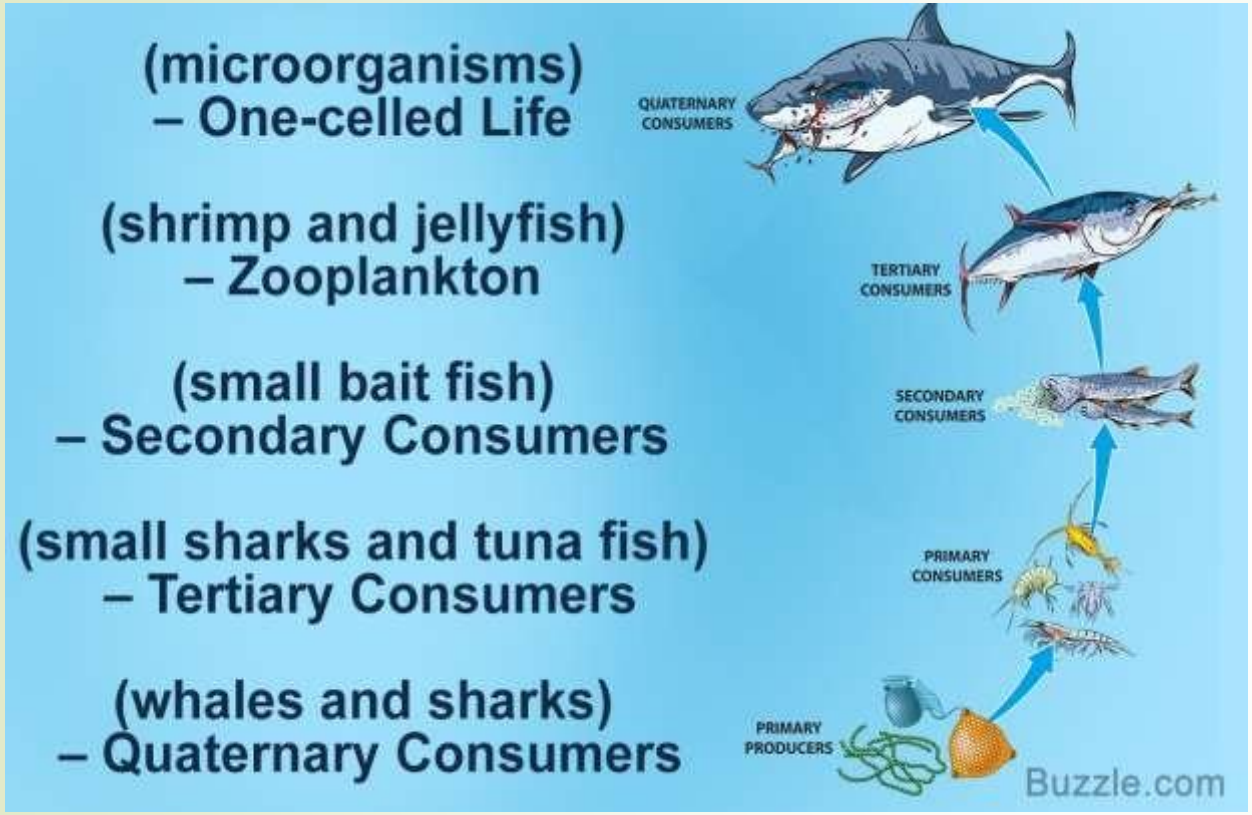
Components of Ecosystem- Biotic

Special feeding groups (Consumers)

Parasites : They live and feed on/in other living organisms called *host*. Parasites not only feed on their host but they also cause lethal or nonlethal disease in it.

e.g: human head lice, bacteria

Components of Ecosystem- Biotic



Consumers in aquatic ecosystem

i. Primary consumers

as microconsumers, zooplanktons and benthos

ii. Secondary consumers:

aquatic predaceous insects, game fish etc

Components of Ecosystem-Biotic

Module 24

Components of Ecosystem- Biotic

Decomposers:

1. feed on dead and decayed plants or animals.
2. make up the final trophic level in food chain.
3. They decompose the dead and decay matter and helps in recycling the nutrients.

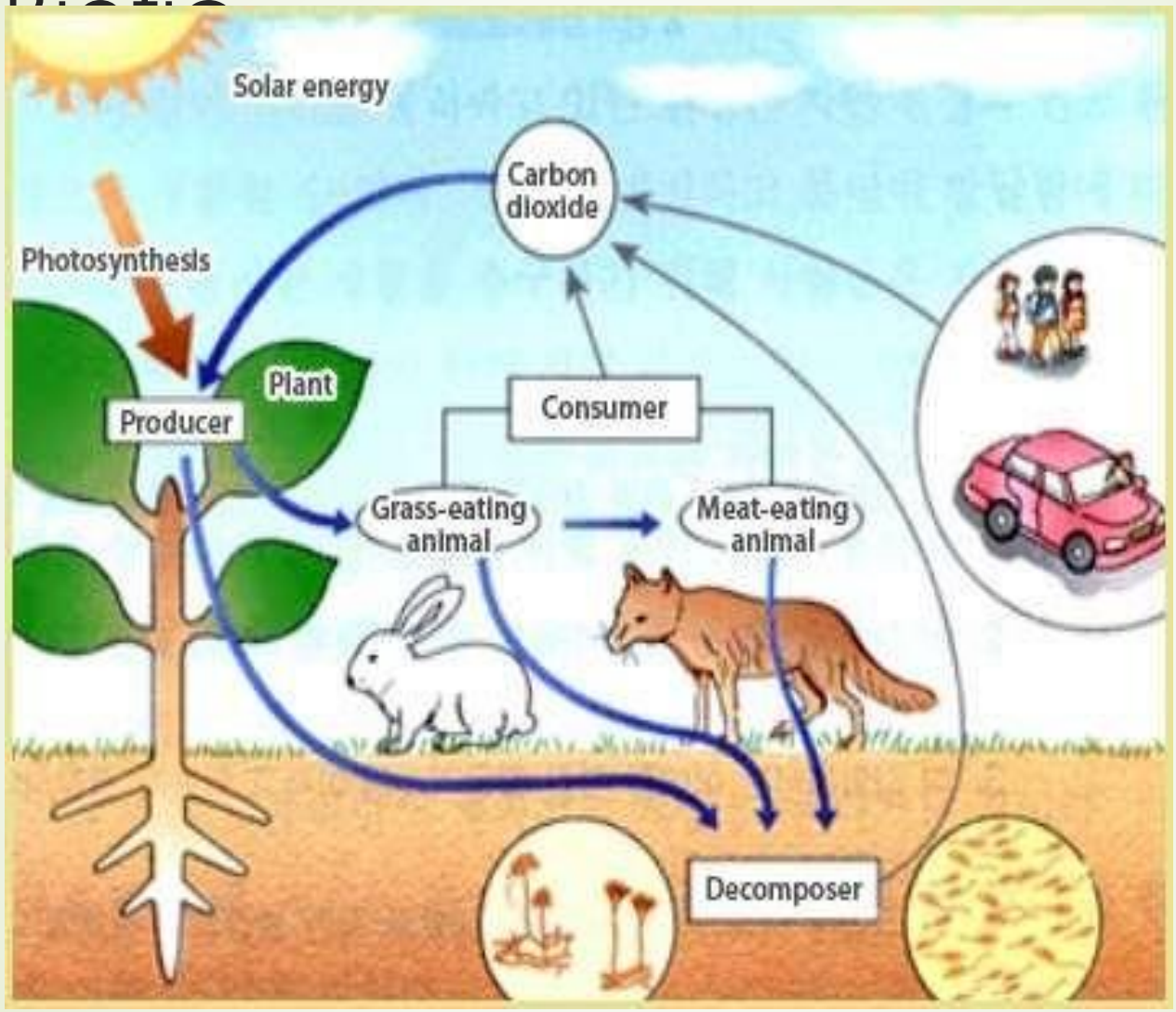
Components of Ecosystem- Biotic

Decomposers:

They are classified into **two class**

a) **Micro-decomposers**: Bacteria, Fungi, Protozoa
b) **b) Macro-decomposers**: Earth worm, Nematodes, Molluscals

Components of Ecosystem-Biotic



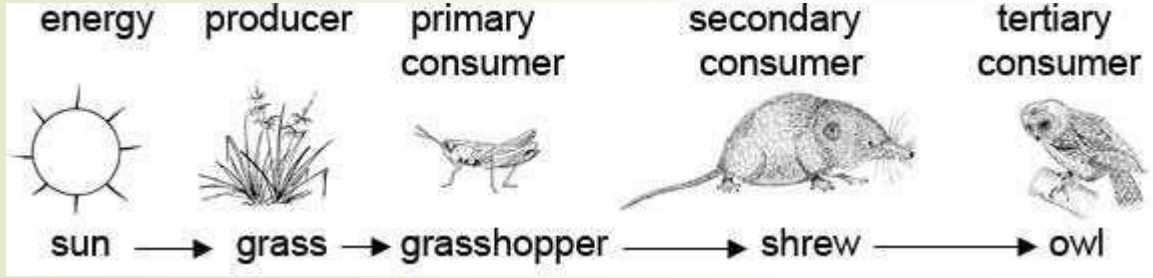
Food Chains in Ecosystem

Module 25

Food Chains in Ecosystem

Food chain

Food Chains in Ecosystem



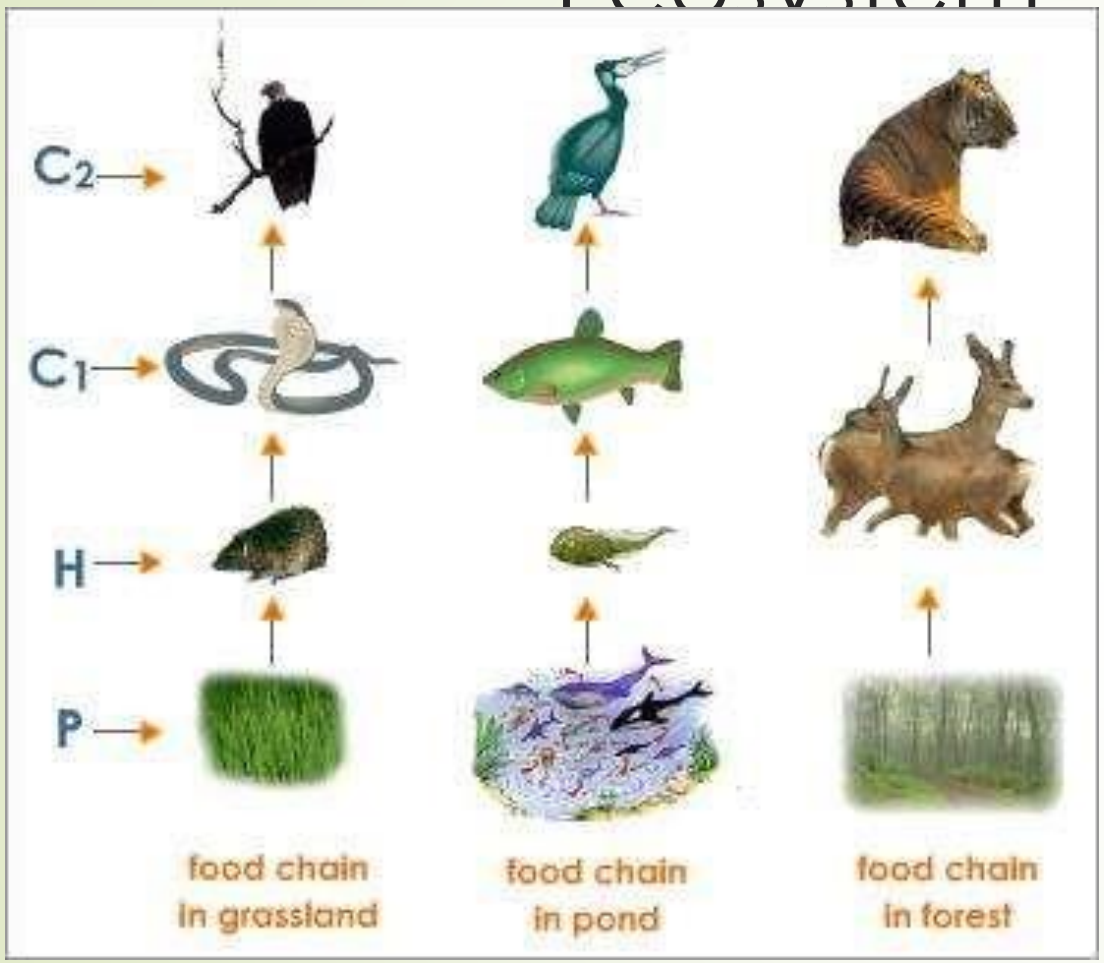
Food chain

"Transfer of food from the plants (producers) through a series of organisms with repeated eating and being eaten is called a food chain"

e.g.

Grasses → Grasshopper → Frogs → Snakes → Hawk/Eagle

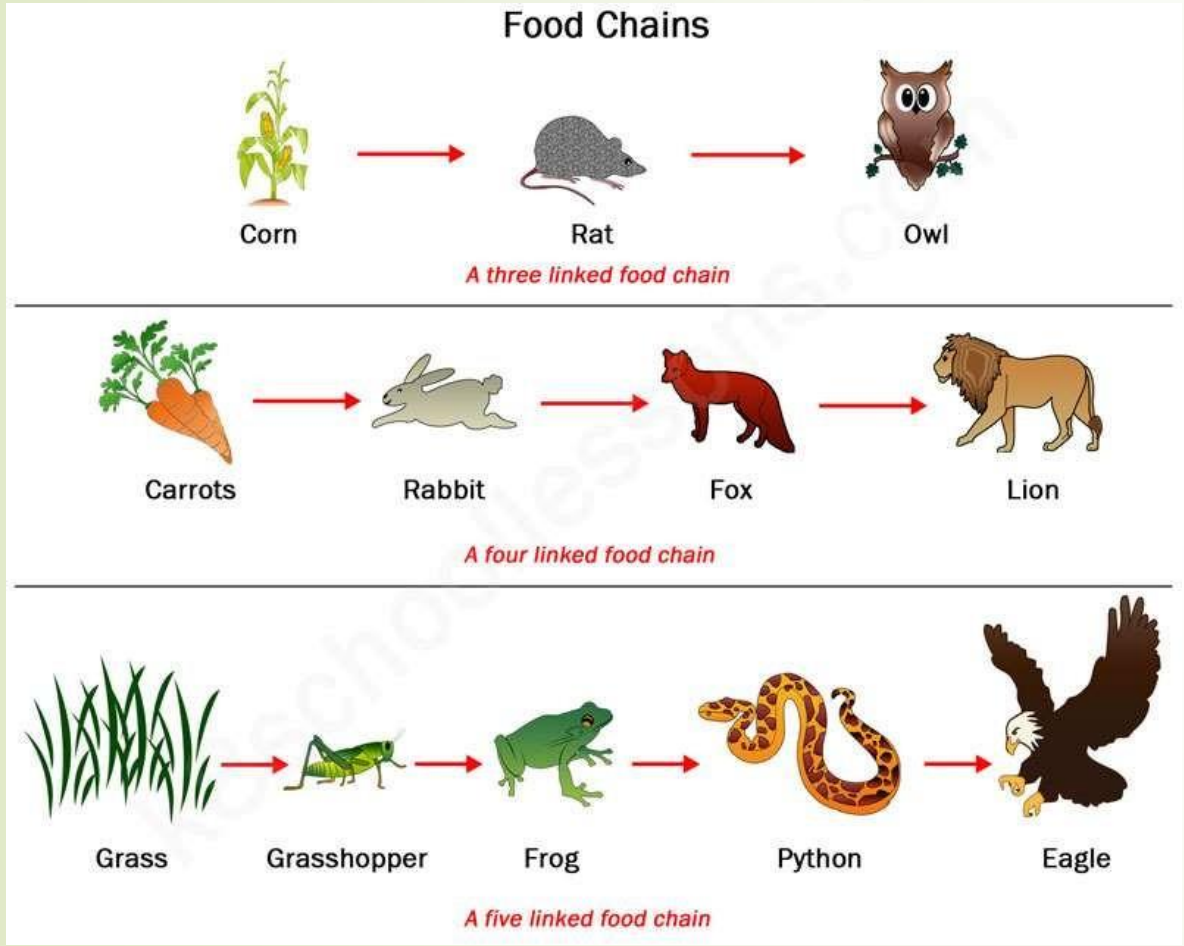
Food Chains in Ecosystem



Food chain and trophic level

1. Each step in the food chain is called trophic level.
2. In this fig grasses are first and eagle, crane and lion represents the fourth and third trophic level.
3. **P=Producers**
4. **H=Herbivores**
5. **C=Consumers**

Food Chains in Ecosystem



Three important features in these **food chains** are :

1. Weaker organisms are attacked by the stronger organisms
2. Number of organisms is reduced at each higher level but the size of organisms is increases.
3. The number of steps in a food chain is limited to 4-5.

Food Chains in Ecosystem

Module 26

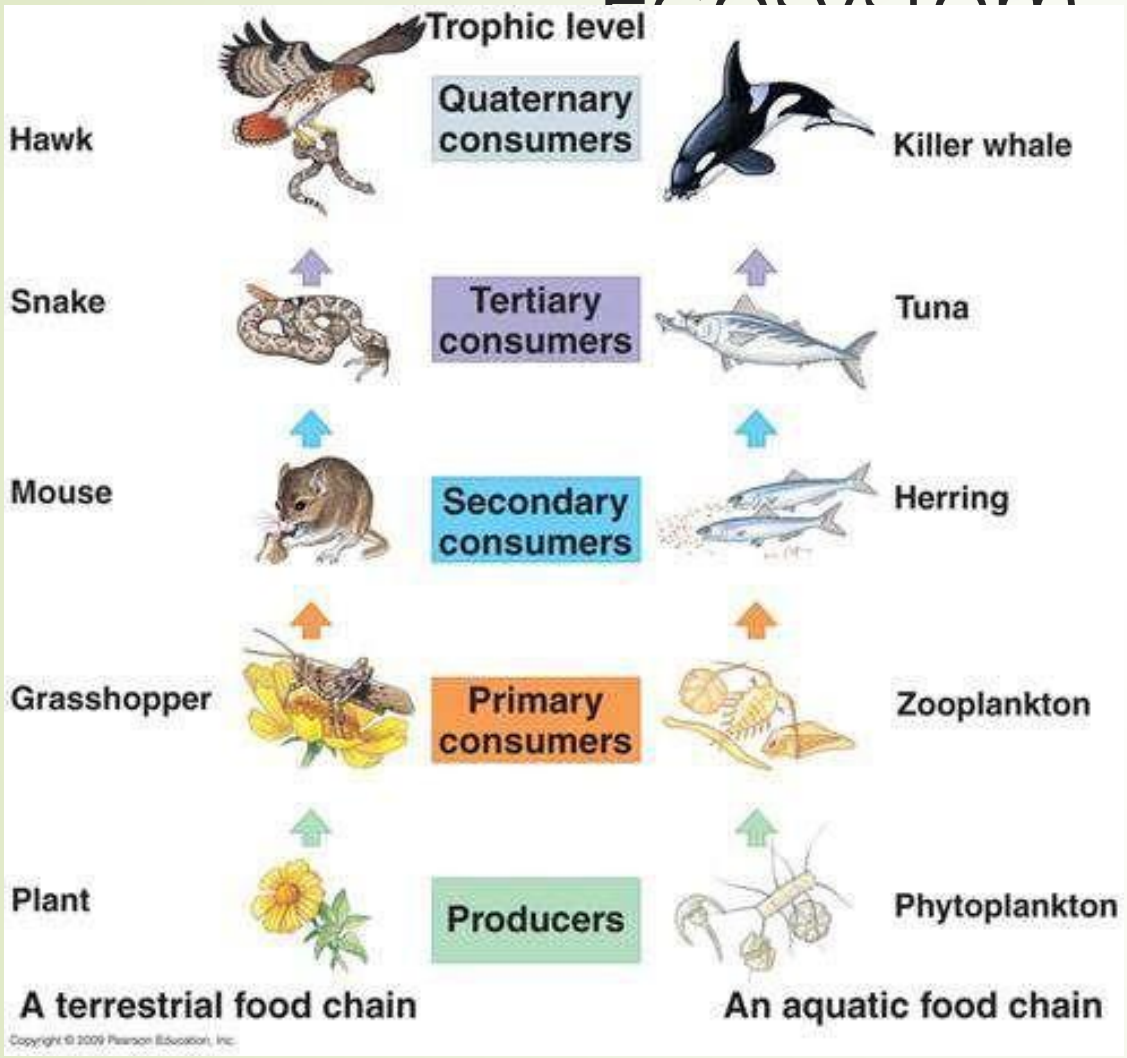
Food Chains in Ecosystem

Food chain and trophic level

FC consists of trophic levels:

1. **(Producers) Autotrophs**
2. first trophic level

Food Chains in Ecosystem



Food chain and trophic level

FC consists of trophic levels:

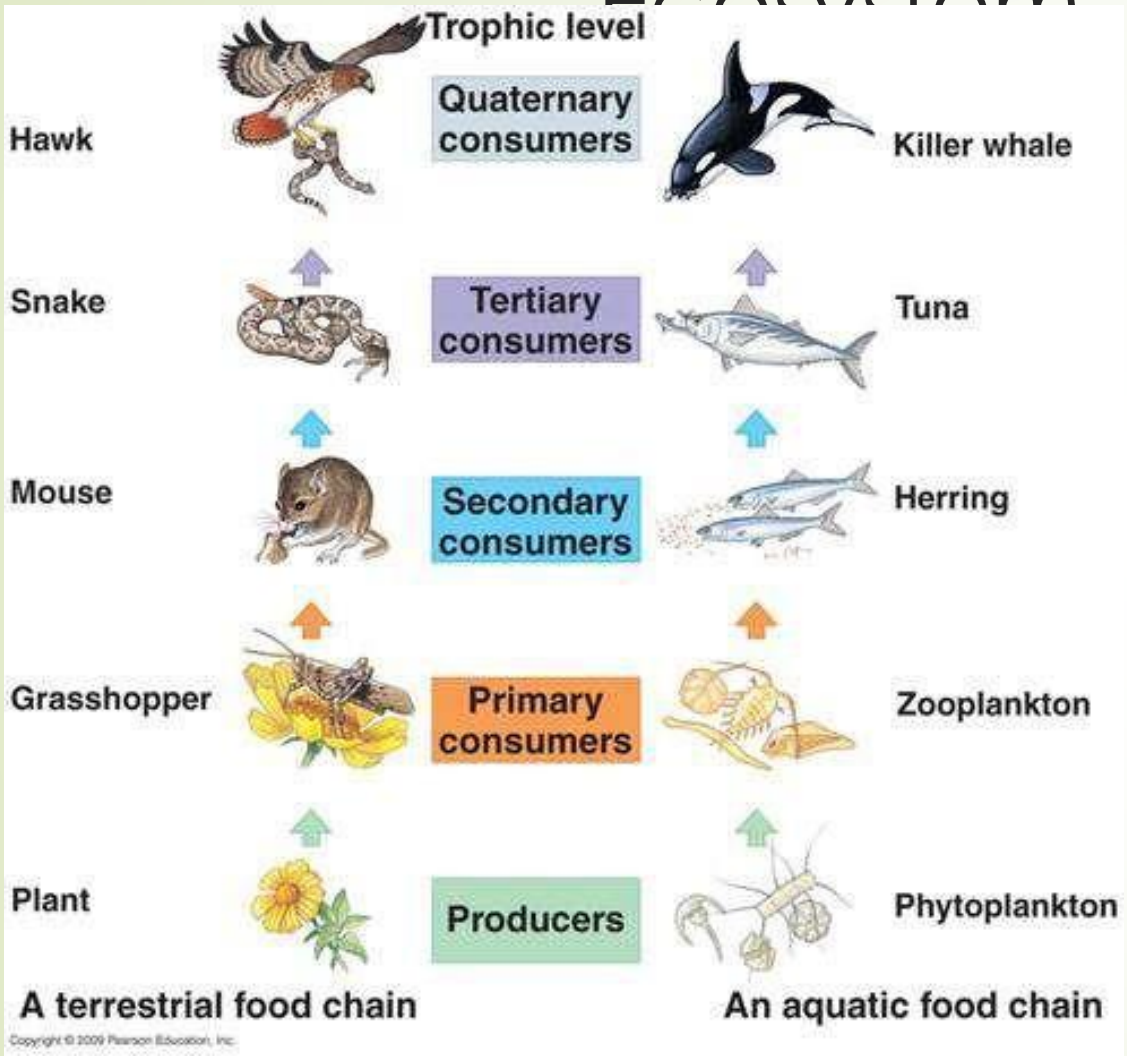
1. **(Producers) Autotrophs**
2. first trophic level

Gross Primary Productivity (GPP)

Net Primary, Productivity (NPP)

GPP = NPP + R or GPP – R = NPP

Food Chains in Ecosystem



Food chain and trophic level

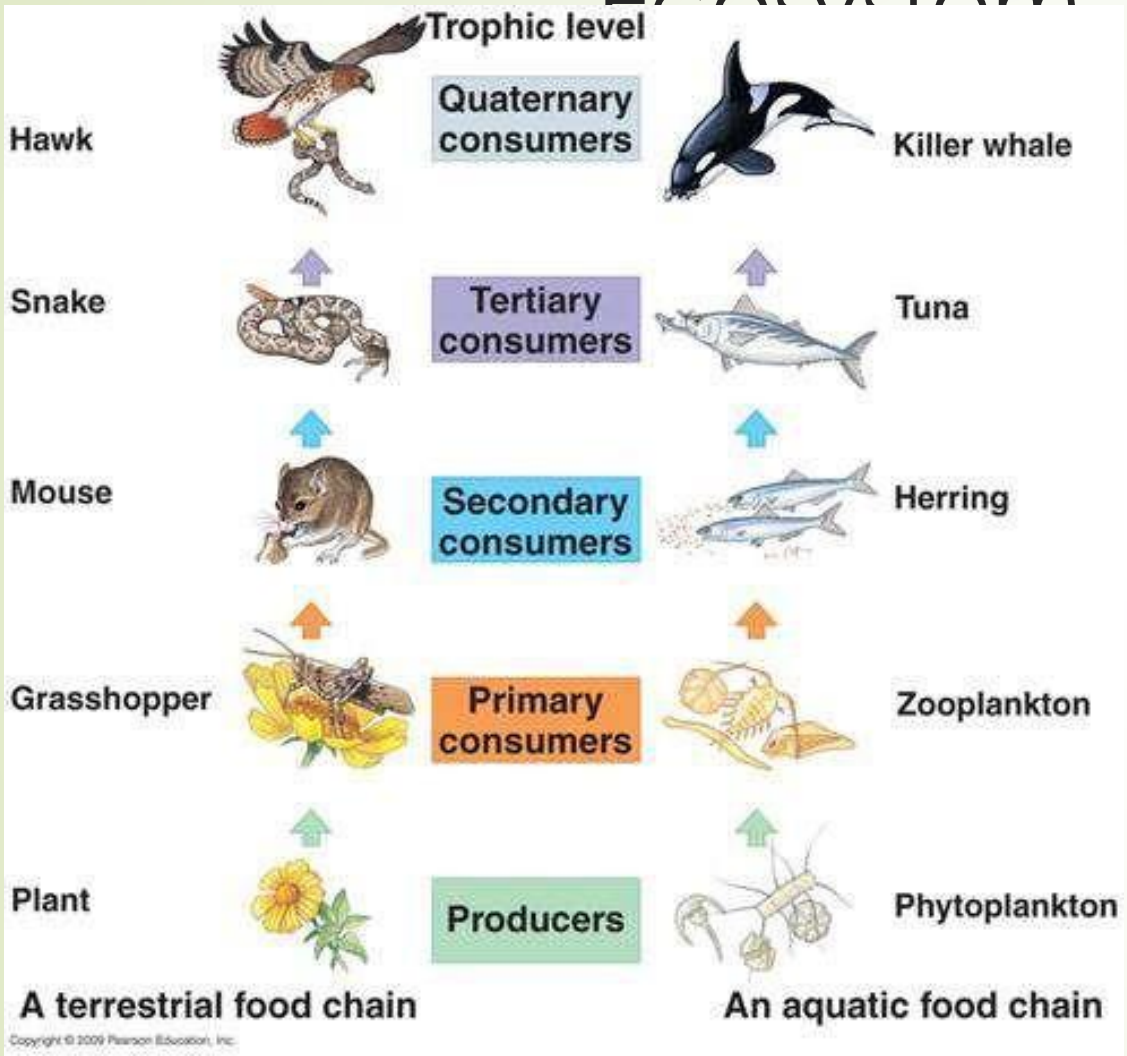
2. Primary consumers Herbivores

a)are at the 2nd trophic level in food chain

b) capable of converting energy stored in the plant tissue into animal tissue

c) They can digest high cellulose diet.

Food Chains in Ecosystem



Food chain and trophic level

2. Secondary consumer (Carnivores)

a)are at the 3rd and 4th trophic level in food chain

b) size of the carnivore/ increases at each trophic level

Food Chains in Ecosystem

Food chain and trophic level

3. Decomposers :

a)They make up the final trophic level in a food chain.

b)feed on dead organic matter called detritus of all the trophic levels

Food Chains in Ecosystem

Food chain and trophic level

3. Decomposers :

c) help in recycling the nutrients.

d) Examples: bacteria, fungi, mites, millipedes, earthworm, nematodes, slugs, crabs and mollusks

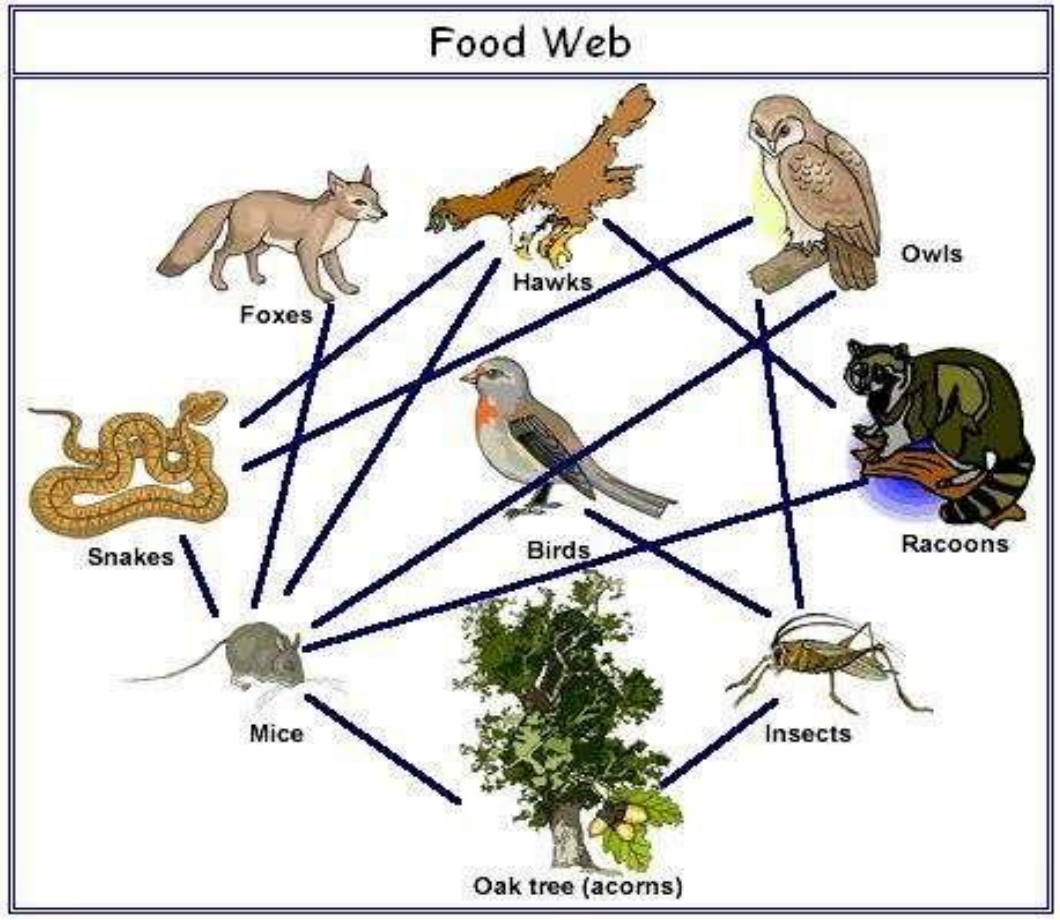
Food Web in Ecosystem

Module 27

Food Web in Ecosystem

Food web

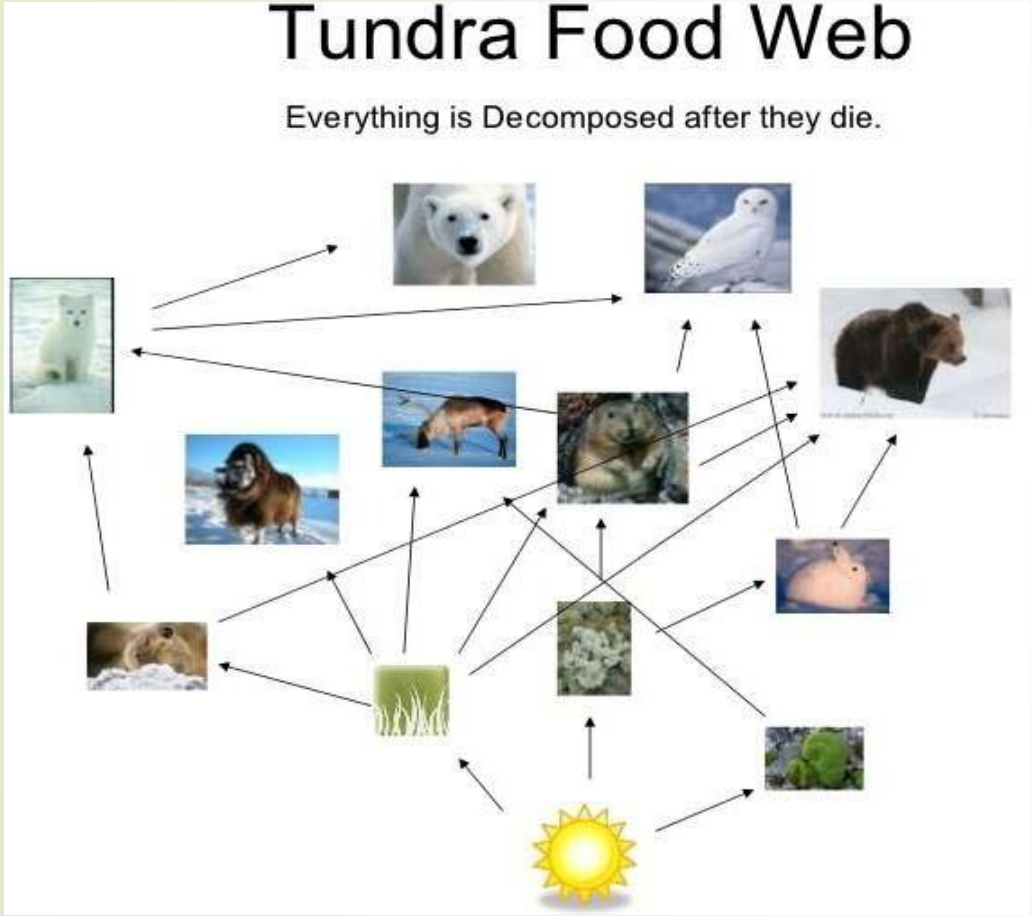
Food Web in Ecosystem



Food web

1. In a food web one trophic level may be connected to more than one food chain
2. Food web is the geographical description of feeding relationship among species in an ecological community

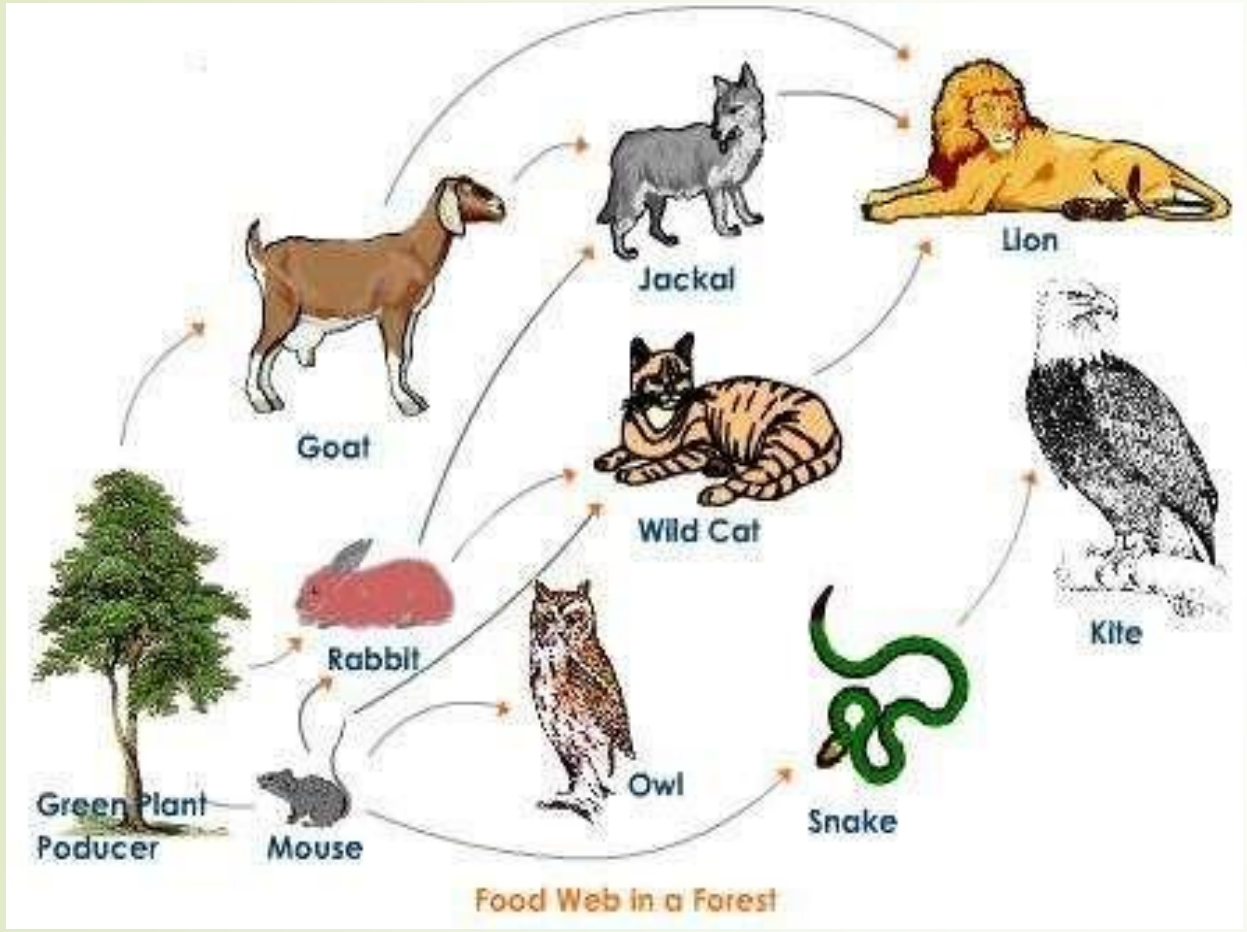
Food Web in Ecosystem

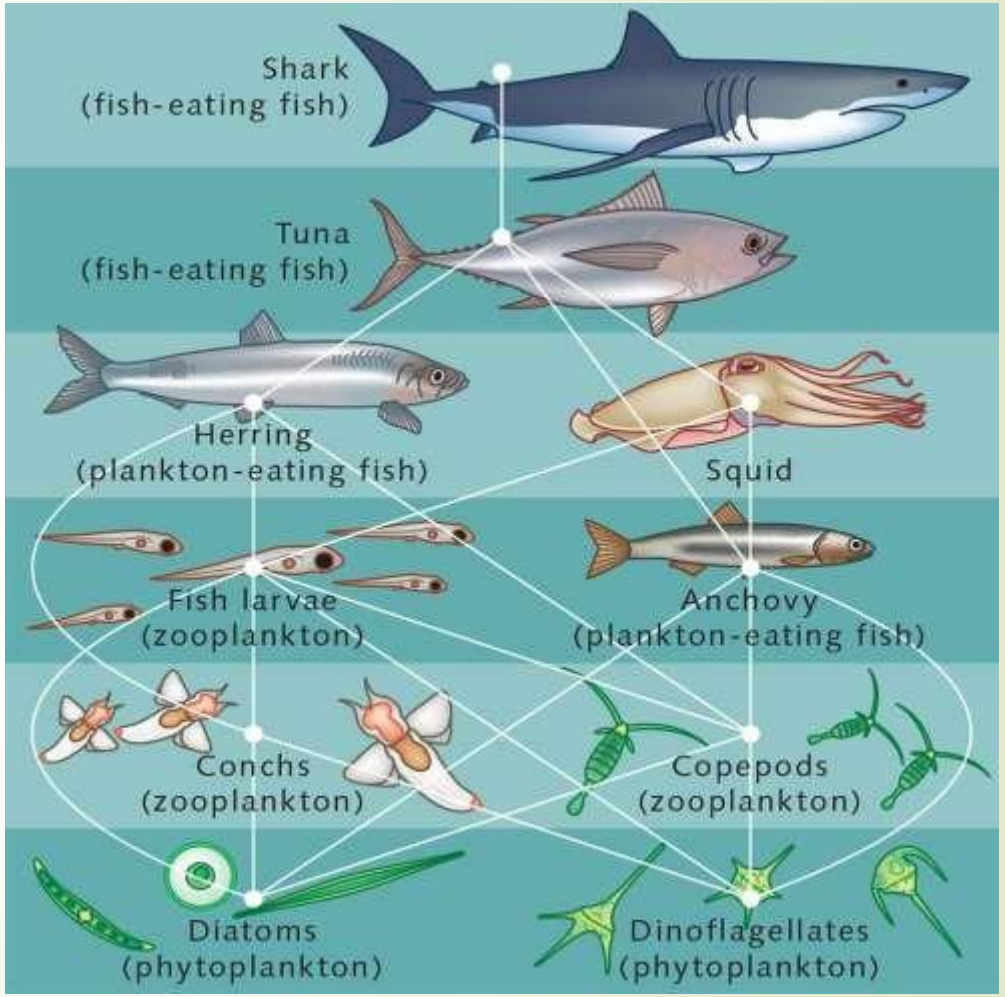


Food web

1. It even specify the energy relation and transfer of energy amongst them
2. Food web is the geographical description of feeding relationship among species in an ecological community
3. **e.g** Grass land food web
4. Tundra Food Web

Food Web in Ecosystem





Energy in Ecological systems

Module 28

Energy in Ecological systems

Thermodynamics

Definition:

"it's the branch of physics which studies energy, its transfer from one place to another and its transformation from one form to another form."

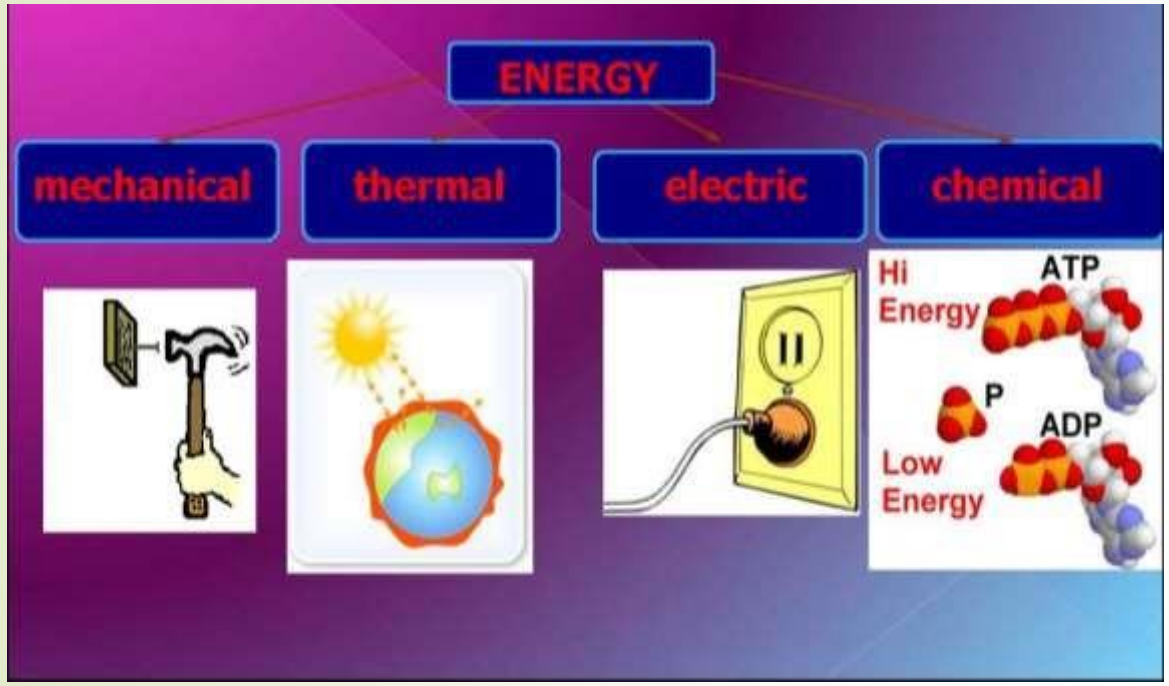
Energy in Ecological systems

Thermodynamics

The laws of thermodynamics describe the relationships between thermal energy, or heat, and other forms of energy, and how energy affects matter

Energy is the ability to do work.
different forms of energy

Energy in Ecological systems



1st law of Thermodynamics Or Conservation of Energy

Energy can be transformed from one form to another but can neither be created nor destroyed

the total *quantity* of energy in the universe stays the same

Energy in Ecological systems

Module 29

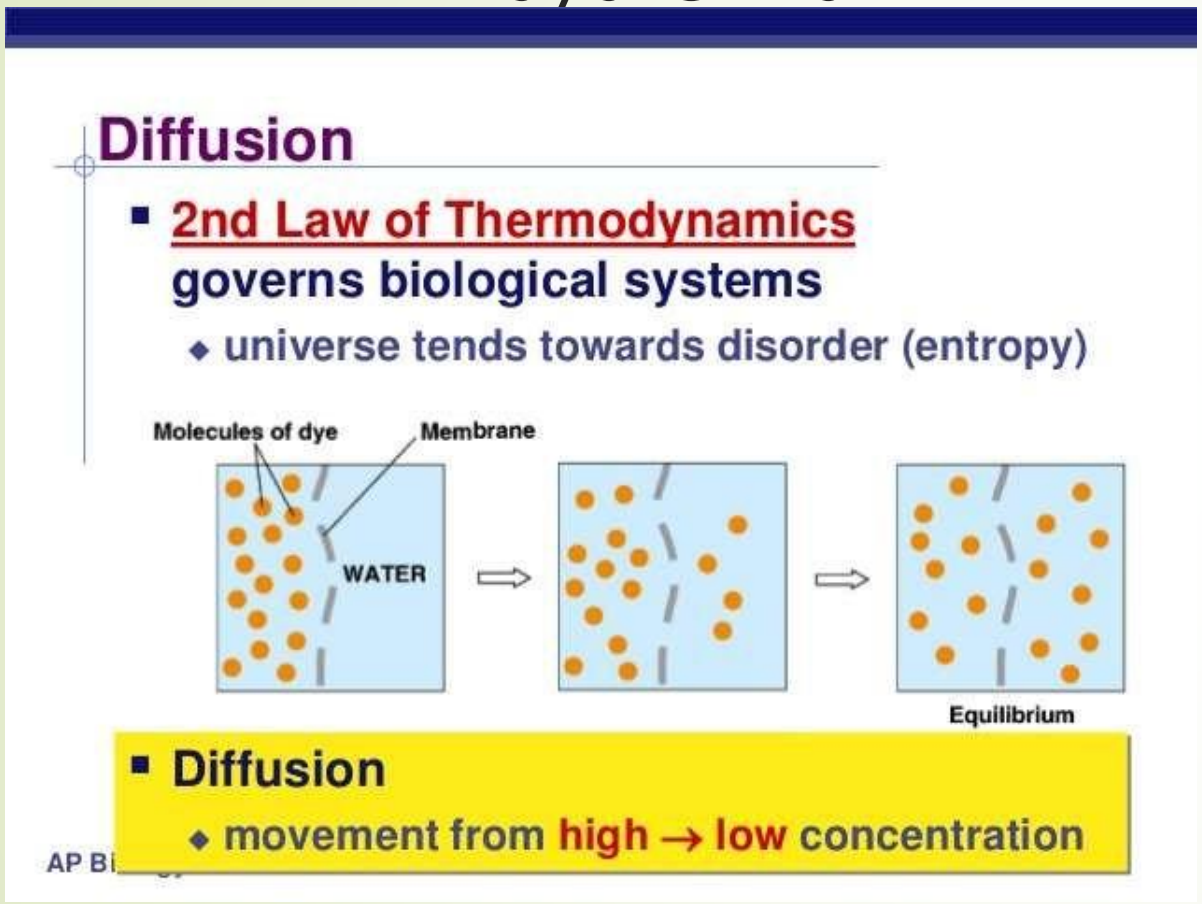
Energy in Ecological systems

2nd law of Thermodynamics Or Law of Entropy

Its about the *quality* of energy

It states that as energy is transferred or transformed, more and more of it is wasted

Energy in Ecological systems

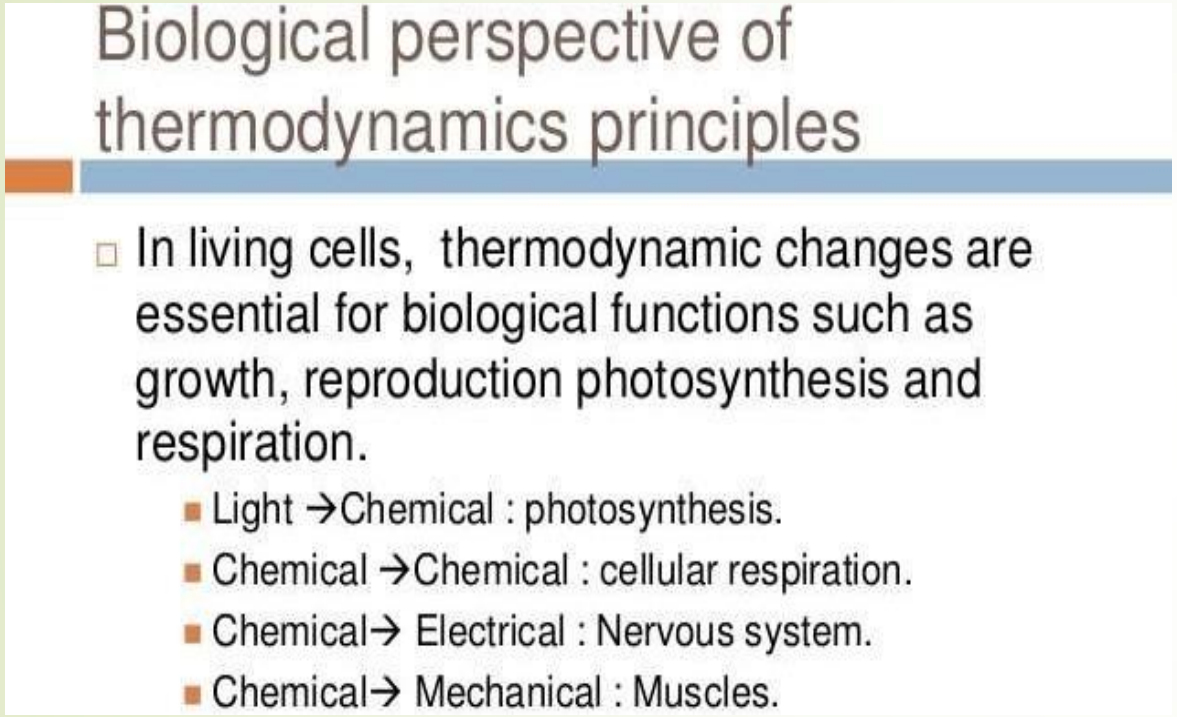


It is concerned with the direction of natural processes.

It asserts that a natural process runs only in one sense, and is not reversible

For example, heat always flows spontaneously from hotter to colder bodies, and never the reverse, unless external work is performed on the system

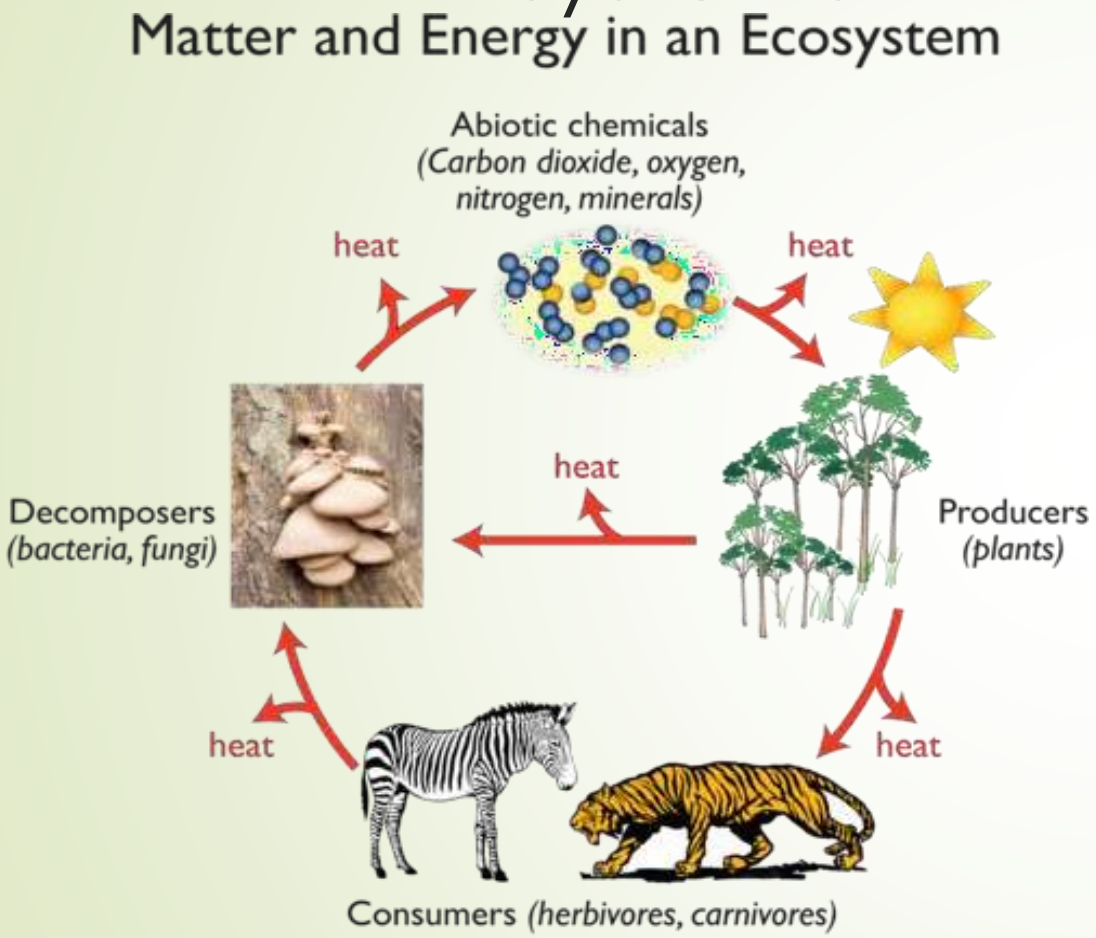
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Biological perspective of thermodynamics principles

In living cell, thermodynamics changes are essential for biological functions such as growth, reproduction photosynthesis and respiration

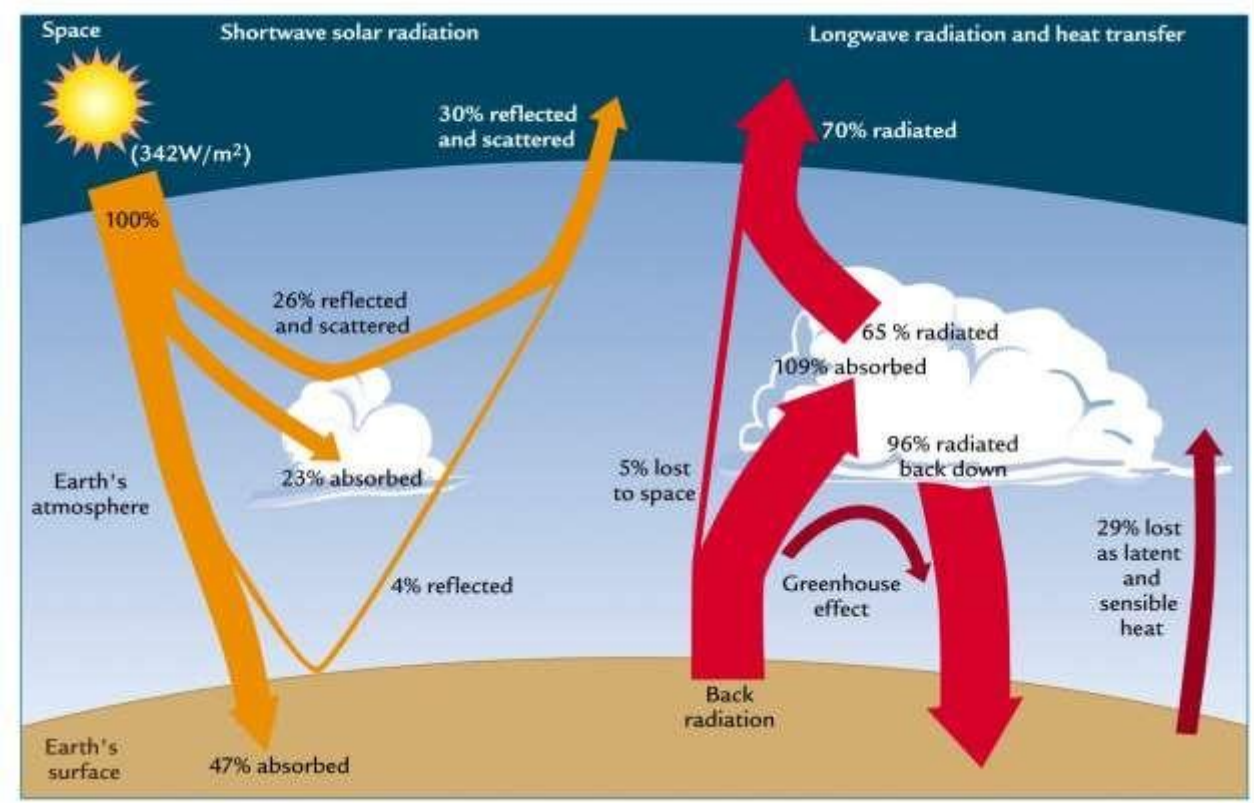
Energy in Ecological systems



Biological perspective of thermodynamics principles

Light-chemical: Photosynthesis
Chemical-chemical: cellular respiration
Chemical-Electrical: Nervous system
Chemical- Mechanical: Muscles

Energy in Ecological systems



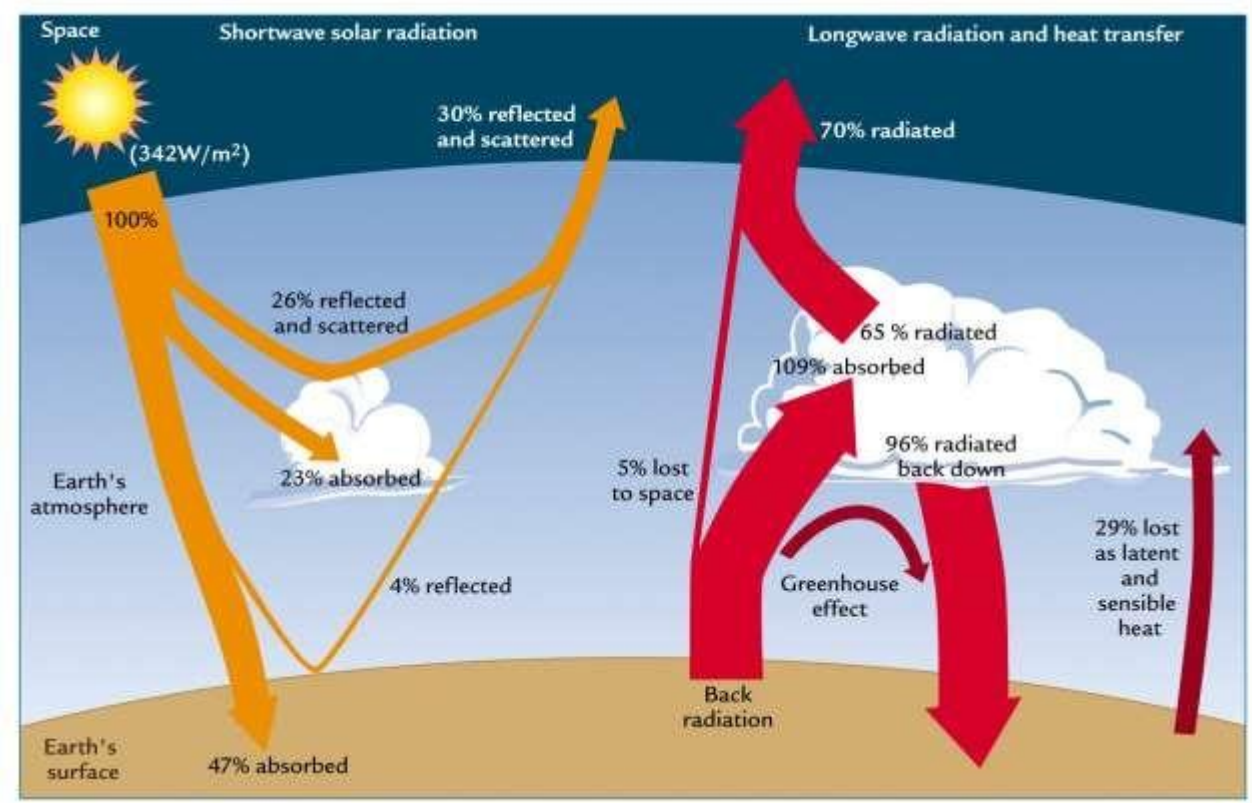
Solar radiation as energy source

sun is a sustainable source of energy

Solar radiation reached on surface of earth on clear day

1. Ultraviolet radiation- 10% - short wavelength-absorbed by Ozone layer

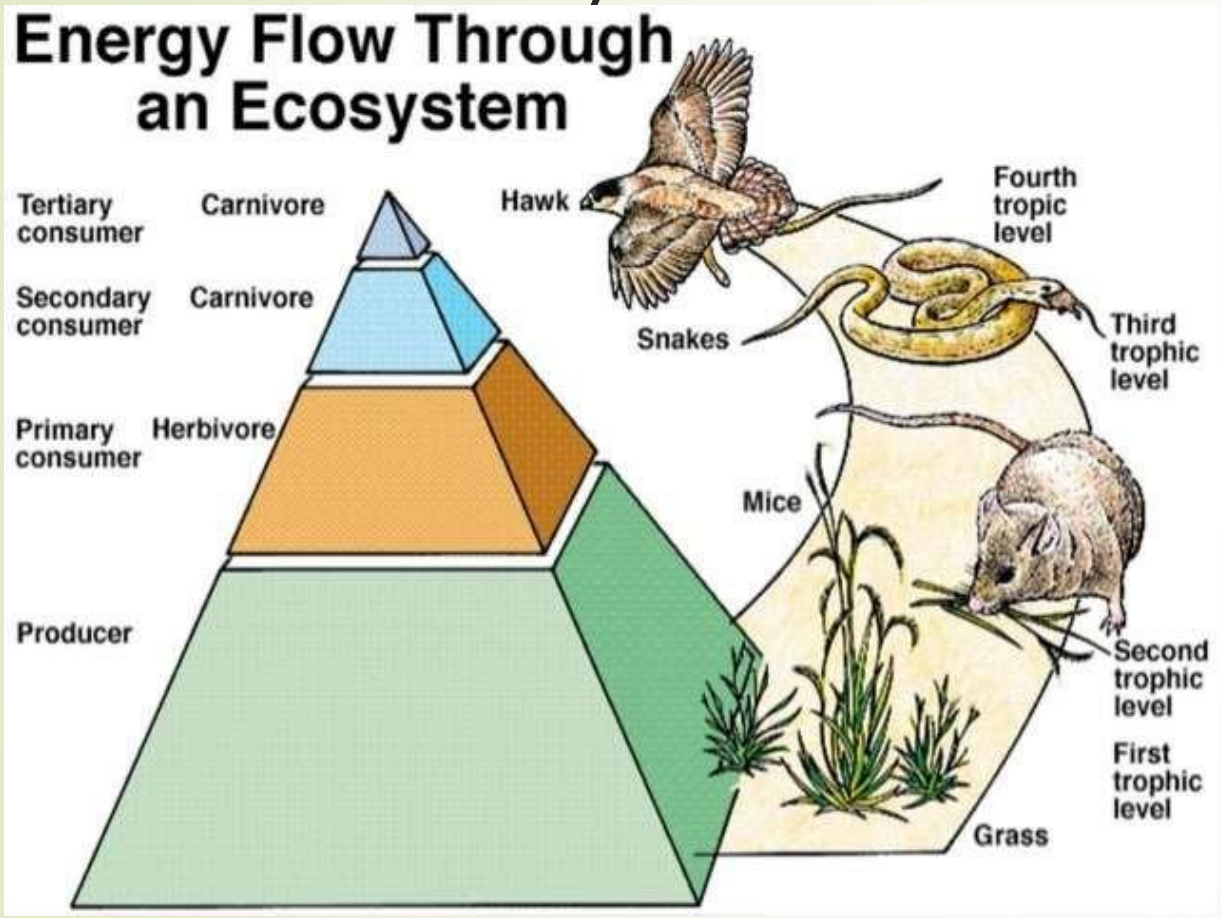
Energy in Ecological systems



Solar radiation as energy source

2. Visible light-medium wavelength-45%- least attenuated by clouds, water
3. infrared rays-45%- longer wavelength-absorbed and reradiated as heat in a complex manner by atmosphere, clouds, and various natural and man – made objects and surfaces

Energy in Ecological systems

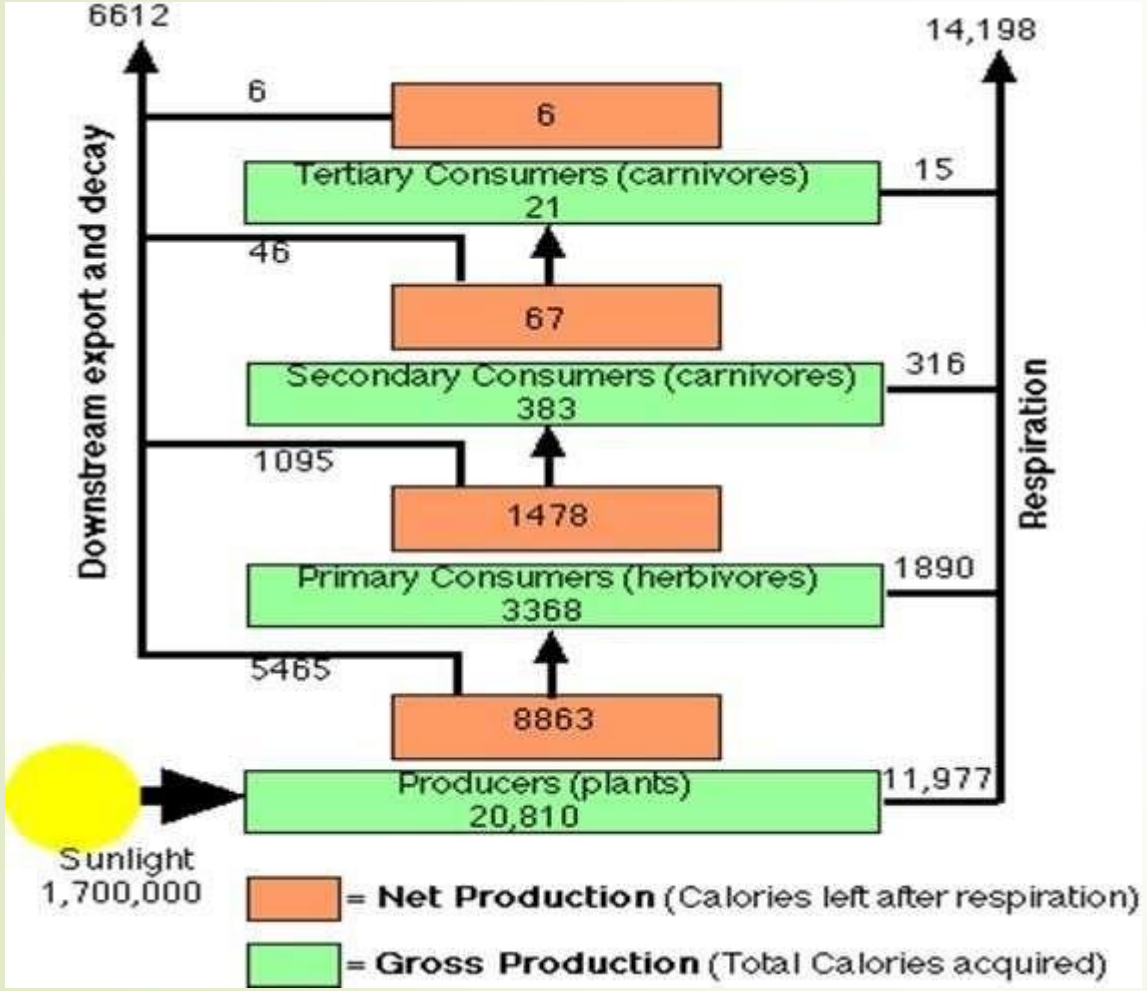


Flow of energy in Ecosystem

Important factor as describe Types and number of organism would live in ecosystem

Ecological Pyramids- relative amount of energy, biomass, number of organisms at each level in ecosystem

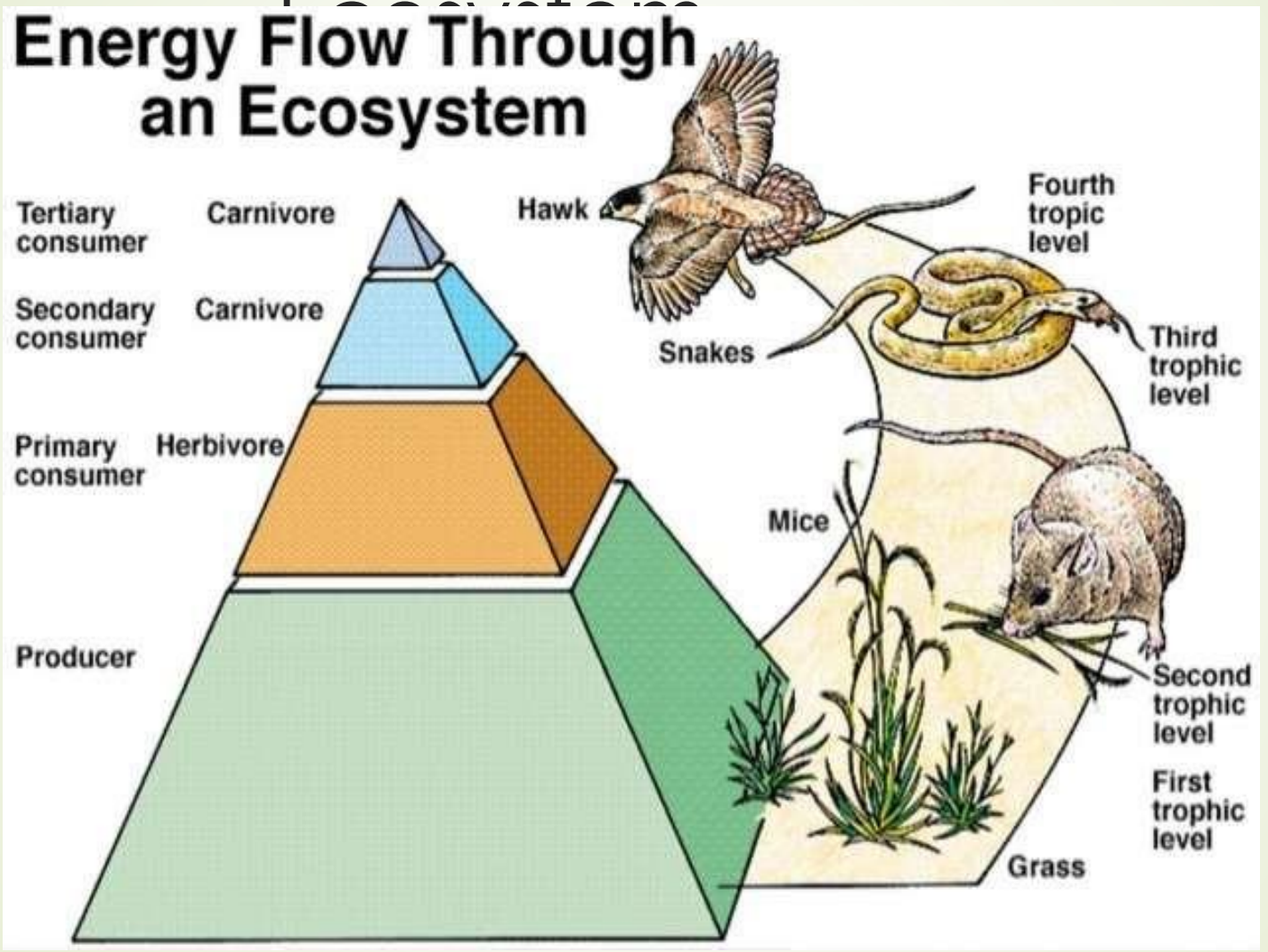
Energy in Ecosystem



Flow of energy in Ecosystem

Ecosystem- energy flows through different trophic levels by food chain following lawof thermodynamics

Energy in Ecosystem



Biogeochemical Cycles

Module 30

Biogeochemical Cycles

Biogeochemical cycles

Biogeochemical Cycles

Biogeochemical cycles

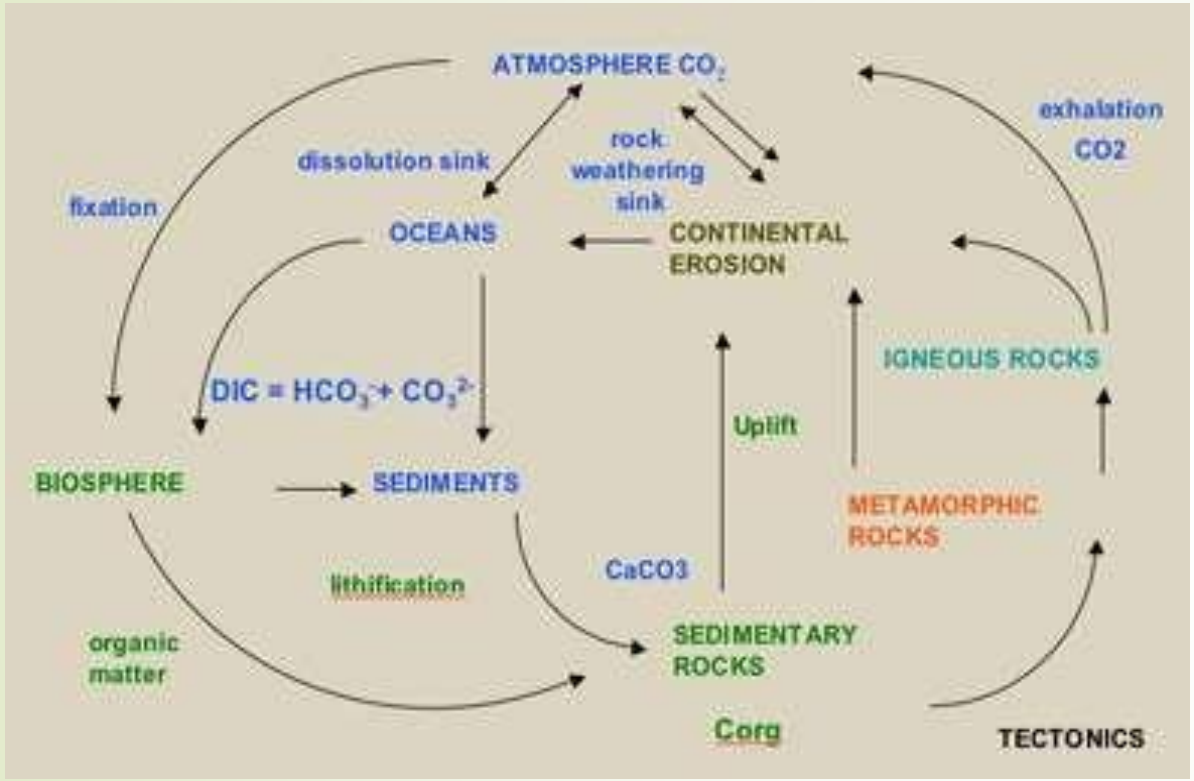
“the chemical elements, including all essential elements of protoplasm, tend to circulate in the biosphere in characteristics pathways from environment to organisms and back to environment” these more or less circular pathways are called biogeochemical cycles”

Biogeochemical Cycles

Biogeochemical cycles

1. **Interaction between Biotic and Abiotic environments**
2. Photosynthesis-Transpiration-Respiration-Decomposition
3. Recycling of material to be used over and over again

Biogeochemical Cycles



Biogeochemical cycles

Elements move through 4 components of earth system

Atmosphere (Air)
Hydrosphere (water)
Lithosphere (Soil)
Biosphere (Living things)

Examples: Nitrogen, Carbon, Water, Phosphorus, Sulphur cycles

Biogeochemical Cycles

Biogeochemical cycles

1. **Nutrient cycles-** movement of elements and minerals
2. Two pools or compartments for nutrient cycle

a) Reservoir Pool

b) Labile or Cycling pool

Biogeochemical Cycles

Biogeochemical cycles

a) **Reservoir Pool**

Large, slow moving, abiotic

a) **Labile or Cycling pool**

Smaller, active and usually biotic

END

Biogeochemical Cycles

Module 31

Biogeochemical Cycles

Types of Biogeochemical cycles

a) **Gaseous type-** in which reservoir is in atmosphere or the hydrosphere (Ocean)

examples.

Nitrogen Cycle
Carbon Cycle
Water Cycle

Biogeochemical Cycles

Types of Biogeochemical cycles

b) Sedimentary type- in which reservoir is in the earth Crust

examples.

Phosphorus Cycle

Sulphur Cycle

END

Biogeochemical Cycles

Lecture 32

Biogeochemical Cycles

Nitrogen (N2) Cycle

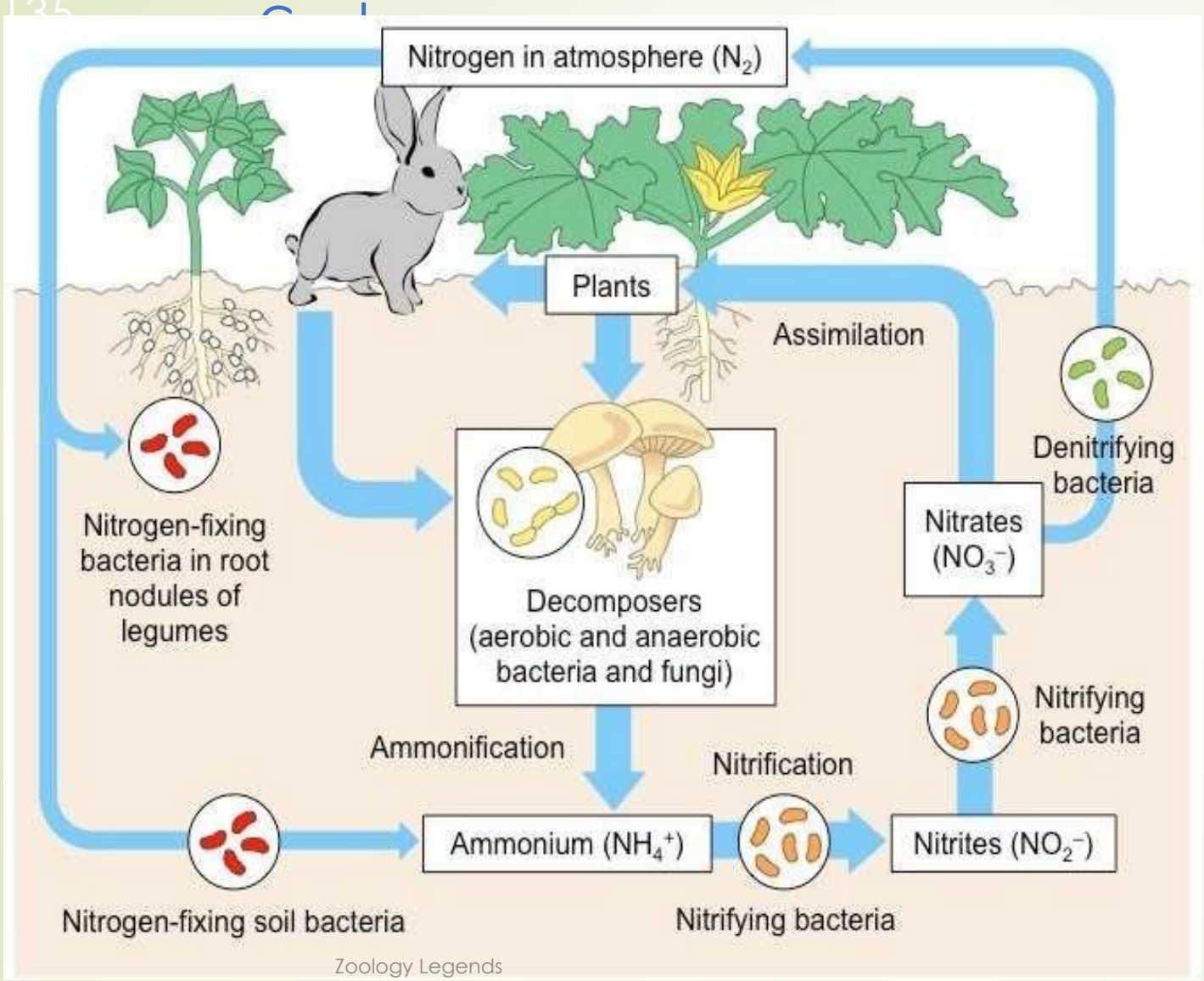
Importance of Nitrogen

Building structure of protein, DNA and RNA- (nitrogenous bases-nucleotides-genetic material and chlorophyll

a) Atmosphere

as reservoir -78%-
safety valve

Biogeochemical Cycles-Nitrogen (N_2)



N_2 Cycle Steps

Nitrogen fixation- conversation of Inert atmospheric nitrogen (N2) into the usable form – ammonia (NH3) or ammonium ion or nitrates.

None-Biological Fixation

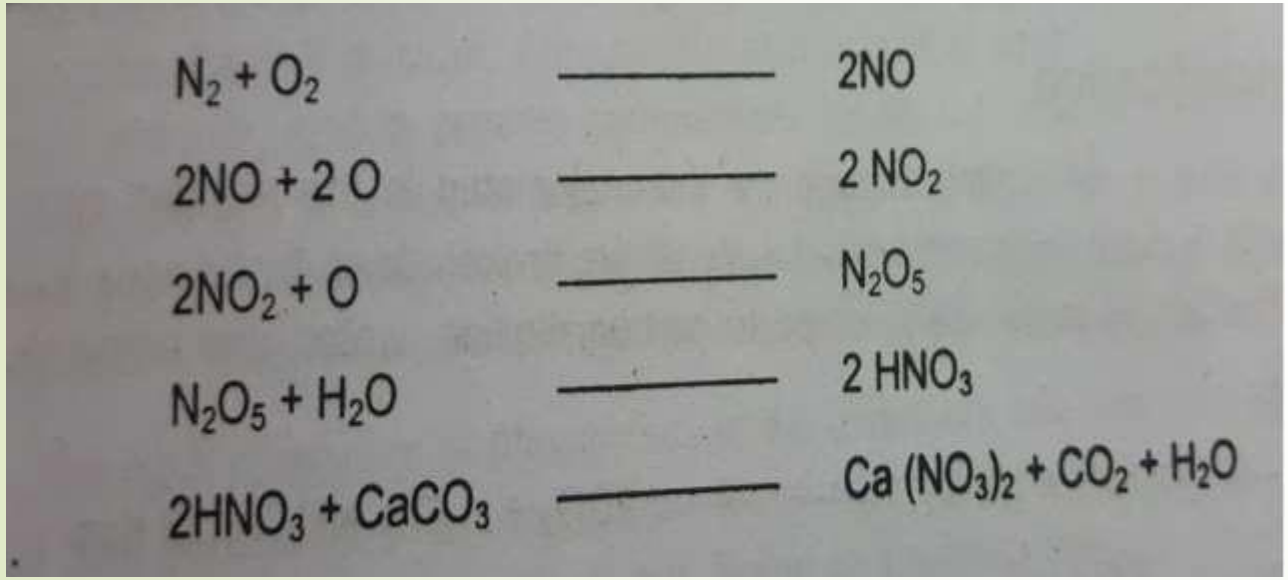
Biological Fixation

Lecture 33

Biogeochemical Cycles-

Nitrogen (N_2) Cycle

Biogeochemical Cycles-Nitrogen (N_2) Cycle



Non Biological Nitrogen Fixation

1. Atmospheric fixation: A natural phenomenon where the energy of lightning splitting up of the gaseous nitrogen, combing with oxygen (ozone layer) into nitrogen oxides and is then used by plants as nitrates.

Biogeochemical Cycles-Nitrogen (N_2) Cycle

2. Industrial nitrogen fixation: Is a man-made alternative that aids in nitrogen fixation by the use of ammonia. Ammonia is produced by the direct combination of nitrogen and hydrogen and later, it is converted into various fertilizers such as urea.

END

Biogeochemical Cycles-Nitrogen (N_2) Cycle

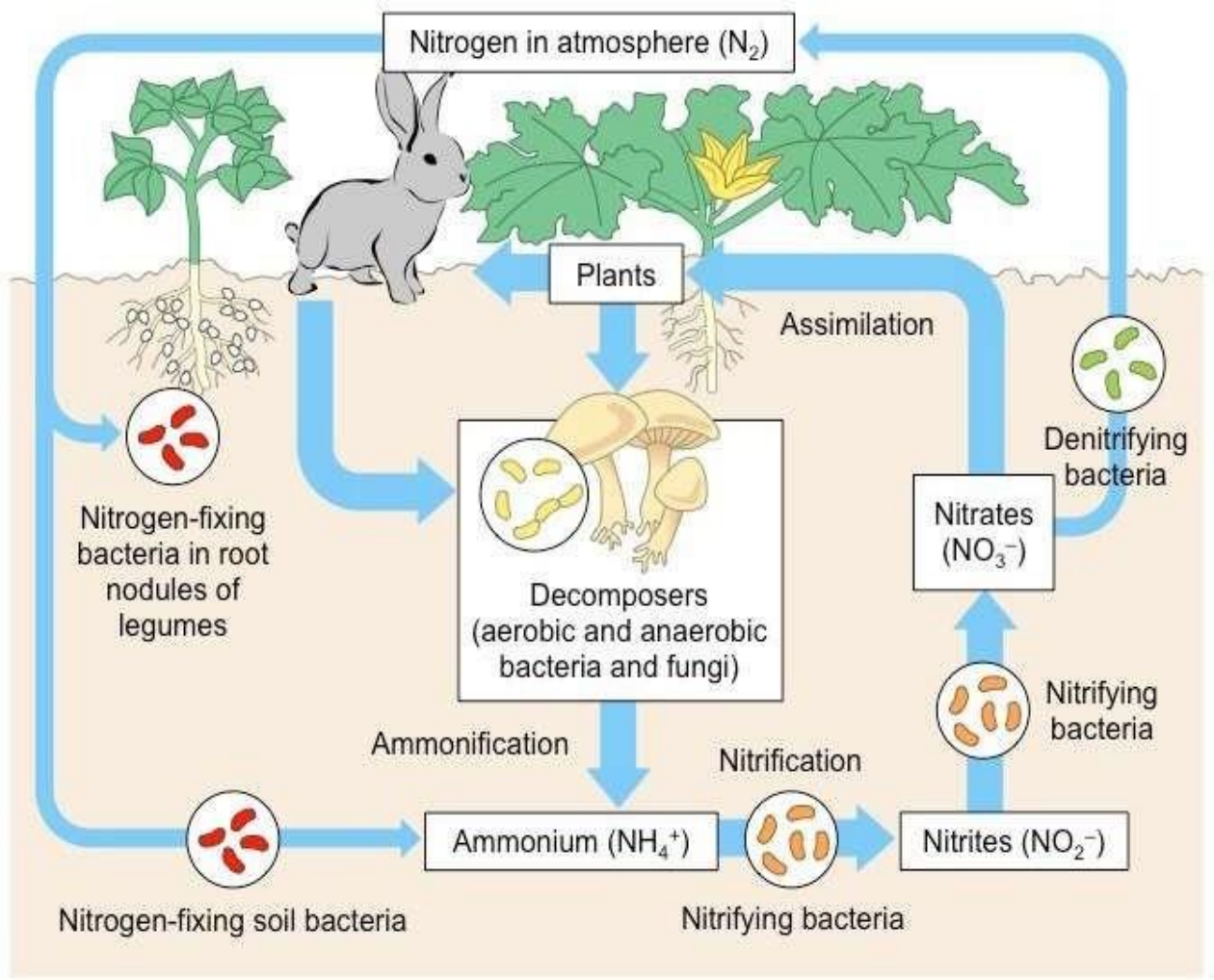
Lecture 34

Biogeochemical Cycles-Nitrogen (N_2) Cycle

Biological Nitrogen Fixation Steps

a) 90% fixed nitrogen to earth (100-200 kg/ha)
b) ammonia (NH3) or ammonium ion or nitrites or nitrates basic useable form by plants
c) Legume nodules plants-10g glucose-fix 1g
d) Free living microbes fixers-100g glucose- 1g

Biogeochemical Cycles-Nitrogen (N_2) Cycle

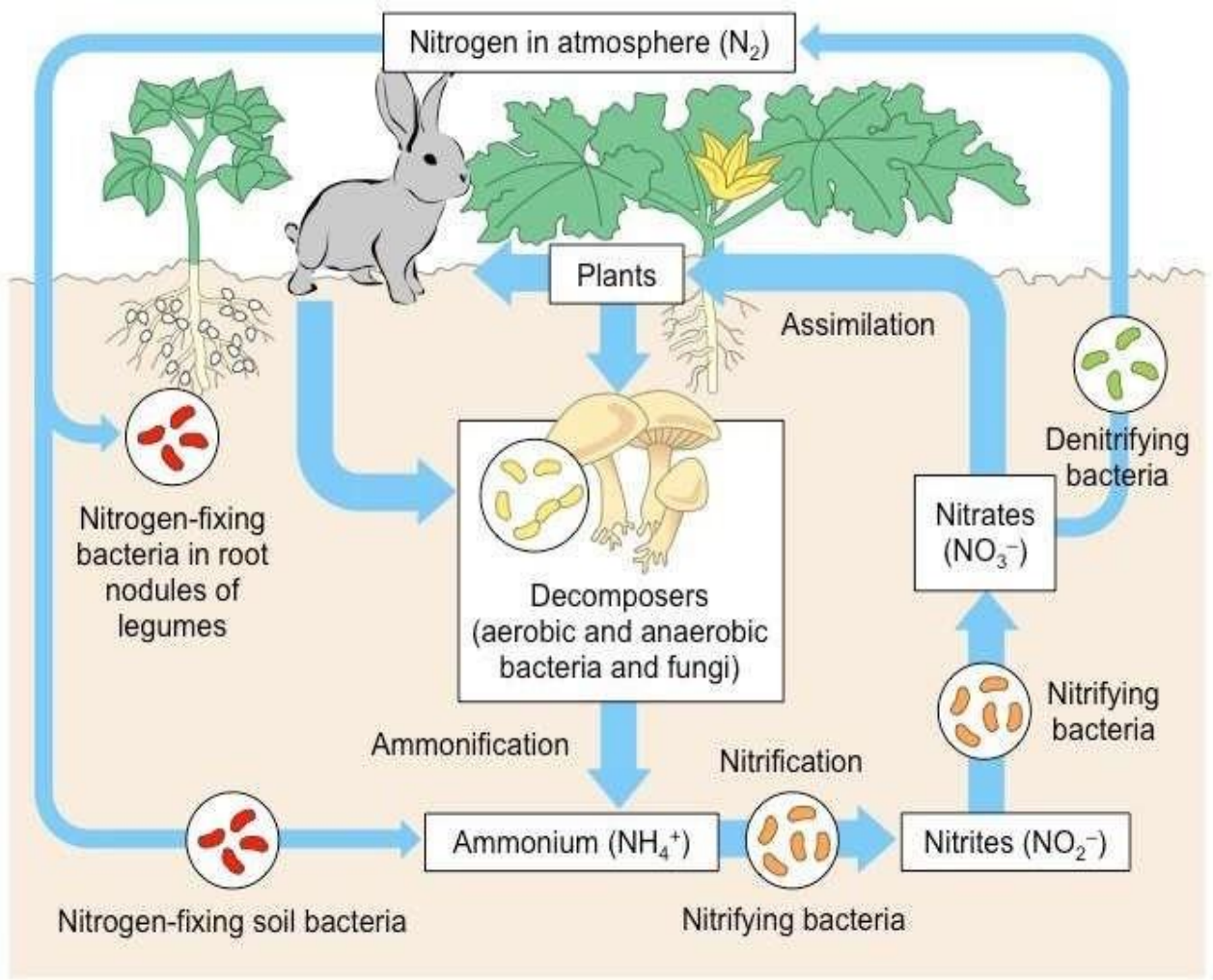


Biological Nitrogen Fixation

Steps

a) Ammonification
b) Nitrification
c) Assimilation
d) Denitrification

Biogeochemical Cycles-Nitrogen (N_2) Cycle



3. **Biological fixation:**

a) Mutualistic bacteria- beneficial association with

b) Leguminous (agricultural system)

c) Root nodules in none-leguminous plants

d) Free living nitrogen fixing bacteria-Soil

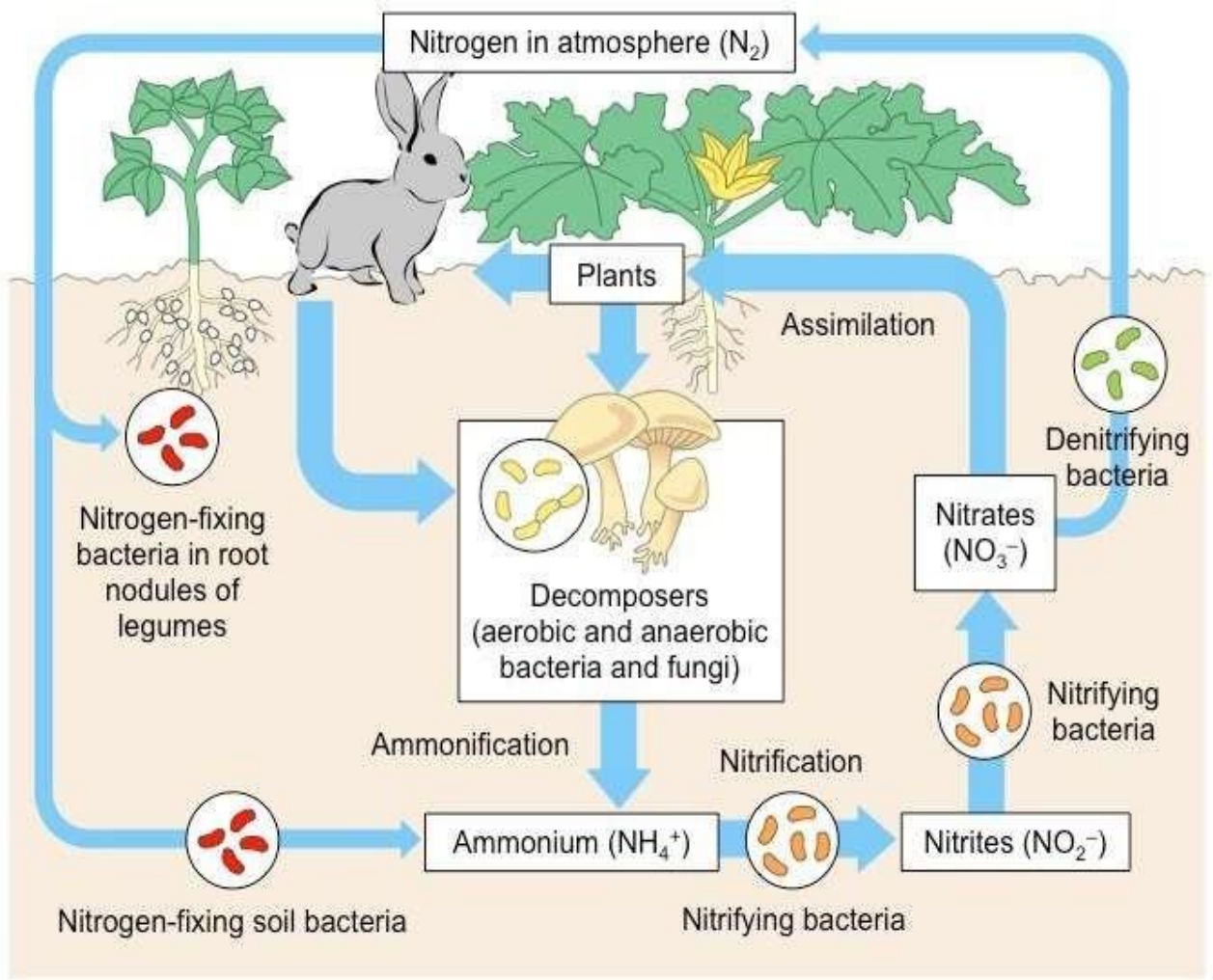
e) Cyanobacteria- Ocean ecosystem

Lecture 35

Biogeochemical Cycles-

Nitrogen (N_2) Cycle

Biogeochemical Cycles-Nitrogen (N_2) Cycle



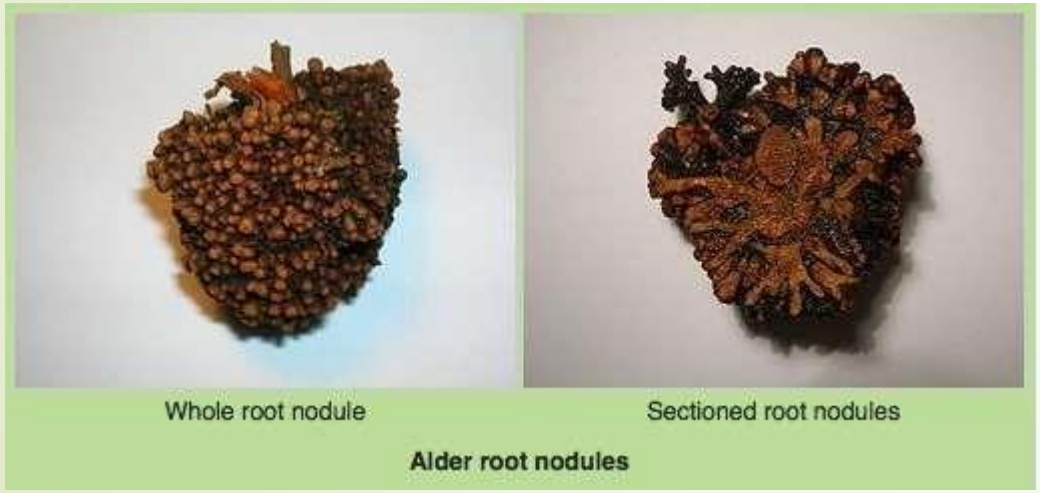
3. Biological fixation: Nodulated legumes and none leguminous plants

mutualistic relationship with aerobic rod shaped soil bacteria of genus Rhizobium- surrounding of plant roots called Rhizosphers- damaged epidermal cells of root hairs- plant response is nodule formation- enzyme is nitrogenase

Biogeochemical Cycles-Nitrogen (N_2) Cycle







Zoology Legends

3. Biological fixation:

Many none leguminous plants-alder, Ginkgo, Russian Olive- grow in wildland-filamentous bacteria- actinomycetes in root nodules- so can fix nitrogen

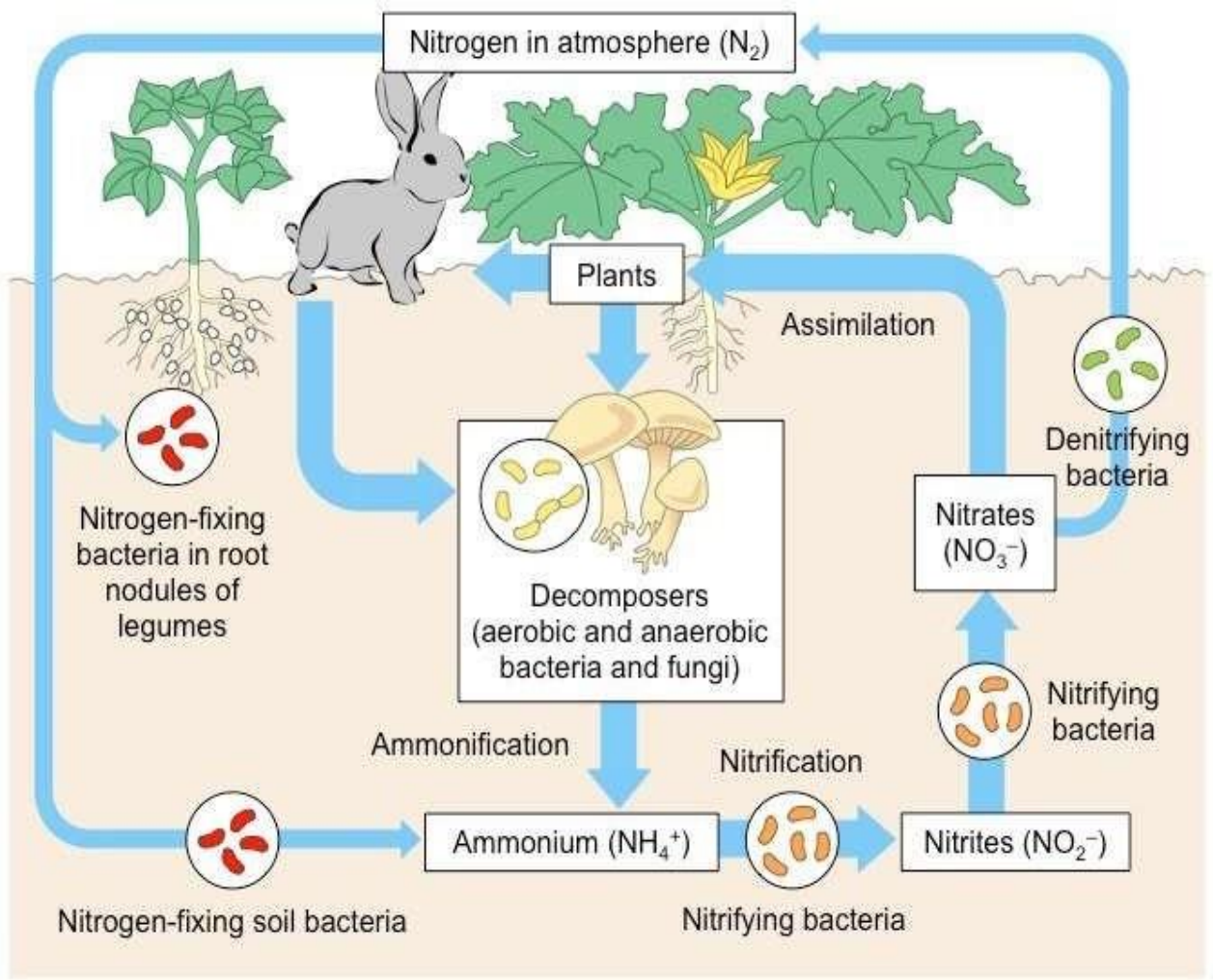
160 spp-8 genera-5 families- of dicotyledons-actinomycetes induced nodules

Lecturer 36

Biogeochemical Cycles-

Nitrogen (N_2) Cycle

Biogeochemical Cycles-Nitrogen (N_2) Cycle



3. Biological fixation:

Free living bacteria- 15 known genera fix gaseous nitrogen- *Azobacter* (aerobic), *Clostridum*- (anaerobic) found nearly all types of soil, pH of soil 6-7, ammonia is first stable and final product

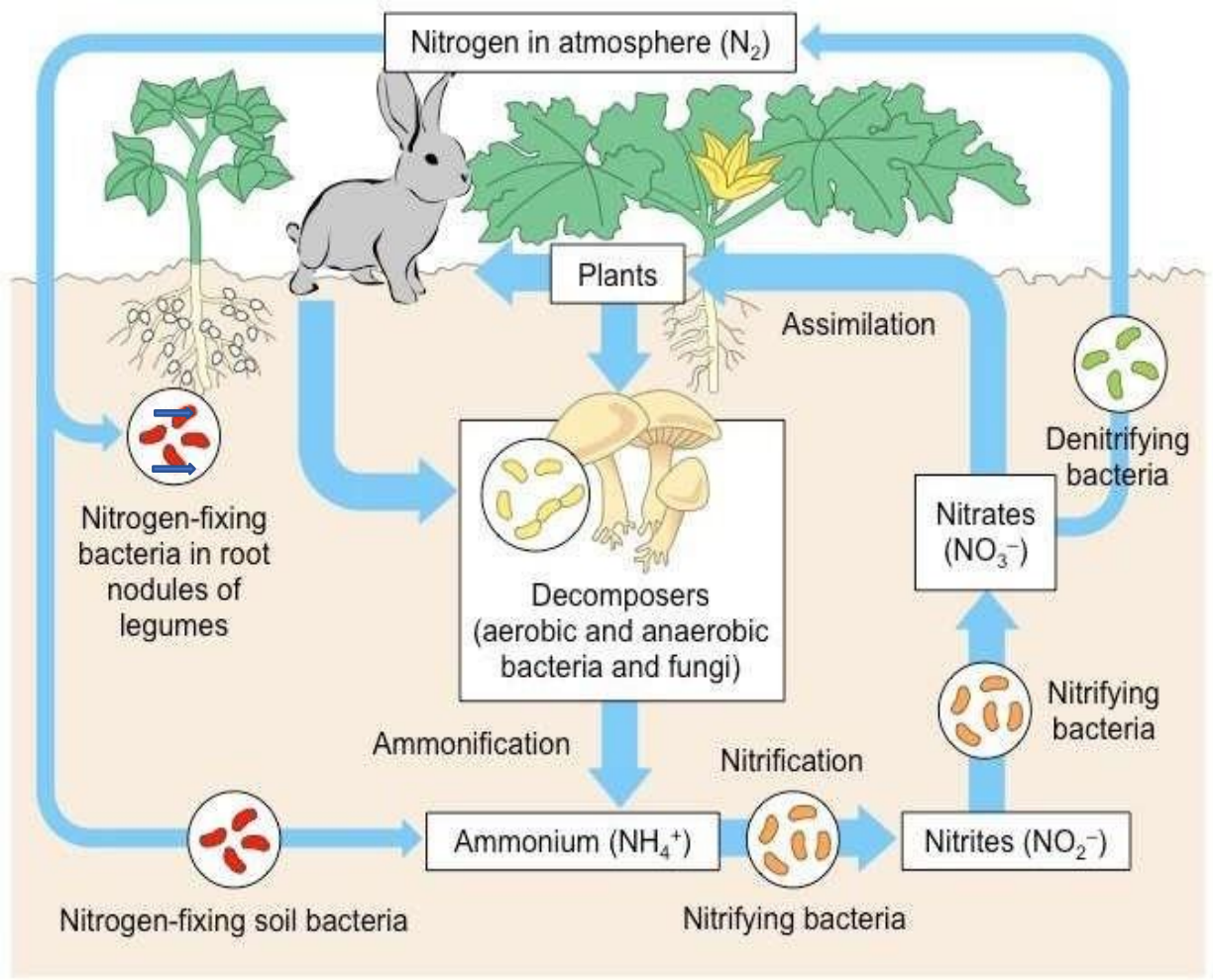
Cyanobacteria- (*Anabaena*, *Nostoc*, blue green algae) fix nitrogen in soil and aquatic habitat. Fix nitrogen in rice paddies in Asia

Lecture 37

Biogeochemical Cycles-

Nitrogen (N_2) Cycle

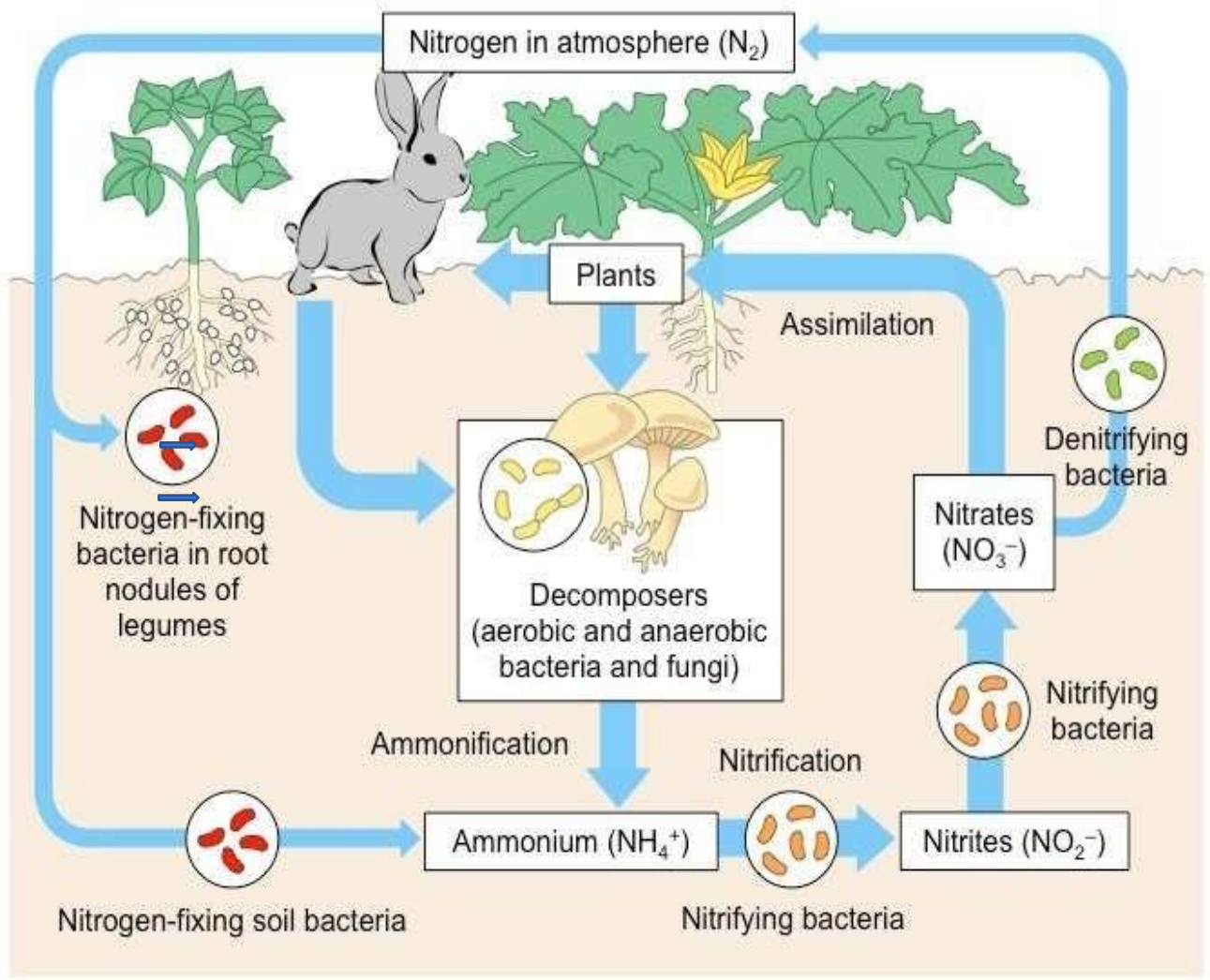
Biogeochemical Cycles-Nitrogen (N_2) Cycle



Ammonification

When plants or animal die, the nitrogen present in the organic matter is released back into the soil.

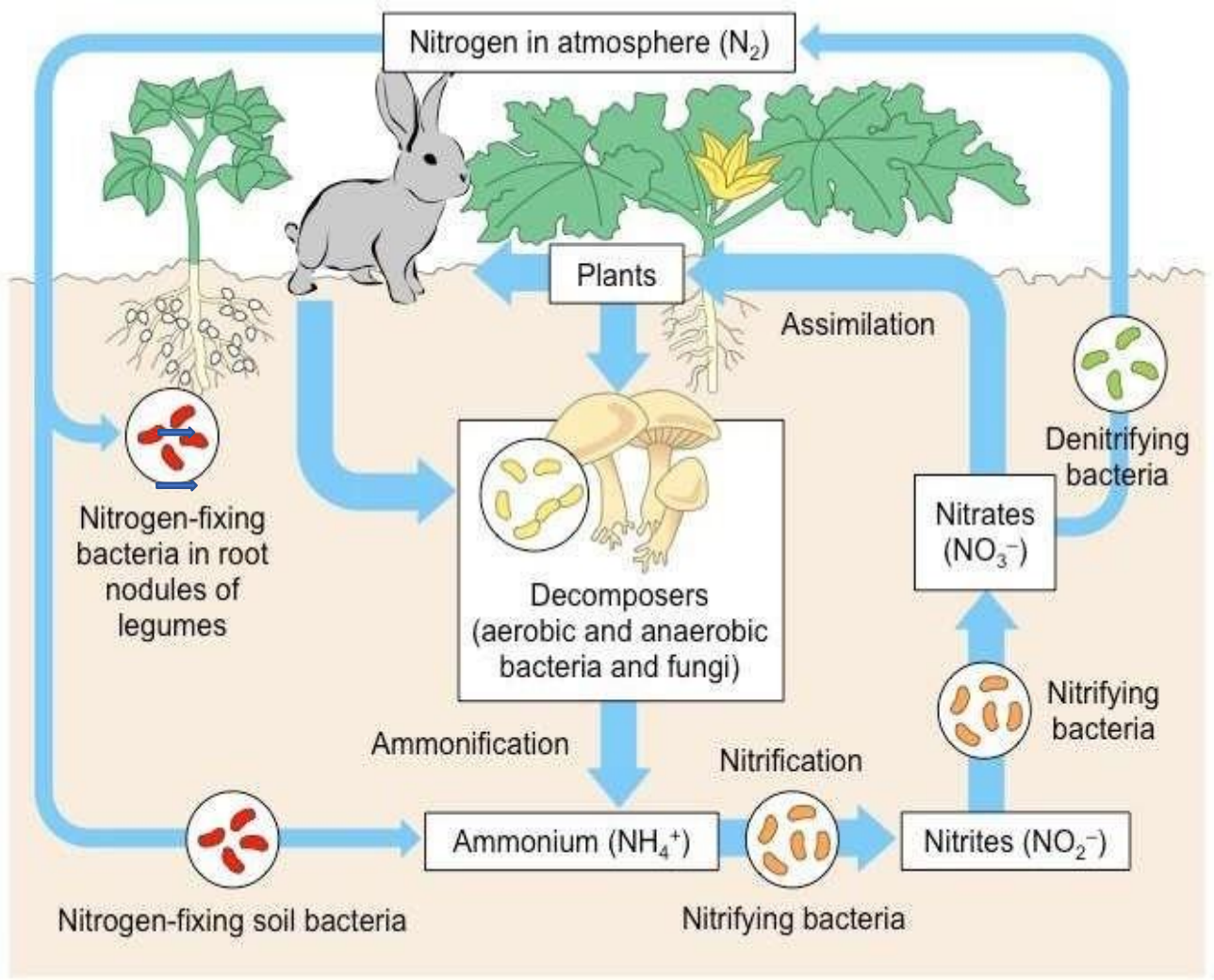
Biogeochemical Cycles-Nitrogen (N_2) Cycle



Ammonification

The decomposers, namely bacteria or fungi present in the soil convert the organic matter back into ammonium. This process of decomposition produces ammonia which is further used for other biological processes.

Biogeochemical Cycles-Nitrogen (N_2) Cycle



Nitrification-ammonia is oxidized into nitrate

Nitrosomonas bacterium in soil

$2NH_4 + 3O_2$ $2NO_{-2}+ 4H^+ + 2H_2O$

nitrites are converted into nitrates by *Nitrobacter*

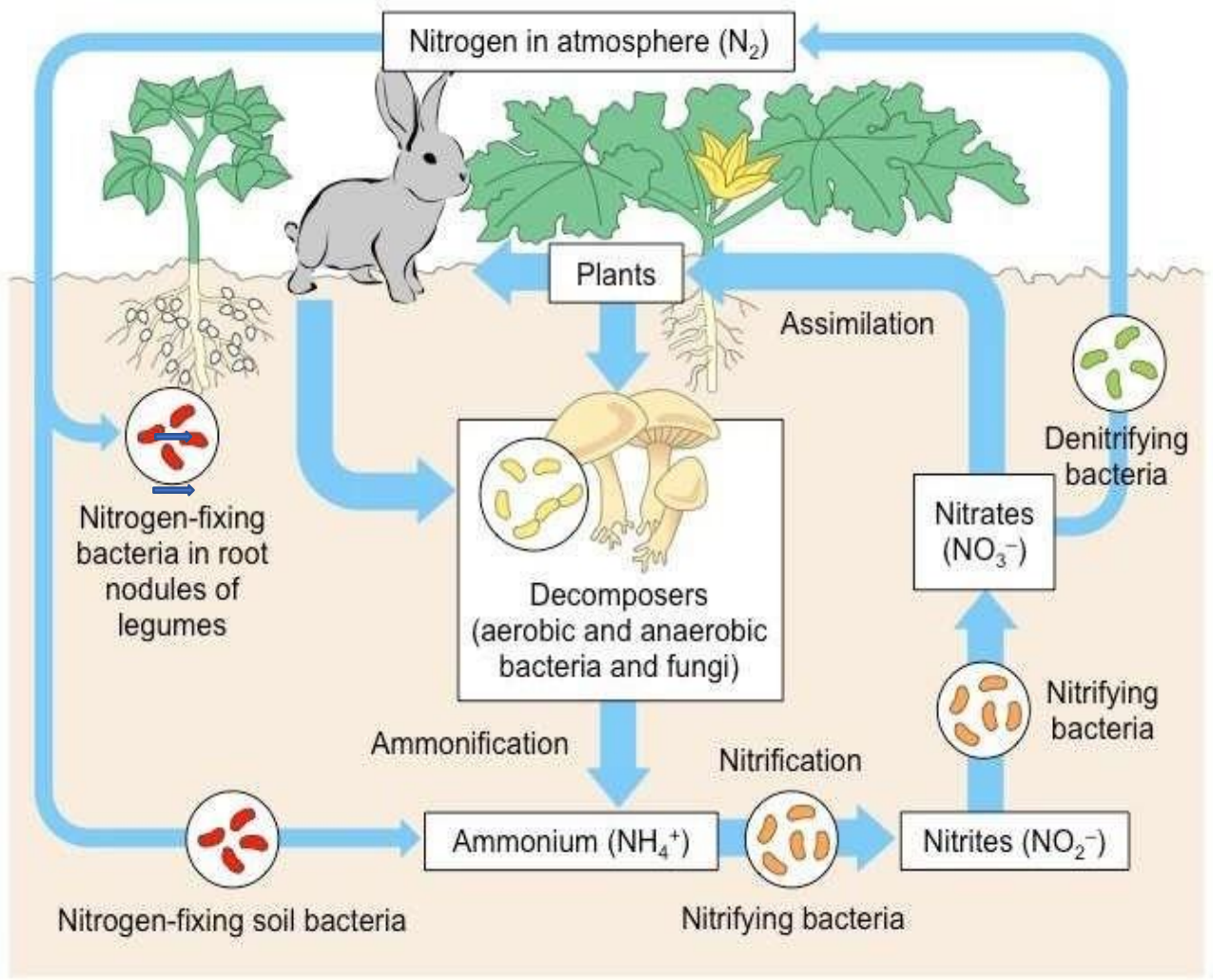
$2NO_{-2} + O_2$ $2NO_{-3}$

Lecturer 38

Biogeochemical Cycles-

Nitrogen (N_2) Cycle

Biogeochemical Cycles-Nitrogen (N_2) Cycle



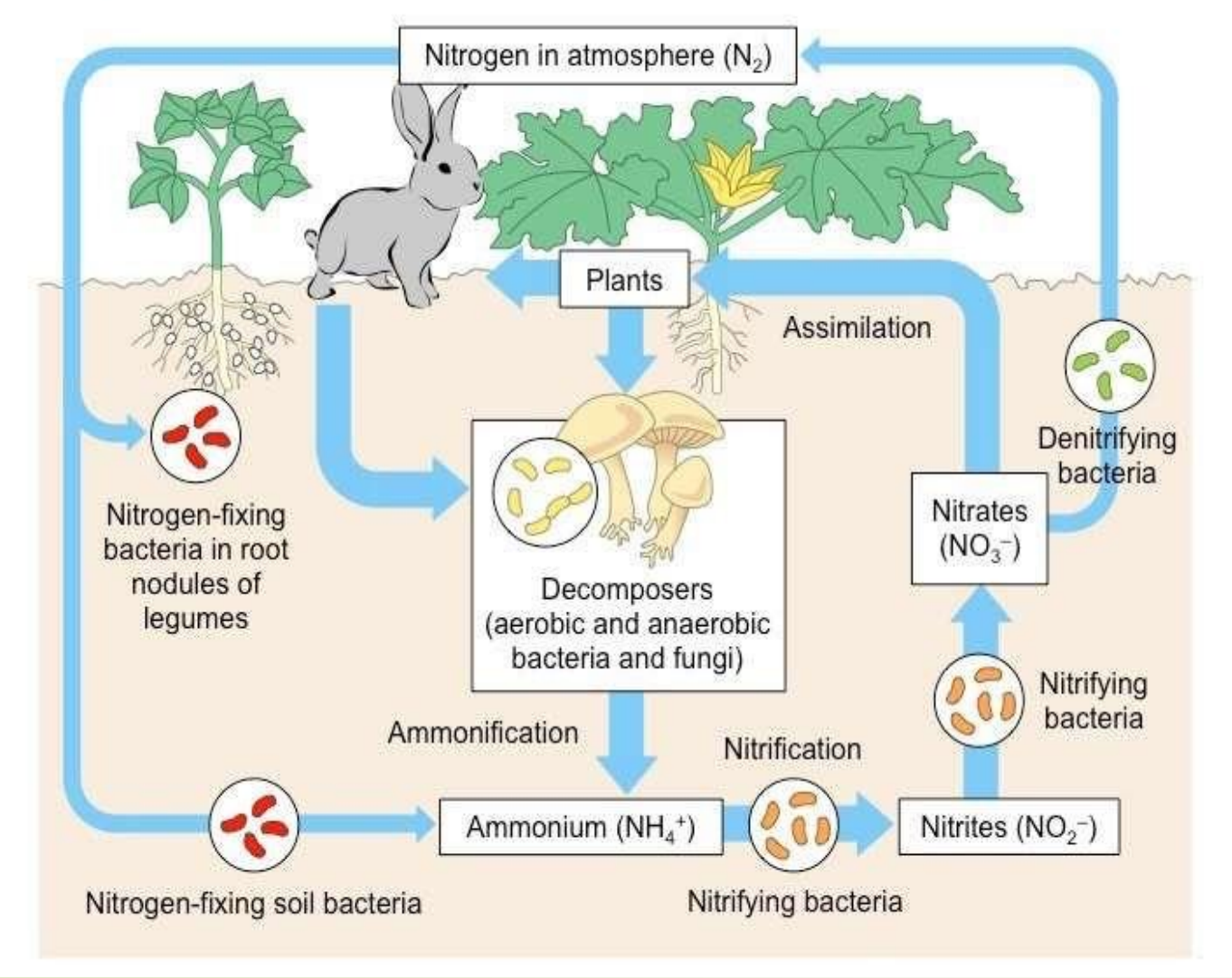
Assimilation

Primary producers- nitrite ions, nitrate ions or ammonium ions- through roots- plant protein

Nitrogen enters in Food chain

When Primary consumers eat plants

Biogeochemical Cycles-Nitrogen (N_2) Cycle



Denitrification

Denitrification is the process in which the nitrogen compounds makes its way back into the atmosphere by converting nitrate (NO_3) into gaseous nitrogen (N).-anaerobic microbes denitrifying bacterial species- *Clostridium* and *Pseudomonas*

Biogeochemical Cycles-Hydrological Cycle

Lecture 39

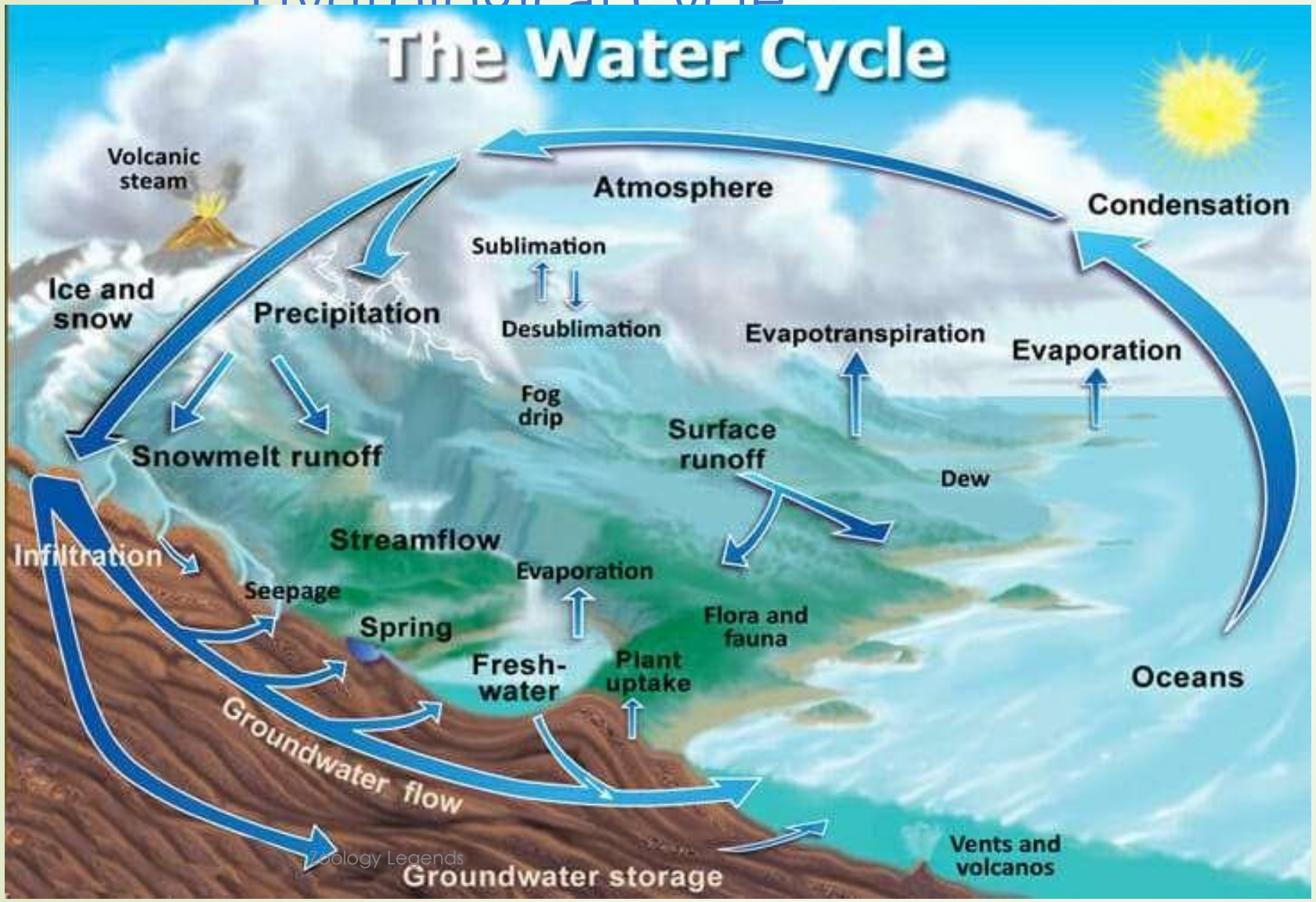
Biogeochemical Cycles-

Hydrological Cycle

Definition

Movement of water from oceans (large reservoir) by evaporation into atmosphere (the smallest reservoir), then by precipitation (rainfall) back down to the surface of earth with infiltration and runoff from the continents and eventually return to the ocean

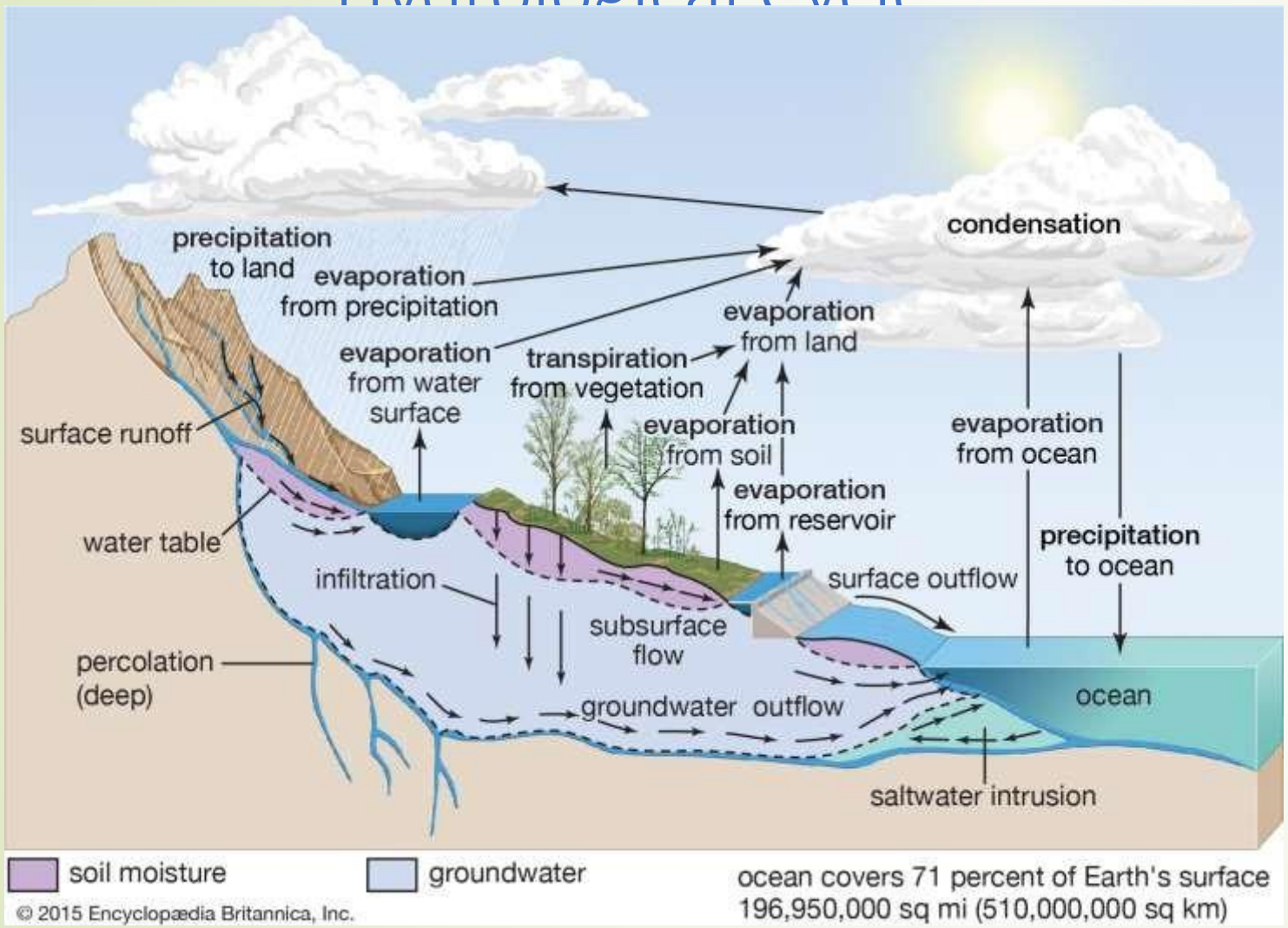
Biogeochemical Cycles-Hydrological Cycle



1. **Evaporation,**
2. **Condensation**
3. **Precipitation**

are the main processes involved in water cycle, these processes alternate with each other

Biogeochemical Cycles-Hydrological Cycle



Facts of hydrological cycle

1.Nearly 97.3% is in the oceans and 2.1% exists as polar ice caps, world's second largest reservoirs.

2.Only 0.6% is present as fresh water in, the form of atmospheric water vapors, ground and soil water

Biogeochemical Cycles-

Hydrological Cycle

Facts of hydrological cycle

1. One third of all solar energy is dissipated in driving the water cycle or In terms of energy, uphill loop (evaporation) of water cycle is sun driven
2. More water evaporates from the sea than returns to it by rainfall and vice versa for land.
3. Human activities tend to increase the rate of runoff

Biogeochemical Cycles-

Hydrological Cycle

Facts of hydrological cycle

3.Down hill loop releases energy, used for electric power generation and used by ecosystem

4.Aquifers- under ground water- largest global water reservoir- porous underground strata mostly limestone, sand, gravel, clay that hold water like elongated tank

Biogeochemical Cycles-Hydrological Cycle

Lecture 40

Biogeochemical Cycles-

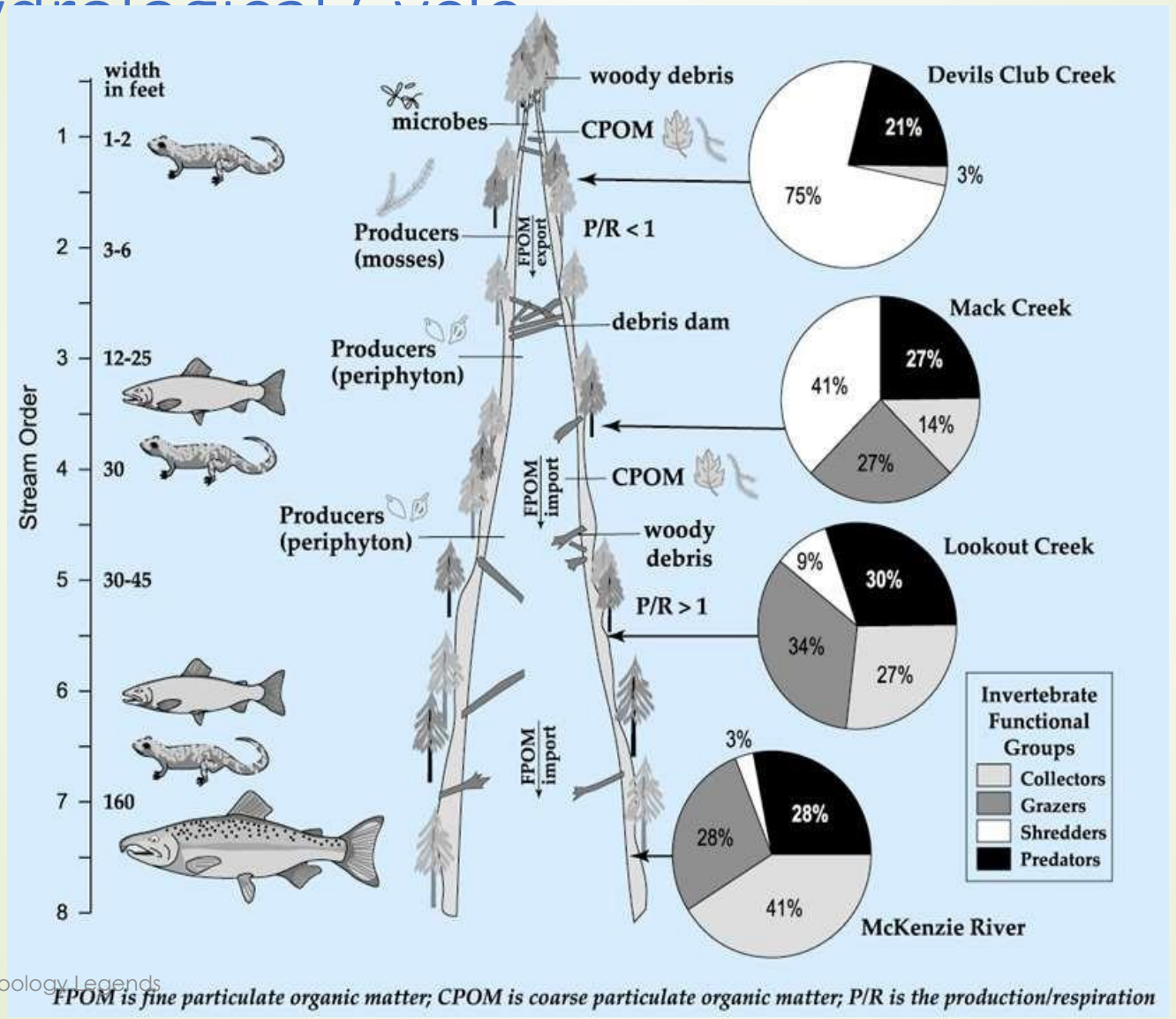
Hydrological Cycle

River continuum concept of hydrologic cycle

Diagram showing stream order depicting organisms by feeding type, change in particulate matter, change in community metabolism, diversity, particle size from headwater streams to large rivers

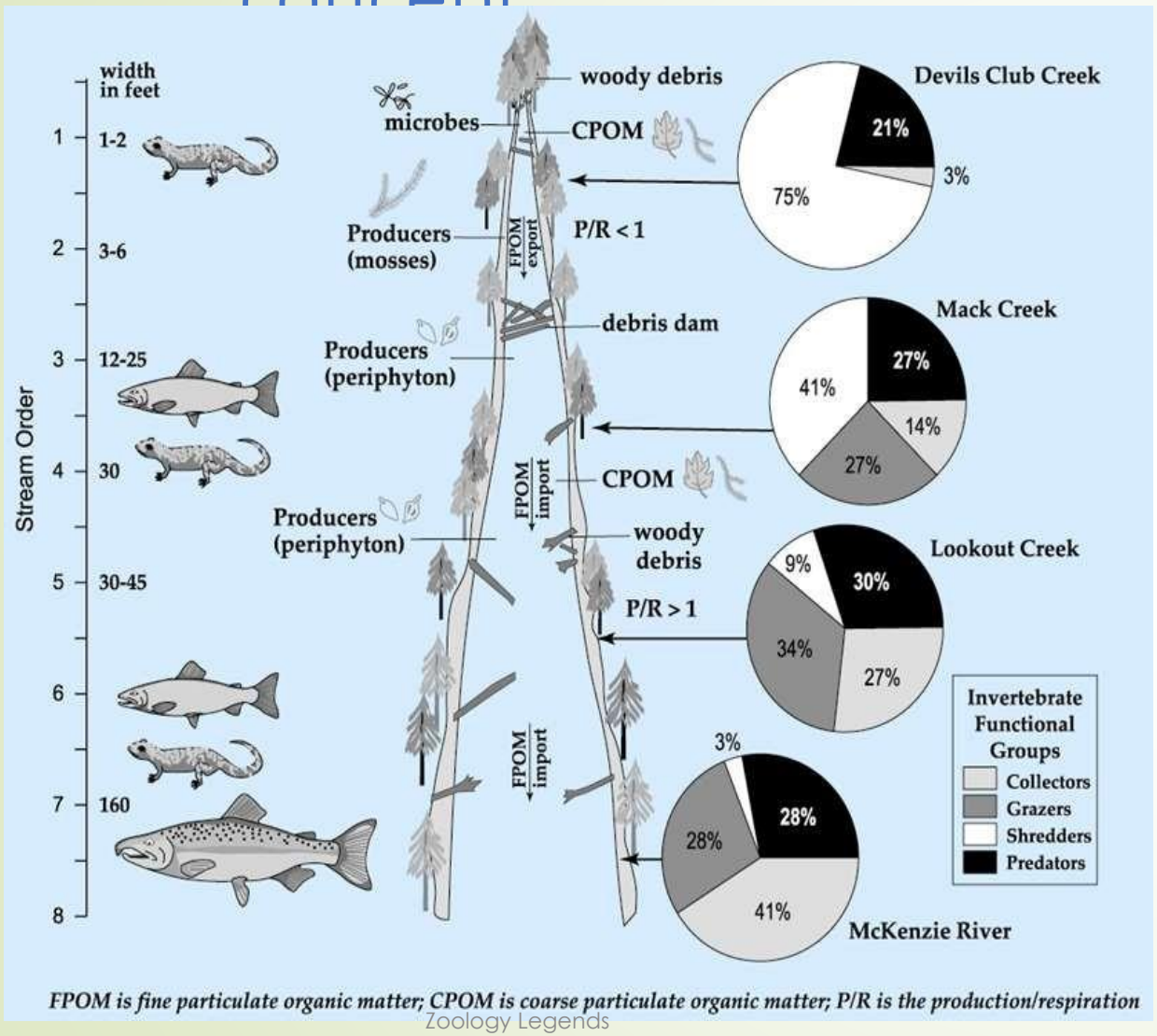
Biogeochemical Cycles-

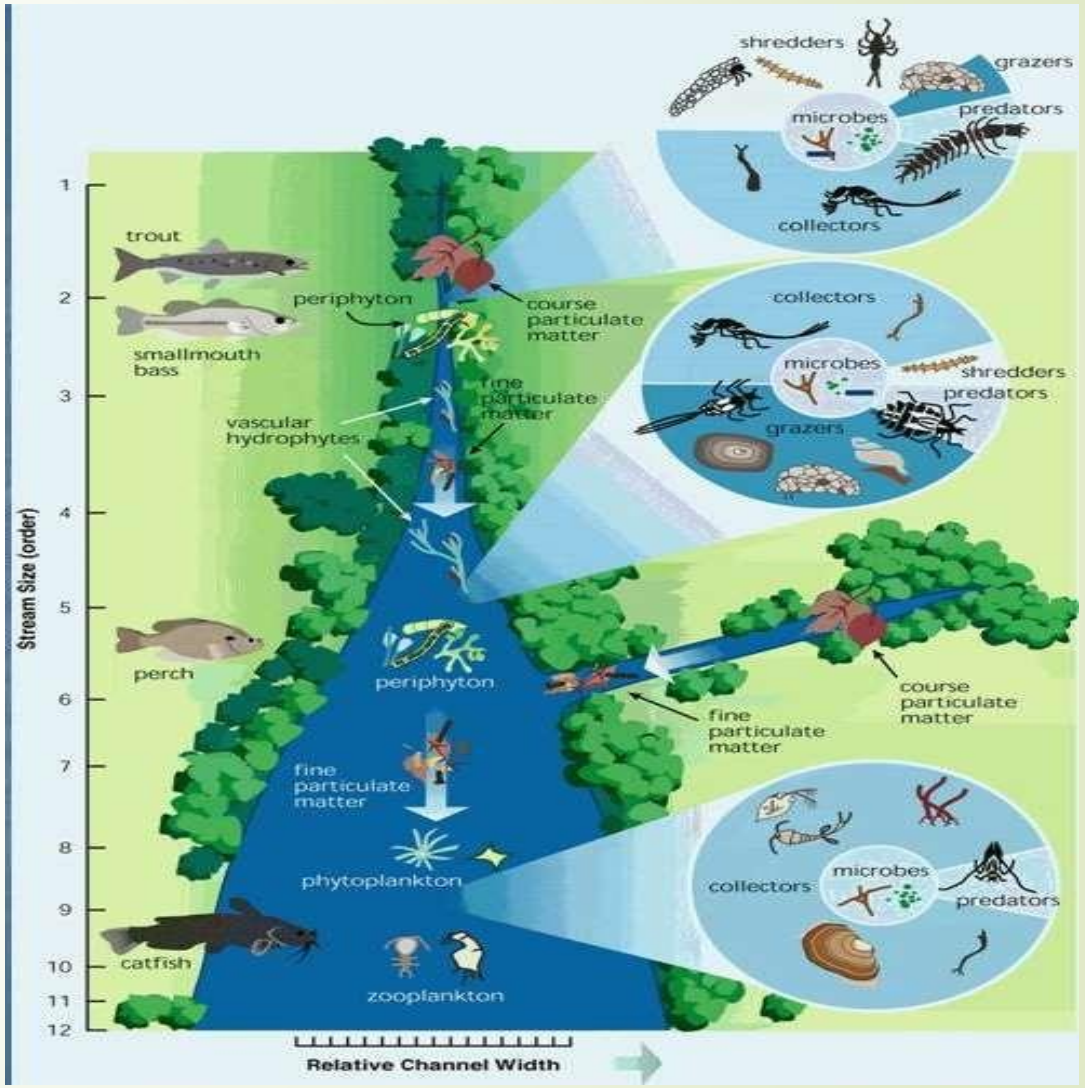
Hydrological Cycle



FPOM is fine particulate organic matter; CPOM is coarse particulate organic matter; P/R is the production/respiration

Biogeochemical Cycles-River Continuum concept

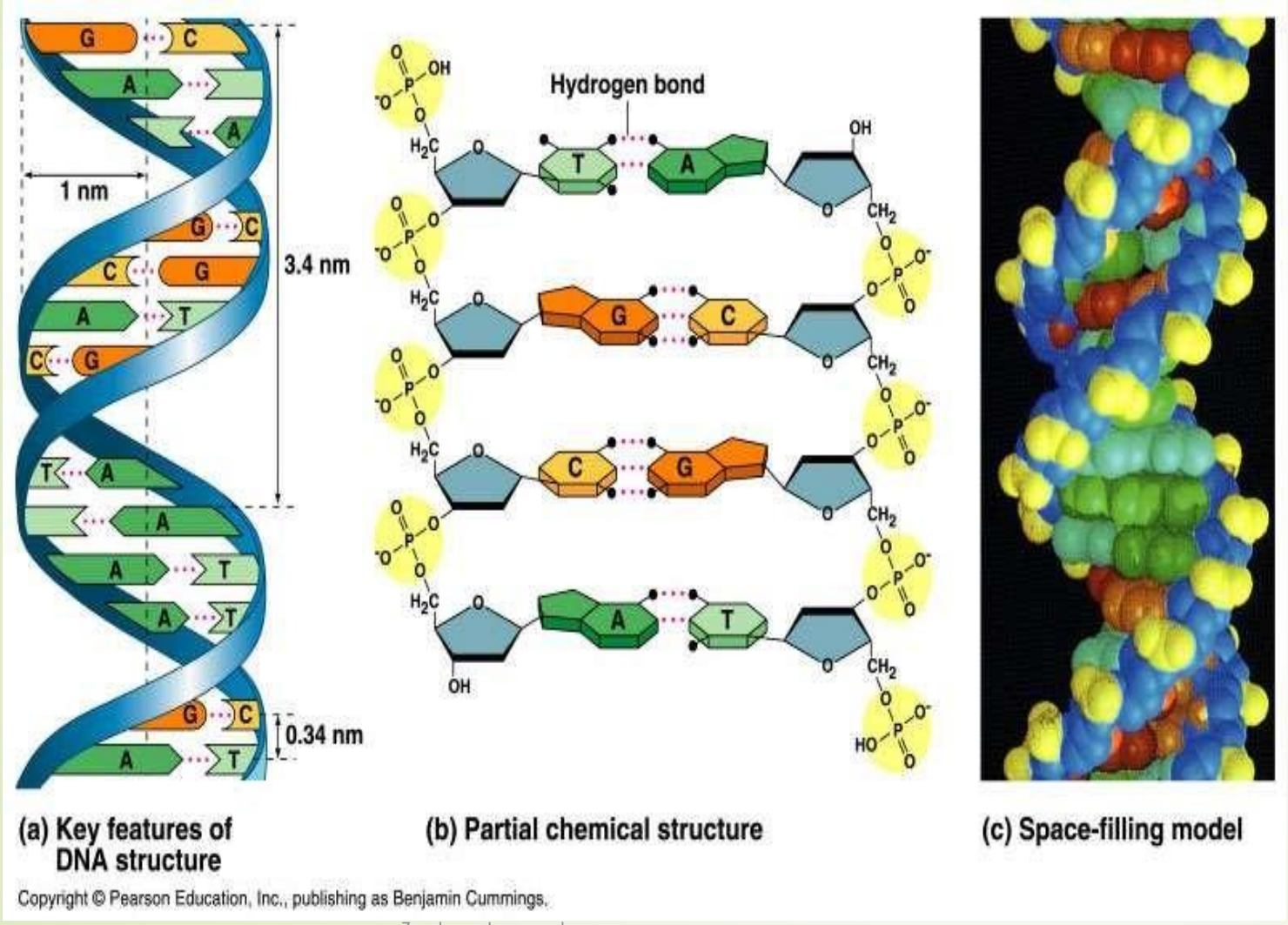




Biogeochemical Cycles-Phosphorous Cycle

Lecture 41

Biogeochemical Cycles-
Phosphorous Cycle

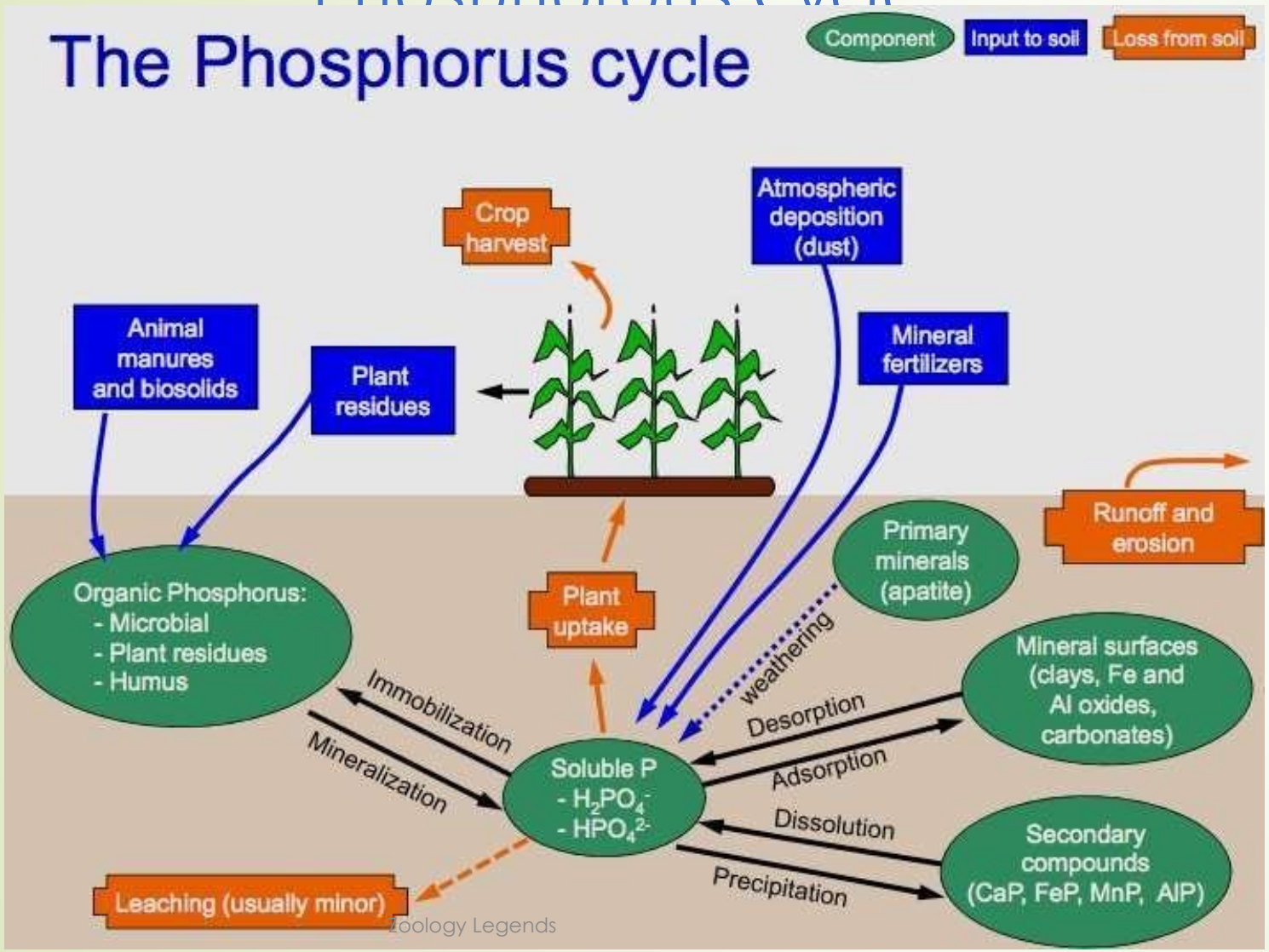


Phosphorus is a necessary and important constituent of the protoplasm in the living organisms

ATP, ADP, AMP, NDPP, Phospholipids of cell membrane
back bone of nucleotide chain, production of DNA

85% P found in bones and teeth's

Biogeochemical Cycles- Phosphorous Cycle



1. Reservoirs of phosphorus are the rocks- formed in past geological ages

2. Erosion of these deposits release phosphates in the ecosystem.

3. Much of it escapes into the sea where part of it is lost to the deep sediments and some of it deposited in the shallow marine sediments

Biogeochemical Cycles-

Phosphorous Cycle

4. Plants take up inorganic phosphate as orthophosphate ions.

5. Available form of Phosphate (PO4) returns to sea

6. Animal consume plants- phosphorous

7. Death and decay (decomposition), excreta of animals, bones and teeth –Phosphorous return to nature

Biogeochemical Cycles

Lecture 42

Biogeochemical Cycles- Carbon Cycle

Sources of Carbon

1. Atmospheric carbon dioxide is the source of all carbon in both living organisms as well as in the fossils (used as fossil fuel).
2. It is highly soluble in water.
3. Oceans also contain large quantities of dissolved carbon dioxide and bicarbonates.

Biogeochemical Cycles- Carbon Cycle

Important Steps in Carbon cycle

1. Photosynthesis
2. Respiration
3. Decomposition
4. Combustion

Biogeochemical Cycles- Carbon Cycle

Impact of human activities

Carbon dioxide is continuously increasing in the atmosphere due to human activities such as industrialization, urbanization and increased use of automobiles. This increase in atmospheric CO2 is bading to green house effect and global warming.

Lecturer 43

Population Ecology- Definition





Any group of organisms of the same species occupying a particular place and functioning part of biotic community, which in turn, is defined as assembly of populations functioning as unit ,showing interactions due to coevolved metabolic transformations in particular physical habitat.

Population Ecology-Indices of Population density

Population density types

Crude density- the number of populations per unit of total space

Ecological density- number of populations per unit of habitat space (actual area where population colonize

Population Ecology-Indices of Population density

Changes in Population density (decrease or increase measurement)

Relative abundance-time related, e.g, no. of birds seen/hr

Frequency of occurrence- % of sample plots occupied by that species or single

Population Ecology-Indices of Population density

Lecturer 44

Population Ecology- Population density

Methods for estimating Population density

1. Lincoln or Petersen index- relies on capturing and marking some fraction of the total population and using this fraction to estimate the total population density

Population Ecology- Population density

Lincoln index Equation

$$\text{Total Population (N)} = \frac{\text{Number of individuals caught \& released 1st time (M)} \times \text{Total of 2nd catch (n)}}{\text{Number of individuals recaptured who were marked (m)}}$$

Population Ecology- Population density

Minimum known alive- another method of mark – recaptured method to estimate population densities over an extended period of time. **Also called calendar-of-catches method**

Population Ecology- Population density

Total count-large or with organisms that aggregate into large breeding colonies

Quadrate or transect sampling-counting of organisms of a single species in plots or samples area.

Population Ecology- Population density

Removal sampling- no. of organisms removed from an area in successive samples is plotted on y-axis and total number previously removed is plotted on x-axis

Plotless methods-applicable to sessile organisms such as trees, random points, distance of nearest individual measured

Lecturer
45

Population Ecology-Population Natality

Natality- ability of population to increase by reproduction-birth rate

Maximum natality- (absolute or physiological)-theoretical maximum production of new individuals under ideal conditions and constant of any population (no ecological limiting factors)

Population Ecology-Population Natality

Ecological or realized Natality- population increase under actual or specific environmental conditions
e.g: 400 births/year among 10,000 couples
Crude birth rate is 400/year while realized natality is 4 per 100

Population Ecology-Population Mortality

Mortality- death rate of individuals in the population**.**

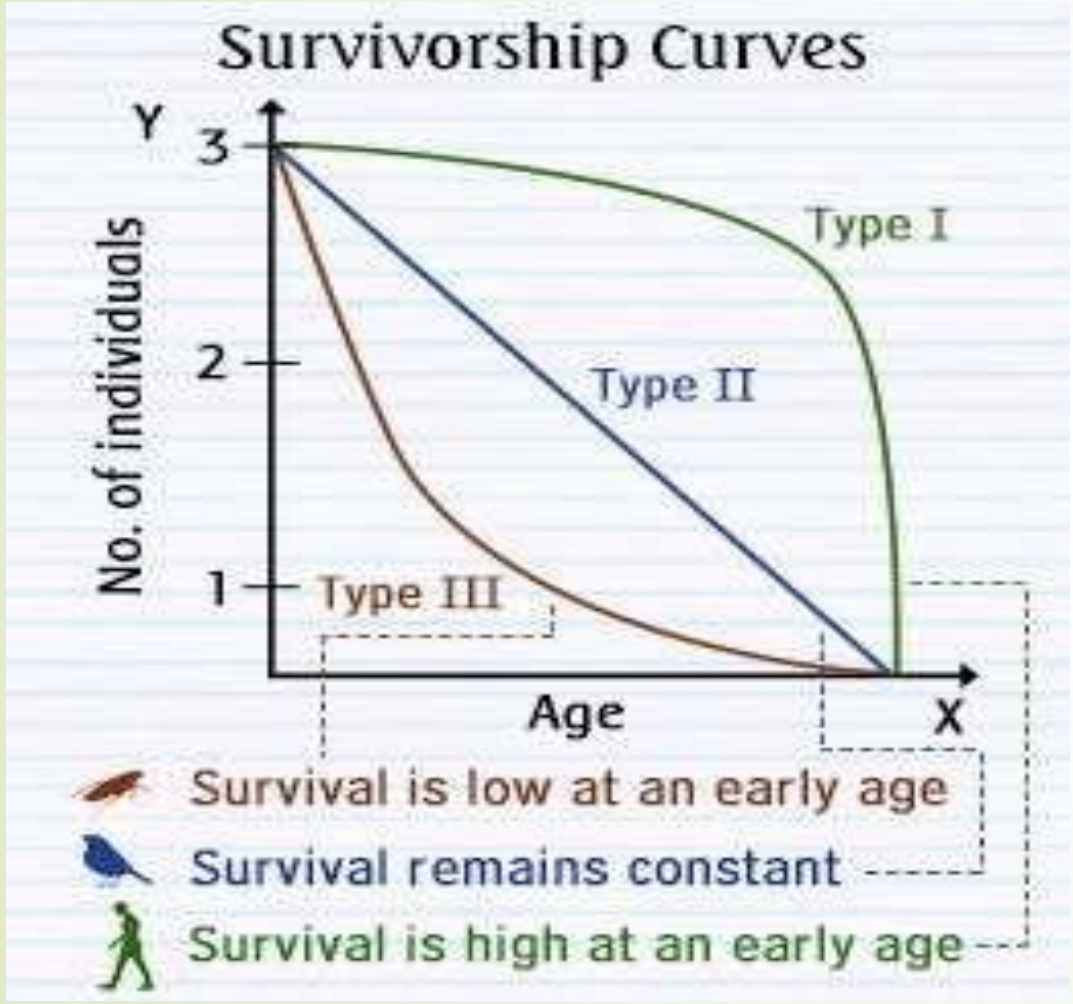
Ecological or realized mortality-the loss of individuals under a given environmental conditions

Population Ecology-Population Mortality

Minimum Mortality- constant for a population- represents the minimum loss under ideal or non-limiting conditions

Lecturer
46

Population Ecology-Survivorship curve

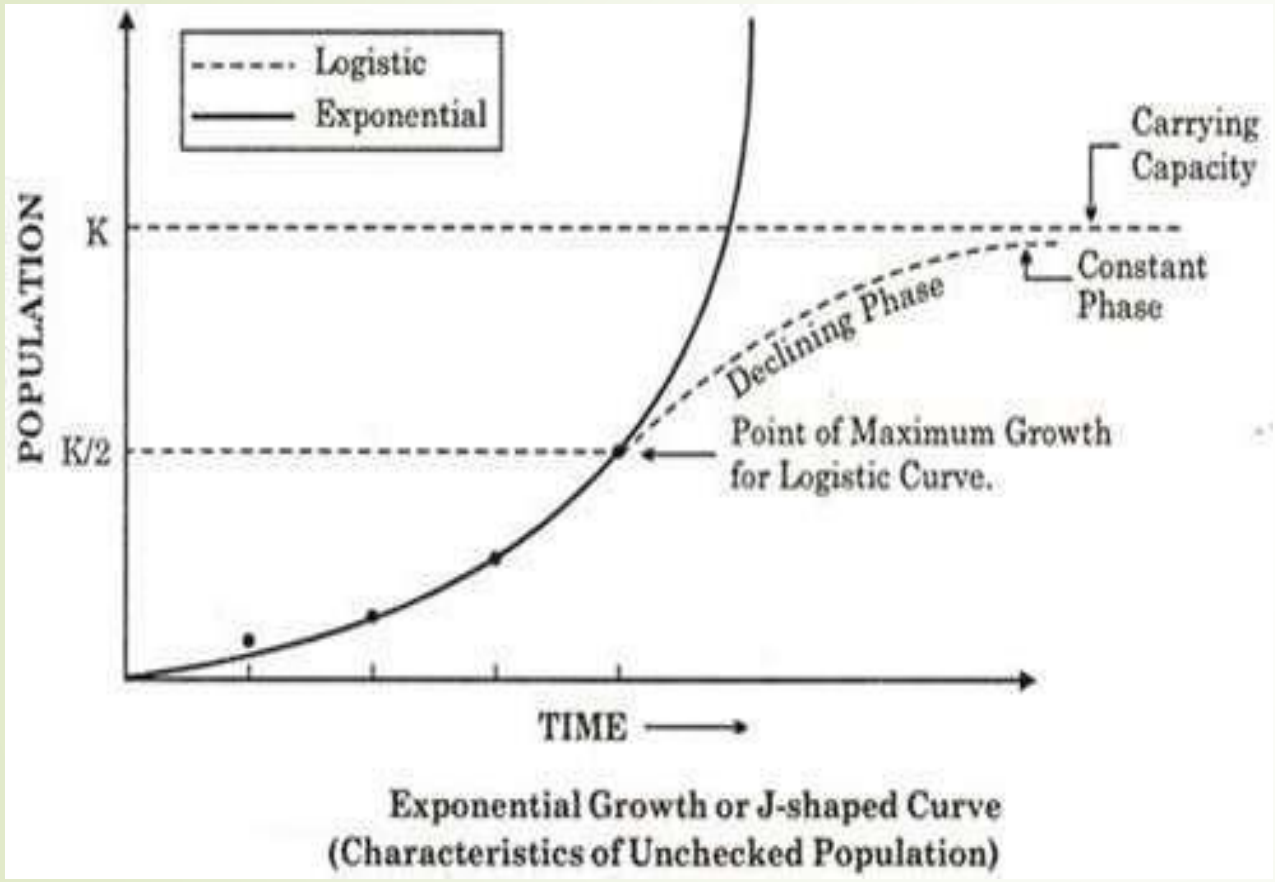


Zoology Legends

Curves plotted from life table data, no surviving at beginning of age out of 1000 on vertical axis and time interval at x-axis, resulting curve is called survivorship curve

Three types of curves

Population Ecology-carrying capacity

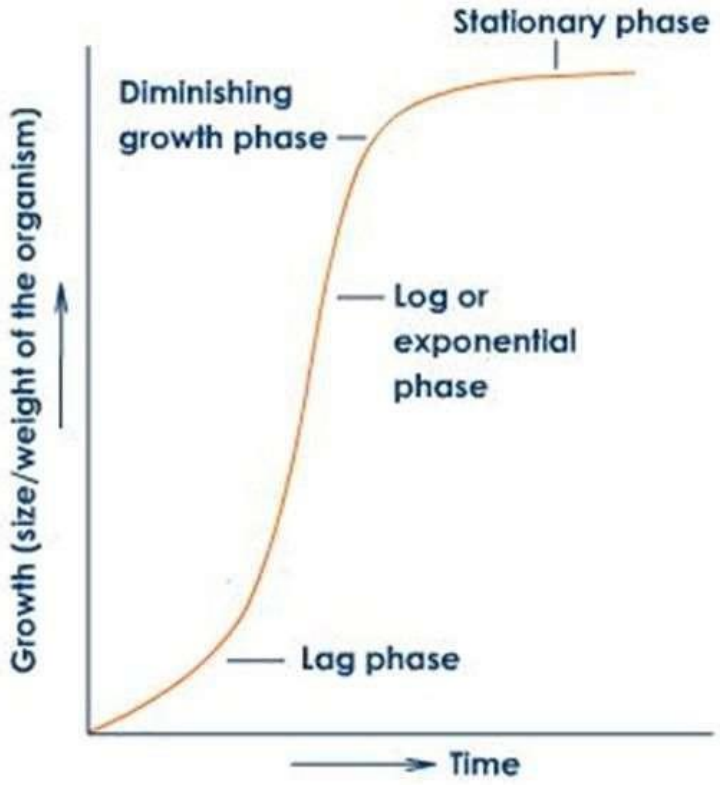


The population size that a particular environment can support is the environment's carrying capacity and is symbolized by “K”

Growth curves can be
J-shaped
S (sigmoid) shaped

Population Ecology-Survivorship curve

Logistic growth form (S-shaped curve)



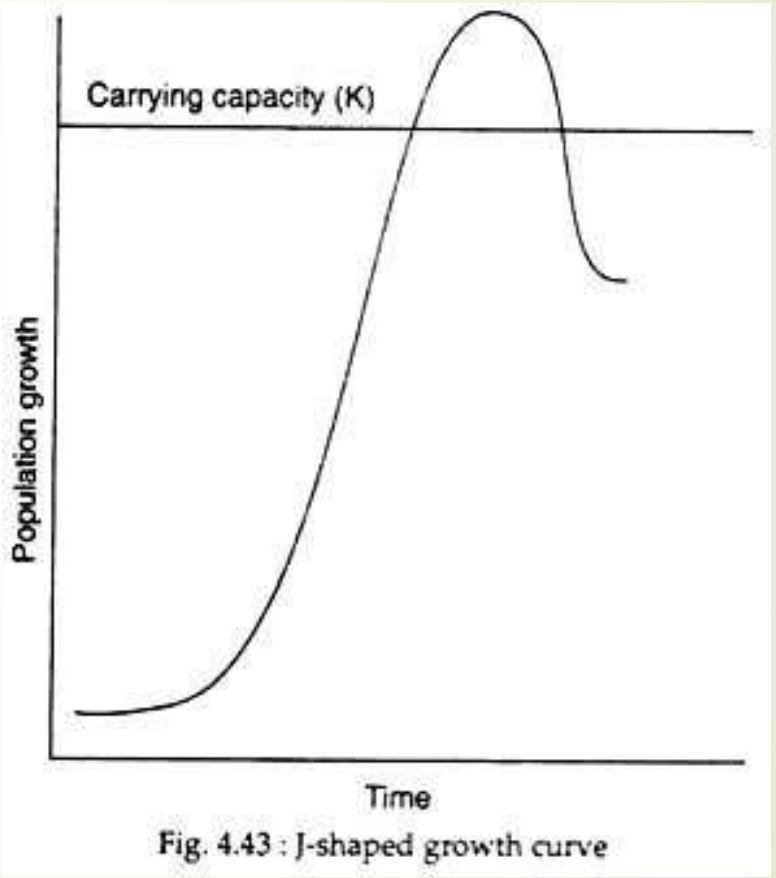
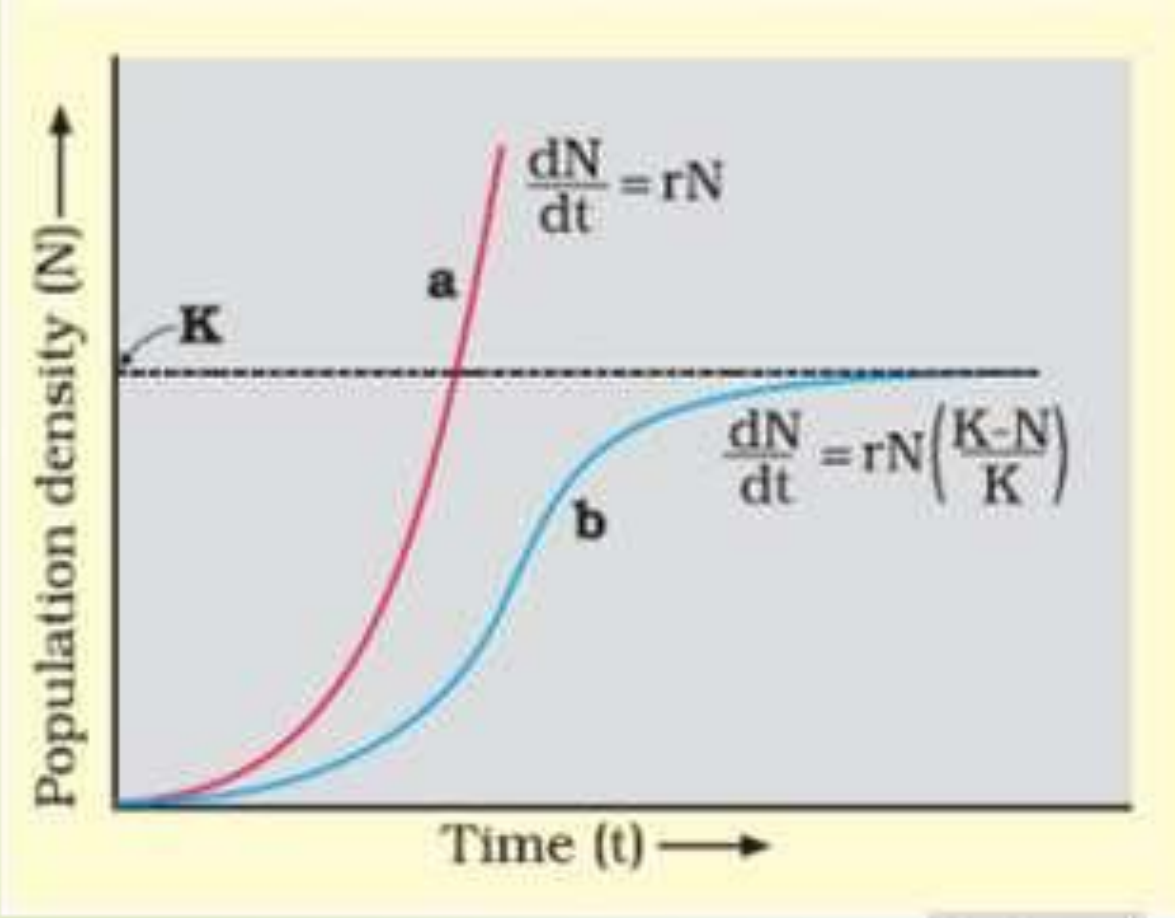
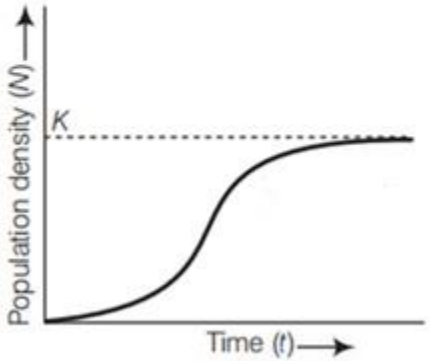
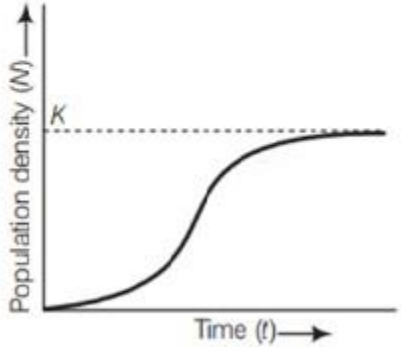


Fig. 4.43 : J-shaped growth curve

Population Ecology-Survivorship curve



J- shaped growth	S- shaped growth
In this growth model, resources are unlimited in the habitat.	Resources are limited in a habitat.
Population occurs in J shaped growth manner	Population occurs in S shaped growth manner
The growth curve is initially lag phase followed by acceleration phase.	The growth curve initially in a lag phase followed by acceleration phase then in a steady state.
The J-shaped growth curve can be represented as:	The S-shaped growth curve can be represented as:
 K = carrying capacity	 K= carrying capacity

Community Ecology-Definition

Lecturer 47

Community Ecology- Definition

An ecological **community** consists of all the populations of all the different species that live together in a particular area

Community Ecology consists of all species that interact in a certain area

Community Ecology- Properties

Properties of Community or community structure

- Species Richness (number of different types of species, more in areas close to equator)
- Species evenness (abundance of each species)
- Species- Area Effect (large areas contain more species)

Community Ecology- Definition

- **Community ecology** seeks to understand how species interact by studying many **different kinds of relationships** between organisms.

For example:

Animal-animal interactions,
animal-plant interactions
plant-plant interactions

Community Ecology-Interactions

Lecturer 48

Community Ecology- Interactions

Each species interactions are analyzed by using **three key** themes

1. Species interactions can affect the distribution and abundance of a particular species

e.g: the effect of disease-carrying tsetse flies on the distribution of cattle in Africa

Community Ecology- Interactions

2.Species act as agents of natural selection when they interact

3.The outcome of interactions among species is dynamic and conditional

Concept of coevolution

Community Ecology-Types of Interactions

200



Interactions between same species in a community are called **intraspecific** —*intra-* means “same.“

- helps nature to keep the population under control
- results in the survival of the fittest

Community Ecology- Interactions

Types of Interactions

Interactions between different species in a community are called **interspecific interactions**—*inter-* means "between.“

Community Ecology- Interactions

The main types of interspecific interactions include

Competition (-/-),
Predation (+/-),
Mutualism, (+/+),
Commensalism (+/0),
Parasitism(+/-).

Community Ecology-Competition

Lecturer 49

Community Ecology- Competition

Competition (-/- interaction),

- where one or both inhibit the population of the other

1. Direct Competition for same recourse

- Competition negatively affects both species

e.g: competition between birds for limited **seeds** for food,
Woodpeckers and squirrels compete for nesting in the same holes and spaces in trees

Community Ecology- Competition

2- Interference competition: occurs when an individual of one species directly interferes with an individual of another species.
e.g: a plant releasing allelopathic chemicals to impede the growth of a competing species.

Community Ecology- Competition

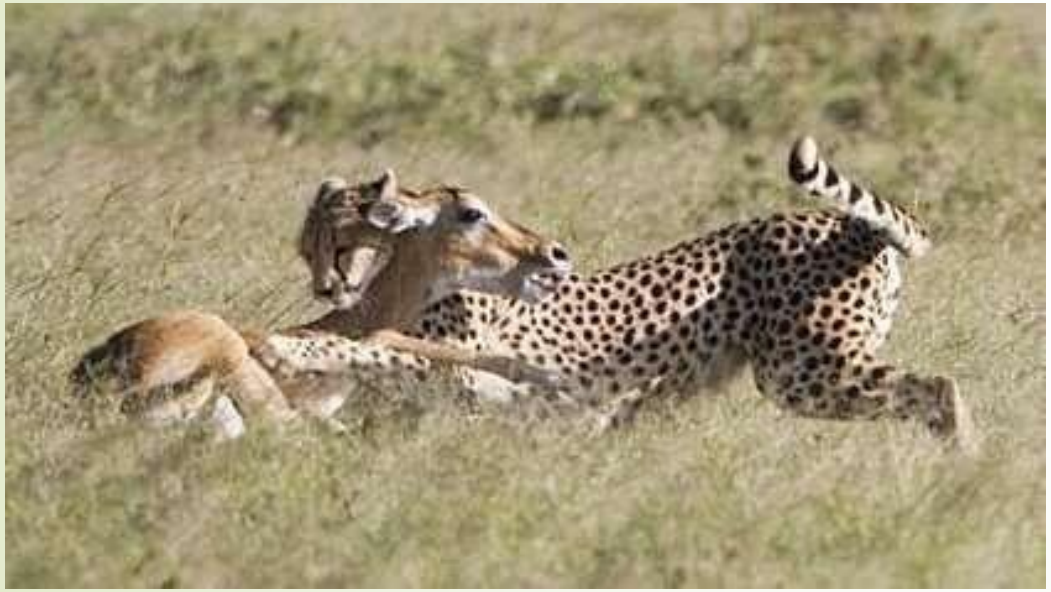
Competition (-/- interaction),

Competition can be minimized if two species with overlapping niches evolve by natural selection to utilize less similar resources, resulting in **resource partitioning**
e.g: five species of warblers that all used the same caterpillar prey spruce tree.

Predation& Herbivory

Lecture 50

Community Ecology- Predation



Predation: (+ –) interaction

- One Wins, One Loses
- predator-prey interaction, in which one species kills and consumes another
- e.g: Lion is predator and deer is prey
- Predation shows cyclical patterns of predator/prey population sizes; predators increase in numbers when prey species are plentiful.

Community Ecology- Herbivory



Herbivory: (+ –) interaction

- When Predation involves involve an animal or insect consuming part of a plant, known as **herbivory**.
- Animals that feed on plants by cropping portions of the plant, but usually not killing the plant, are herbivores
- Plants have spines and toxins-defensive

Symbiosis

Lecture 51

Community Ecology-Symbiosis

Symbiosis:

Interspecific interactions in which two species live together in a long-term, intimate association

- Symbiotic relationships keep a delicate balance
- can help individual species to evolve or change and even thrive

Community Ecology-Symbiosis

Symbiosis: mutualistic association between termite and their intestinal flagellates. Termites can eat wood but have no enzymes to digest it. However, their intestine contains certain flagellate protists (protozoans) that have the necessary enzymes to digest the cellulose of the wood eaten by termites and convert it into sugar.. Both termite and flagellates cannot survive without each other

Community Ecology-Symbiosis

Symbiosis:

- provide people with food, populate the planet with trees and plants, and keep animal and plant populations in balance
- Without symbiotic relationships, no coral reefs, trees might not proliferate as far and wide, aided by birds and insects that transport seeds afar

Parasitism

Lecture 52

Community Ecology-Symbiosis

Symbiosis:
Adaptations arising through coevolution come from two different species living showing various relationships or interactions

Parasitism
Commensalism
Mutualism

Community Ecology-Symbiosis

Parasitism: (+/- interaction) two species have a close, lasting interaction that is beneficial to one, the **parasite**, and harmful to the other, the **host**

- Most of the time, the parasite feeds on the host's body but does not kill the host

Community Ecology-Symbiosis

Parasitism:

Two types of hosts

- definitive host (A definitive host provides a home to an adult parasite or sexual stages of parasite)
- intermediate host (host unknowingly offers a home to a juvenile parasite or asexual stage of parasite)

Community Ecology-Symbiosis



Parasitism:

Ecto-parasites:

Ticks- blood-sucking insects that thrive on the blood of its victims

Also transfer infectious diseases to healthy organisms

Endoparasites-tapeworm in human intestine

Community Ecology-Symbiosis

Parasitism examples

Plant parasite: Dodder (*Cuscuta)* plant is a parasitic weed that obtains moisture and nourishment by attaching to a green, living plant.

Animal parasite: *Ascaris* or round worms are internal parasites found in the human intestine

commensalism

Lecture 53

Community Ecology-Symbiosis

Commensalism:
(+/0 interaction)

Two species have a long-term interaction that is beneficial to one and has no positive or negative effect on the other

Community Ecology-Symbiosis





Sucker fish, often attaches to a shark by means of its sucker which is present on the top of its head. This helps remora get protection, a free ride as well as a meal from the left over of the shark's meal

Community Ecology-Symbiosis



Commensalism: example- grazing cattle and cattle egrets. As the cattle graze in the grass, they stir up the insects living there, allowing the cattle egret a tasty meal. The cattle egrets get a meal, but the cattle receive nothing in return from the long-necked birds, nor are they harmed by the relationship.

Mutualism

Lecture 54

Community Ecology-Symbiosis



Mutualism: (+/+) interaction

- two species have a long-term interaction that is beneficial to both of them
- **Example-**
- Lichens **and** Mycorrhizae

Mutualism: examples

There are many different examples of mutualistic relationships:

- Plants and microbes *e.g.* rhizobium in root nodules
- Protists and fungi *e.g.* lichen
- Terrestrial plants and insects, *e.g.* pollination
- Animals and protists/bacteria *e.g.* ruminants, corals
- Animals and other animals *e.g.* crocodile and plover bird







Amensalism

Lecture 55

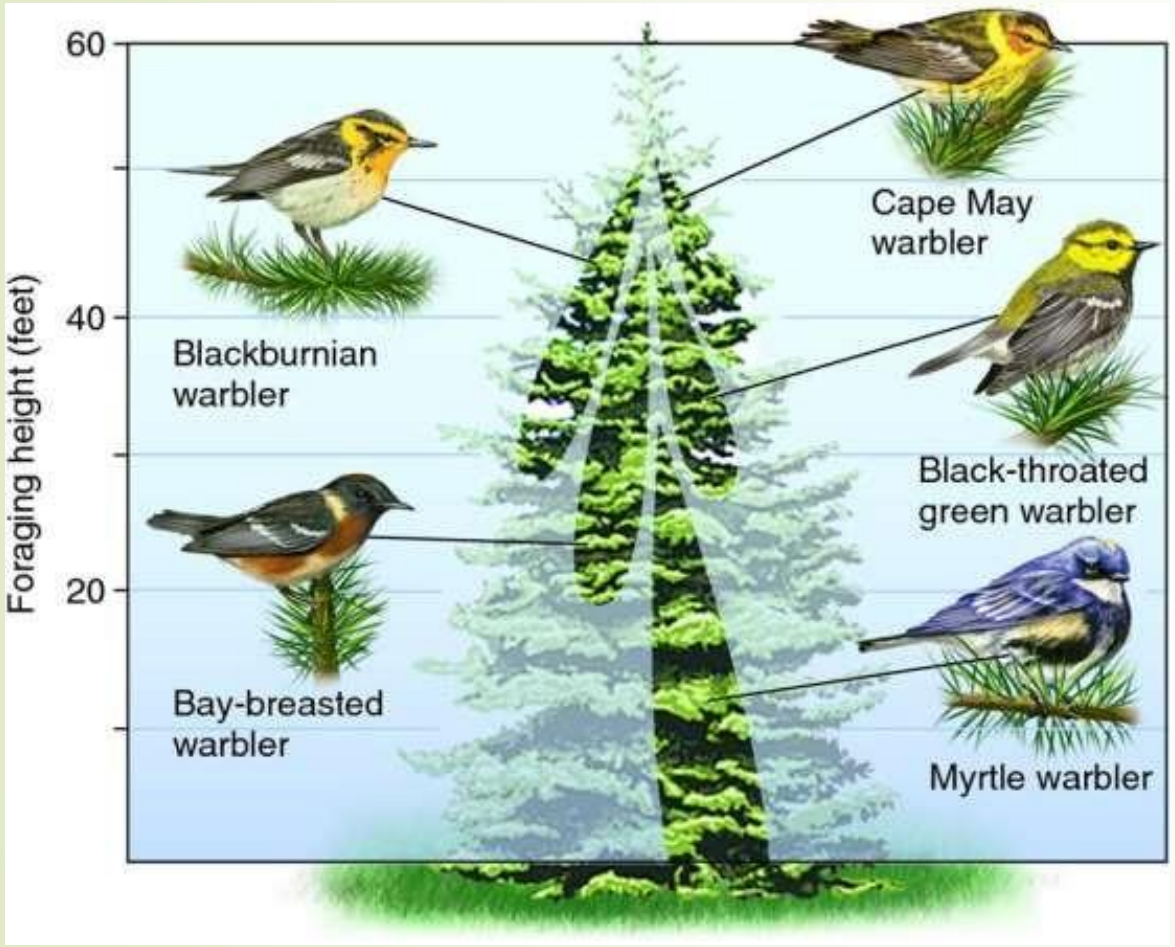
Community Ecology-Symbiosis

Amensalism: (-/0,Interaction)
It describes an interaction in which the presence of one species has a negative effect on another, but the first species is unaffected.
Example: a herd of elephants walking across a landscape may crush fragile plants

Ecological Niche

Lesson No. 56

Community Ecology-Ecological Niche



- an important concept of community structure.
- Ecological niche involves both the habitat of organism, type of food it eats, where it lives, reproduce and its relationship with other species, the role its plays in its habitat.

Succession

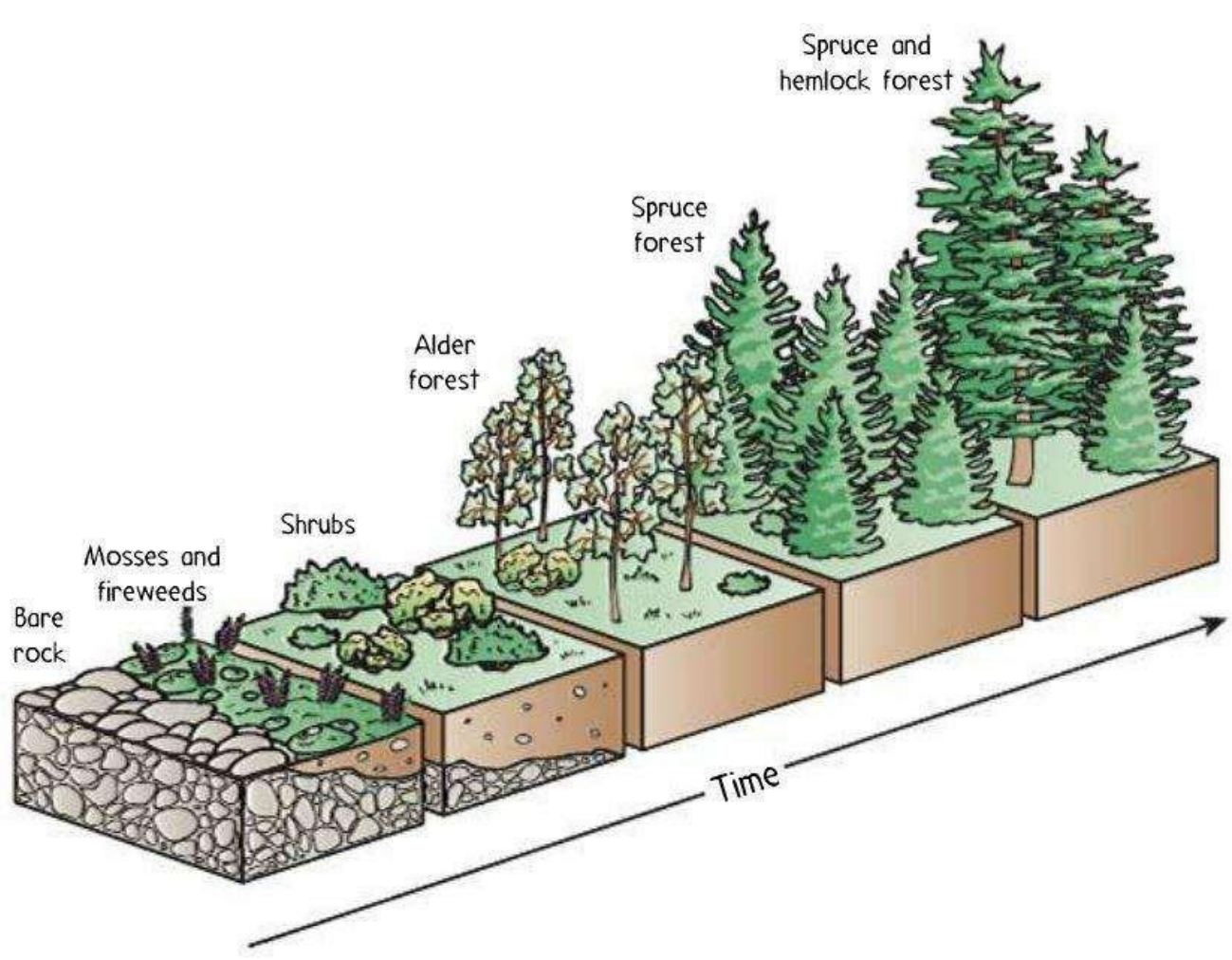
Lesson No.57

Community Ecology-

Succession

Succession • The process by which communities of plant and animal species in an area are replaced by another over a period of time is known as ecological succession

Community Ecology-
SUCCESSION



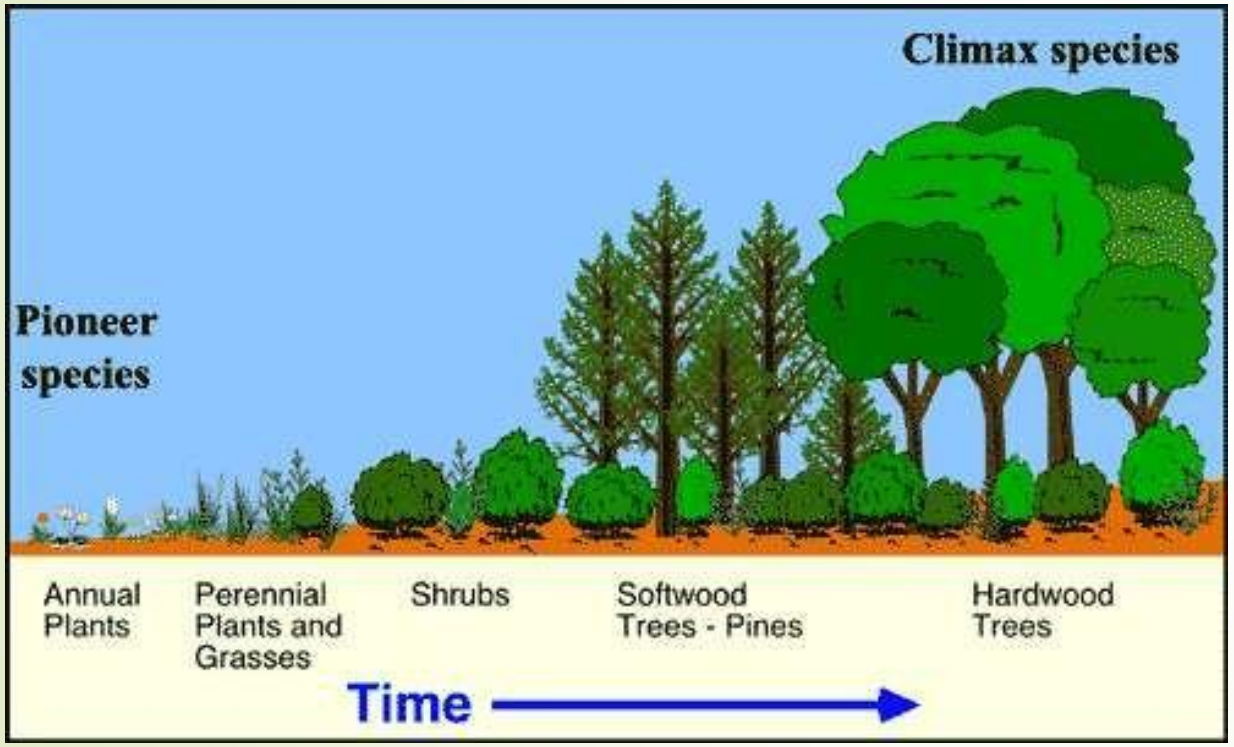
<u>SUCCESSION</u> • Sequential and gradual growth of a community **Or** The dominant members of a community often change a community in predictable ways in a process called **succession**

Pioneer community

Seral stage

Climax community

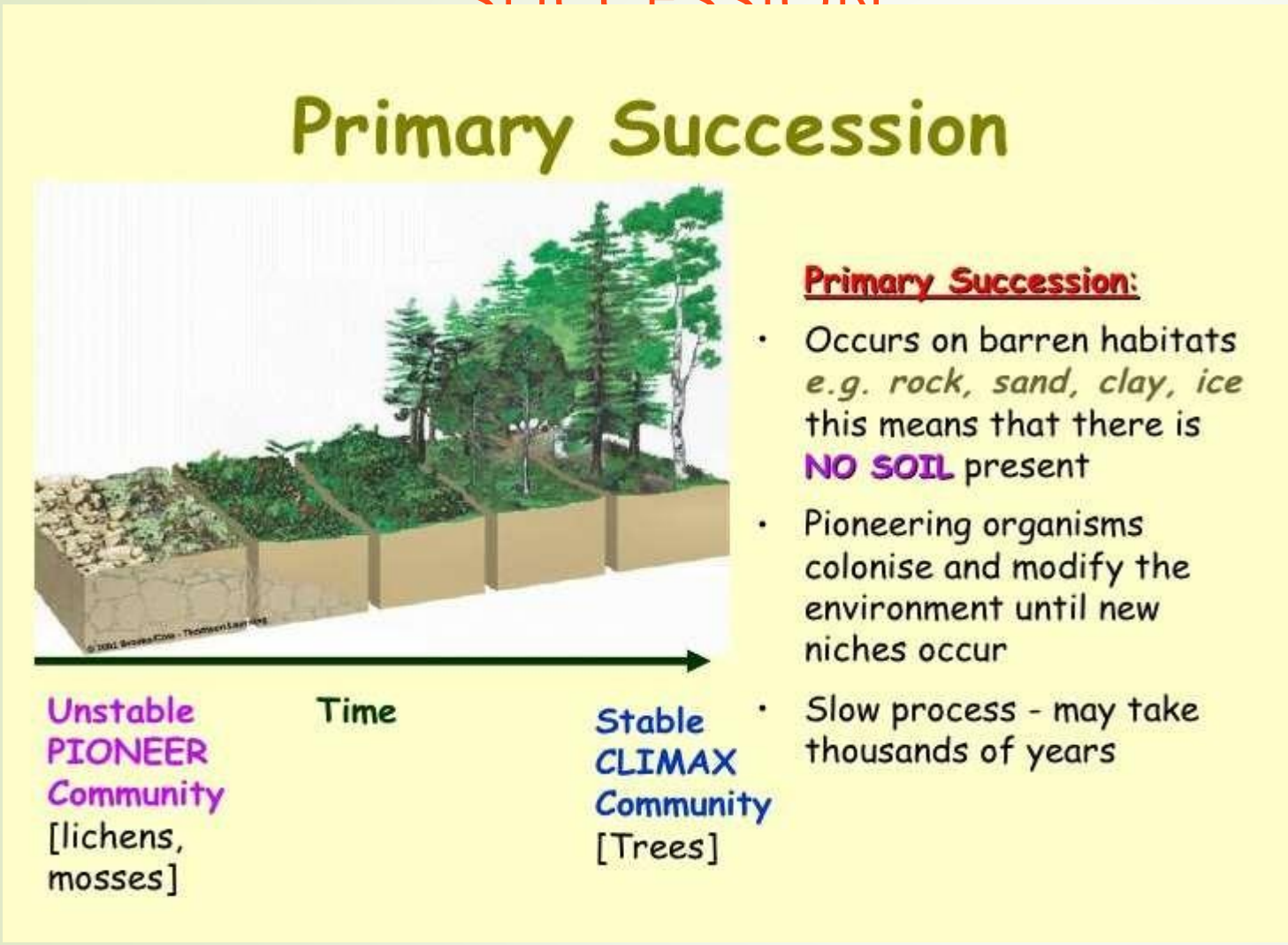
Community Ecology- SUCCESSION



Pioneer species and Primary succession

- Communities may begin in areas nearly devoid of life. The first community to become established in an area is called the **pioneer community.**
- Microbes, lichens and mosses create soil before other plants can grow

Community Ecology- SUCCESSION



Seral stage

Over thousands of years nutrients accumulate, and the characteristics of the ecosystem change.

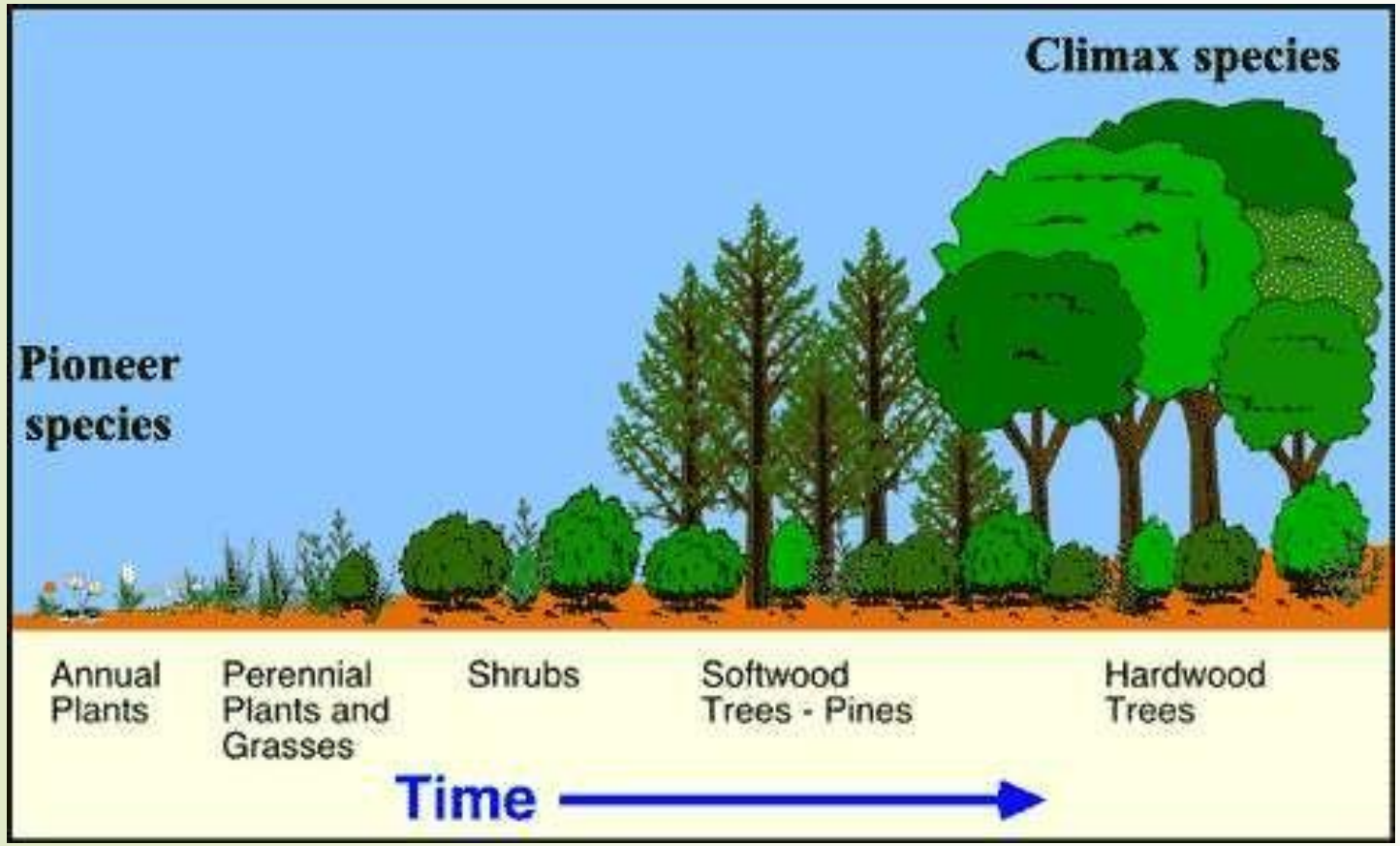
Each successional stage is called a **seral stage,**

and the entire successional sequence is a **sere**

Succession

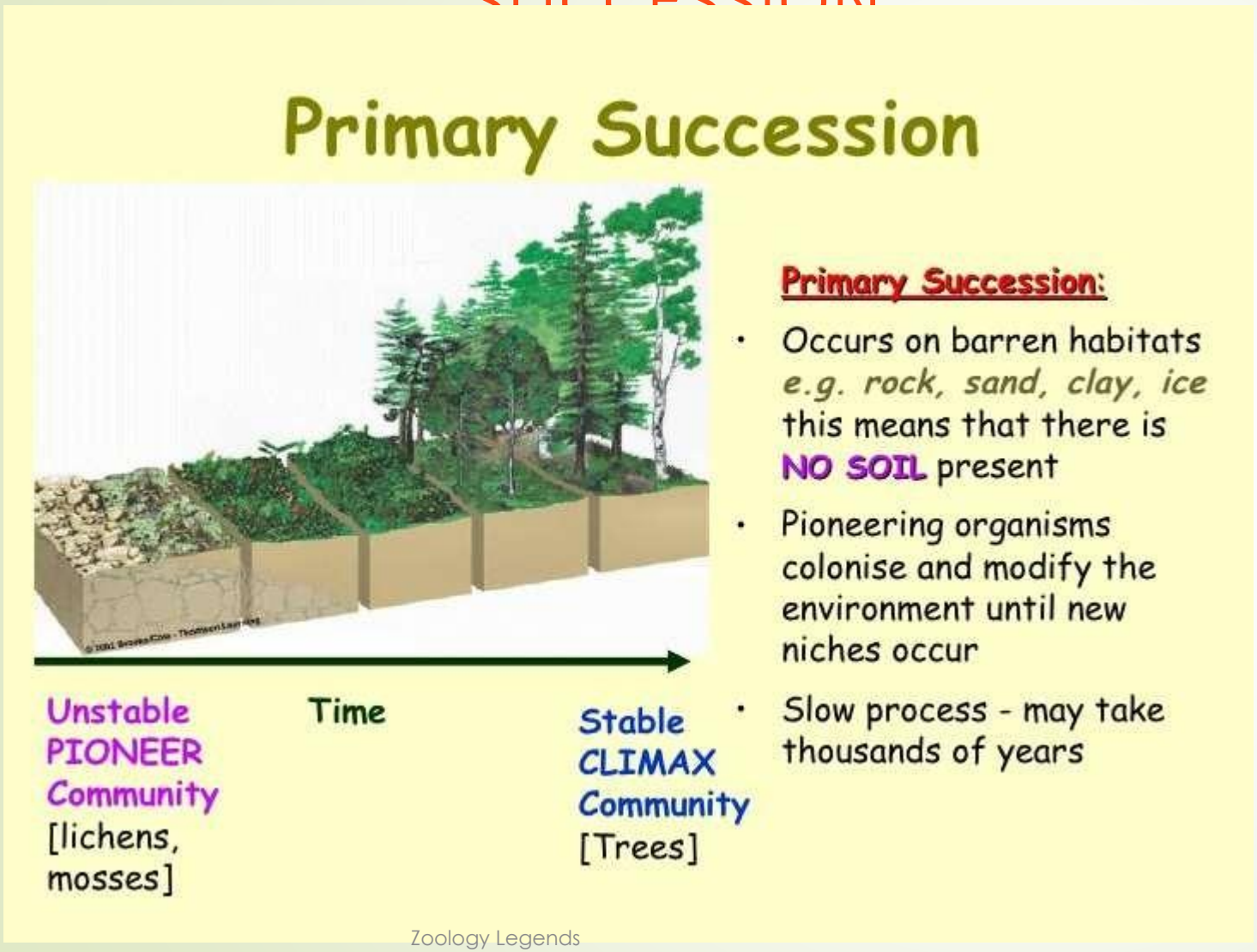
Lesson No.58

Community Ecology- SUCCESSION



Succession occurs because the dominant life-forms of a sere gradually make the area less favorable for themselves, but more favorable for organisms of the next successional stage. The final community is the **climax community,** usually have complex structure and high species diversity

Community Ecology- SUCCESSION



Two types:
Primary and secondary
Primary succession occurs when land is first formed.
1. Microbes, lichens and mosses must create soil before other plants can grow. 2. Pioneer species: first species to dominate early in succession (small and fast growing)

Community Ecology- SUCCESSION

Secondary succession occurs when succession must start over after the destruction of a climax community (marked by the domination of long-lived species and great diversity).
- takes place on sites that have already supported life

World Biomes

Lesson No.59

World's Biomes

The natural broad biotic zones of the biosphere called, **Biomes**.
Each biome is characterized by uniform life form of vegetation such as grass, desert plants, deciduous trees or coniferous trees.

World's Ecosystems

A Biome is a large ecosystem which is embracing the large land scape, characterized by specific flora and fauna. Ecosystem is defined as a functionally independent unit (of nature) where living organisms interact among themselves as well as with their physical environment. Ecosystem is a self sustaining unit of nature.

World's Ecosystems

Lesson No.60

World's Ecosystems

Types of Biomes/Ecosystem

Natural and Artificial

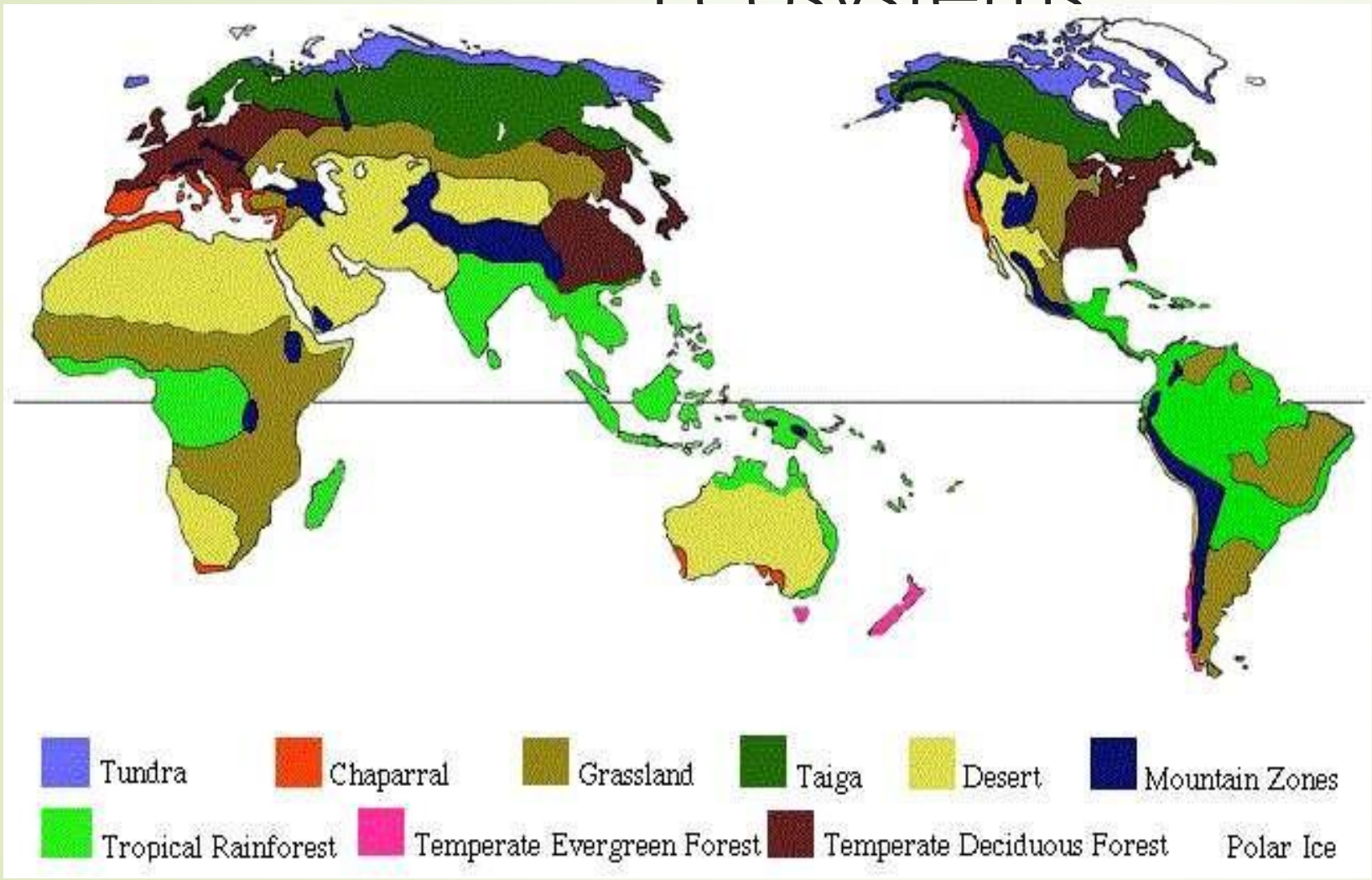
Natural-

a) Terrestrial ecosystem
b) Aquatic ecosystem

Artificial or Man-made

e.g: Crop
lands and aquarium

World's Ecosystems



a) Terrestrial ecosystem

Terrestrial :

These are the biomes found on land

e.g„ Tundra, forest, deserts, grasslands

Thanks You!